The Pennsylvania State University

The Graduate School

SOCIOECONOMIC AND TECHNOLOGICAL FACTORS INFLUENCING TECHNOLOGY ADOPTION IN CACAO FARMS OF TWO POST-CONFLICT REGIONS IN COLOMBIA.

A Thesis in

Rural Sociology and International Agriculture and Development

by

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Submitted in Partial Fulfillment
of the Requirements
for the Degree of

Master of Science

August 2021

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ABSTRACT

After more than 50 years of conflict, in late 2016, the Peace Agreement was signed between the Colombian government and the guerrilla group FARC-EP, the main illegal armed group that funded many of its operations through drug trafficking. This agreement represents an unprecedented opportunity for peace, not only because it promotes actions against illicit drugs and their devastating effects on the rural population, but also because it revitalizes and gives ground to the land reform postponed many times by the government during the last century.

Among the strategies proposed by the Agreement is the Rural Development Plans with a Territorial Approach (PDET), which will be implemented for 10 years in 170 municipalities of special attention in the post-conflict period, and the National Comprehensive Plan for Substitution of Illicit Crops (PNIS) which focuses on eradicating coca plantations and supporting those involved, especially farmers and leaf collectors, to move towards licit activities.

In this context, the cultivation of cacao stands out as a productive alternative for these communities. The national government, international aid organizations, and the private sector, including the National Federation of Cacao Growers (FEDECACAO), have made favorable statements regarding the potential (in terms of markets, quality, and added value) of this product to appropriately replace illicit crops and promote rural development in post-conflict areas. However, among the concerns of this proposal is the low capacity of the current cacao sector to materialize this potential, especially regarding achieving production volumes and quality that make the business viable enough to allow for the abandonment of coca.

Among other factors associated with the poor results, researchers have highlighted the low usage of technologies necessary to generate the expected production. Although some available studies evaluate the adoption levels in Colombia, few studies investigate how social and economic factors influence the technical decisions of cacao farmers for the management of their plantations. This is especially important in cacao areas where farmers also face the effects of conflict (47% of the planted area and 46% of cacao production in Colombia is located in 149 municipalities of the 170 recognized by the Government as areas highly affected by the conflict). To address this shortcoming, this research focuses on identifying and analyzing the effects of socio-economic and technological factors on the technological adoption of cacao farms in two post-conflict regions in Colombia, Montes de María and Sierra Nevada de Santa Marta-Perijá.

Through a document review process, data collection in the field (through interviews, focus groups, and workshops), and the application of a survey during 2018 and 2019, it is concluded that both individuals and contextual factors affect the adoption of technologies and practices those farmers consider relevant to improve the quality and productivity of cacao.

In conclusion, the implementation of maintenance pruning, monilia control, sowing density, and harvest and post-harvest practices in cacao in the study regions are influenced differently by socioeconomic factors. Among them stand out mainly factors such as cacao relevance into farmers' income, cacao cultivation size, training, technical assistance access, mechanism or channel utilized to commercialize the cacao beans, membership in a farmer's organization, and cacao prices. Other factors were also identified but require additional research to provide more conclusive results. Among them are transportation costs, credit, labor availability, historical roots of cacao in the region, and participation in cacao quality awards. Other practices and technologies, such as irrigation, the selection of planting material from the application of the compatibility matrix, were also analyzed.

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ACKNOWLEDGEMENTS[1]

There are many people to thank, without whom there would not have been completed this thesis. First, I would like to express my gratitude to Dr. Leland Glenna for his support and encouragement during this process. He is a kind and care guide I could always go with any question or concern around my education as a rural sociologist. I also want to acknowledge and sincerely thank Dr. Siela Maximova for leading the Cacao for Peace project and encouraging me to participate in this enriching experience. Her continuous support has been heartwarming. I also thank Dr. Leif Jensen for his valuable contribution to my learning process and in my committee.

Thanks to my family and friends for encouraging me and for being my permanent support during this process, especially to my husband Jose, my little Gabriel, and my mom for being so tolerant and patient, and companying me in good and bad moments, including the horrible Covid-19 pandemic that constantly attempted to undermine my enthusiasm.

To Agrosavia, special thanks for its generous support. Primarily to Dr. Juan Lucas Restrepo and researchers Anais Hernandez, Rafael Novoa, Darwin Martinez, Ledy Ropero, Wilman Londoño, Cesar Baquero, and Marlon Yacomelo who collaborated reviewing materials and collecting data. Finally, gratitude to Daniel Macías for his willingness to guide me in the Sierra Nevada de Santa Marta and connect me to the communities. I hope his courage and braveness allow him to continue helping his people.

Finally, thanks to Drs. Deanna Behring and Melanie Miller-Foster from International Programs in the College of Agricultural Sciences who constantly provided me their support through the INTAD program.

[1] Funding for this research was provided by Fulbright Colombia, USDA-FAS, and USAID via Cacao for Peace project and the College of Agricultural Sciences at Penn State University; findings and conclusions do not necessarily reflect their views.

CHAPTER 1: INTRODUCTION

According to *Monitoring of Territories Affected by Illicit Crops* from the United Nations' Office on Drugs and Crime, Colombia reached 169,000 hectares of coca by the end of 2018 (UNODC, 2019b). Beyond showing a 16% increase in the area compared to 2016, this figure is an indicator of the complexities faced by the country toward the implementation of the Peace Agreement signed by the government and the principal guerrilla group *Fuerzas Armadas Revolucionarias de Colombia (FARC-EP)* (Garzón, Gelvez, & Bernal, 2019; UNODC, 2019b).

Given the economic and social impacts[2] of this phenomenon and its relationship with violence and crime[3], the Peace Agreement incorporated commitments associated with illicit crops eradication through the operation of the *National Comprehensive Plan for Substitution of Illicit Crops (PNIS)* (Gobierno Nacional de Colombia, FARC-EP, & Paises Garantes, 2016). This program operates by voluntary substitution agreements between households involved in coca and the government. According to the official announcements, signing the agreement implied a commitment of households to no relapse on cultivation, or involvement with any activity associated with inputs commercialization or processing (Alta Consejería para el Posconflicto, 2017). In return, the government would provide them with a package of assistance for at least two years.

In addition to illicit crop substitution, the Agreement also proposed a set of programs and plans aimed to contribute to the structural transformation of rurality, offering conditions to

[2] Illicit crops in Colombia correspond 99% to coca and generates about 3% of agricultural GDP. The on-farm value of coca leaf and its byproducts was estimated at US$560 million in 2016 (UNODC, 2017). Likewise, UNODC (2017) calculated 106,000 households involved with the activity, including those dedicated to planting, harvesting ("raspachines") and primary processing. Other analyses estimated this would be between 123,000 and 150,000 families (CINEP & CERAC, 2018b; Garzón et al., 2019).

[3] 80% of homicides associated with the conflict in Colombia occurred in municipalities with coca plantations in 2018 (UNODC, 2019b).

alleviate poverty, promote equality, and ensure the enjoyment of citizens' rights. This set of actions was named *Integral Rural Reform (RRI)* (Gobierno Nacional de Colombia et al., 2016).

The measures anticipated in the RRI can be divided into two groups. The first consists of actions focused on democratizing land access to rural communities and guaranteeing property rights. Therefore, the focus is on land titling, registration, redistribution, and restitution of land. It also addresses the development and operation of a new rural cadaster and the agricultural frontier's delimitation to define the division between farming areas and those for conservation and other uses. (Gobierno Nacional de Colombia et al., 2016). The second group are National Plans aimed at providing public goods and services like road infrastructure, education, health, housing, electricity, irrigation, and research, among others (Gobierno Nacional de Colombia et al., 2016).

The Agreement also drew specific mechanisms to deploy all mentioned measures but focused on the most affected areas through the *Rural Development Plans with a Territorial Approach (PDET)* (Gobierno Nacional de Colombia et al., 2016; MADR, 2017). Thus, with Decree 893 of 2017, the government declared 170 municipalities, organized in 16 zones, as priority regions for attention. Those areas were identified based on their indicators of poverty, violence incidence, weakness in local administration, and coca areas (MADR, 2017). Most of them coincide with the areas where substitution of the illicit crops would be undertaken through PNIS.

While the willingness to eradicate illicit crops by communities represents a major step, the primary concern of the government and stakeholders is about the identification of feasible alternatives to replace coca and achieve economic and social stability in the affected regions. In that regard, Colombia has a history of diverse results in previous attempts of replacement within the so-called "alternative development" approach (Unidad de Consolidación Territorial, 2014, 2015; UNODC, 2010), involving not only agricultural activities but also sustainable use of forests

(Mican-Florez, 2015). Thus, identifying alternative activities is also a relevant milestone for programs oriented to victims or returned communities' hopes to rebuild their lives (Aguilera-Díaz, 2013; Bocchi, 2011; PNUD, 2010).

These initiatives generally have been undertaken by the government in collaboration with international aid agencies and multilateral cooperation organizations focused mainly on the promotion of oil palm, rubber, black pepper, cacao, and livestock as alternatives for economic development (UNODC, 2010; Zorro, 2005). Due to these experiences, during and after Peace Agreement negotiation, official and stakeholders' declarations highlighted cacao cultivation as one of the activities with the highest potential for this purpose, arguing not only opportunities in the national and international market but also with the advantage of having 95% of cacao recognized as fine flavor by ICCO[4] (Aparicio & Marín, 2017; Bautista, 2019; FEDECACAO, 2015a, 2017; MinAmbiente, 2018; Presidencia de la República, 2016a; Sociedad de Agricultores de Colombia, 2016; US Embassy in Colombia, 2016; Zuluaga & Vera, 2019).

Despite these expectations, the main concern is the weakness of cacao for displaying their productive potential in Colombia, and therefore, its limitations for effectively contributing to leverage the transition toward licit activities in the post-conflict. According to data from the Ministry of Agriculture and Rural Development (MADR), domestic production of cacao[5] increased by 5.4% on average annually between 2008 and 2018, reaching 101,020 tons. However, such a result is mainly due to the increase in harvested area (5.6% annual average), which is currently 190,470 hectares, while the national average yield decreased by 0.2% on average each year. Figures also reflect that, although the national average yield had improvements in some years,

[4] International Cocoa Organization (ICCO).

[5] According to MADR (2019), 548 municipalities had cacao planted area, although just 535 reported harvest area and production in 2018.

those were counteracted with deficient performances in subsequent years. During the analyzed period, yields fluctuated between 506 and 563 Kg/hectare/year, ending by 2018 with barely 530 Kg, although its potential reaches 2,000 Kg[6] (FEDECACAO, 2015b).

From the perspective of post-conflict regions, 149 of the 170 municipalities involved in PDETs have cacao cultivations concentrating 46% of the national area and 43% of production. The average annual yield of this group reached 492 Kg/hectare in 2018, which is 14.5% less compared to the 563 achieved by non-PDET municipalities producing cacao. Regarding municipalities with illicit crops, particularly those with coca, data shows that 167 (88%) of them also have cacao; although their area and production are like PDET municipalities, their performance is more limited. The cacao average yield for them reached 473 Kg/hectare in 2018, while those without coca areas obtained 581 Kg/hectare (MADR, 2017, 2019; ODC & Minjusticia, 2019)

Other sources reveal more discouraging results. Based on FAOSTAT, the national productivity of cacao cultivation decreased by 1.3% annually between 2008 and 2018 and barely reached 363 Kg/Hectare in the last year. Likewise, calculations with FEDECACAO data showed that the average yield for 2018 reached 323 Kg/Hectare (FAO, 2019). In general, when comparing with countries of the region, the performance of local cacao shows serious lags in productivity. For 2018, Peru achieved 839 Kg/Hectare and a growth rate of 2% when compared to 2017; Mexico obtained 487 Kg/Hectare corresponding to an increase of 5%; Ecuador's growth was 6% and reached 468 Kg/Hectare; and Brazil increased 4% and had a yield of 415 Kg/Hectare (FAO, 2019).

Regarding these figures, FEDECACAO highlights those differences in technological levels, and economic aspects among farmers result in different productivities (interview with staff

[6] According to Fedecacao, plantations implemented as agroforestry systems that use high-performance cacao clones, with 1,200 to 1,300 trees per hectare and adequate management practices can produce more than 2,000 kilograms/hectare/year.

of FEDECACAO, 2018). Thus, the study of Abbott et al. (2018) classified farmers into four groups: 1) *marginal* farmers which obtain annual yields of 300 Kg/Hectare or fewer; 2) *traditional* farmers get yields between 300 and 500 Kg/Hectare, 3) *technified* farmers obtain productivities between 1,200 and 1,800 Kg/Hectare, and finally, 4) *diversified* farmers, who make a multipurpose use of soil, reach between 300 and 600 Kg/Hectare (Abbott et al., 2018).

Sectorial diagnoses pose the low cacao yield as a problem caused by multiple factors; however, the low technological adoption in the farms stands out as the most relevant, mainly when they refer to small-scale cultivations (Abbott et al., 2018; Acosta & Villarraga, 2006; Arboleda & González, 2010; Castellanos, Torres, Fonseca, Montañez, & Sanchez, 2007; Fedecacao, 2015; García-Cáceres, Perdomo, Ortiz, Beltrán, & López, 2014; Mantilla, Argüello, & Mendez, 2000; Rios, Rehpani, Ruíz, & Lecaro, 2017; SIC, 2012; Technoserve, 2015; Torres & Rodriguez, 2015). Despite research efforts, studies are insufficient and limited in understanding the nuances of this problem. Most of the investigations are concentrated on the main cacao regions (Santander and Arauca), neglecting non-traditional areas, and even dismissing post-conflict regions. Analyses are typically focused on technology acceptance and effects over yields and income of a limited number of innovations (planting material, fertilization, or fermentation practices). Finally, they do scarce research about factors influencing cacao farmers' decisions around technology utilization.

For that reason, this research aims to explore the social and economic factors that interact with technological options to influence farmers' decisions. I applied this study to municipalities included in PDET regions since they are often indicated as places to foster cacao as an alternative to illicit economies and poverty. Finally, I incorporated into the analysis a more extensive set of innovations identified through conversations with farmers and stakeholders. Thus, the questions that guided this research are:

1. *What are the socioeconomic and technological factors that influence technology adoption in cacao farms of post-conflict regions in Colombia?*

2. *How have these factors influenced the technology adoption process in that context?*

3. *How might those factors constraining technology adoption in cacao farms be counteracted in such regions?*

Understanding the constraints that farmers face to adopt technologies in cacao cultivation should contribute to defining more suitable actions in the implementation of programs associated to PDETs and PNIS. To achieve this research, I used a mixed-methods approach with data collected by multiple methods in six municipalities of two PDET regions (San Jacinto, San Juan Nepomuceno, and El Carmen de Bolívar, in the Montes de María region; and Santa Marta, Ciénaga and Dibulla in Sierra Nevada-Perijá region).

In addition to the introduction, this thesis includes five more chapters. Chapter 2 introduces the background of the problem and provides a detailed situational framework about the illicit crops substitution and the socioeconomic and security constraints faced by farmers in Colombia. Chapter 3 develops the theoretical and conceptual framework for addressing a comprehensive analysis of socio-economic and technological factors influencing technology adoption decisions. Chapter 4 covers the research design, explains the methodology, and describes the methods for data collection and analyses applied in the three phases of this research. Chapter 5 presents the analysis and interpretation of qualitative and quantitative data for the six municipalities. Finally, Chapter 6 offers conclusions and recommendations, puts together findings and suggestions for addressing obstacles for technology adoption in post-conflict regions, and offers directions for a future research agenda.

CHAPTER 2: BACKGROUND AND CONTEXT

Colombian Rural Context

Colombia, with an area of 114 million hectares and 48.3 million inhabitants, is the fifth largest and the third most populated country in Latin America (DANE, 2019b; OCDE, 2015). Out of this population, 77% live in urban areas of 1,122 municipalities while the remaining 23% is located in rural areas integrated by populated centers (7.1% of people) and the so-called scattered rural areas (15.8%)[7] (DANE, 2019b). Thus, nearly one-fourth of the population remains in the countryside despite the forced displacement of about eight million people toward cities due to the armed conflict (Unidad Nacional de Victimas, 2019).

Although productive activities have diversified since the end of the 20th century, the main rural occupation continues to be mostly agricultural (DNP, 2015b; Fedesarrollo, 2017). By the end of 2018, 64% of the rural population was dedicated to farming, livestock, forestry, fishing, and aquaculture activities, while the remaining percentage was dedicated to trade (11%), services (8%), industry (7%), construction (4%), transportation (3%) and others (DANE-GEIH, 2019).

Not surprisingly, by 2014 about 43 million hectares (38% of the country area) were used for agricultural activities (DANE, 2016). Of these, 34.4 million were occupied with pastures and stubble fields, which indicates that livestock accounts for most of the agricultural land in Colombia. It is followed by crops that occupy 8.5 million hectares (7.1 million in plantations and 1.4 million in fallow) and infrastructure that barely occupies 0.1 million hectares (DANE, 2016). Such areas are settled down in 2.37 million Units of Agricultural Production (UPAs or farms)

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[7] According to DANE (2018), municipalities are composed by head, populated centers, and scattered rural areas. The municipal head is considered the urban perimeter. A populated center is a concentration of at least twenty semi-detached houses in rural areas with urban characteristics. Finally, scattered rural area is the remaining area with the dispersed disposition of dwellings and farms.

across the country. Of these, 70% have a size of fewer than five hectares representing only 2% of the whole area, followed by a group equivalent to 25% of the UPAs that have sizes between five and 50 hectares, while the remaining 5% have more than 50 hectares. In this last group, 0.2% of UPAs (approximately 4,740 farms) stands out as having sizes equal to or greater than 1,000 hectares, concentrating 74% of the total agricultural area (DANE, 2016).

Despite the area dedicated to agriculture, the sector represented only 6.3% of the total Gross Domestic Product (GDP) in 2018 and has exhibited a growth rate (3.3%) below the general economy (3.8%) across the decade (DANE, 2019e). Crop production and associated activities contribute 69% of agricultural GDP and had an average annual growth of 3.5% between 2010 and 2018, while livestock contributed 25% and grew 2.2% in that period. The remaining 6% of production was provided by forestry, fishing, and aquaculture sectors (DANE, 2019e).

Although the sectoral growth rate is over the average growth of the world agricultural product (2.8%) or Latin America's performance (3.0%) (World Bank, 2019), Colombia has significant limitations to achieving its agricultural full potential. According to the last Agricultural National Census (CNA 2014), scarcely 16.5% of the UPAs received technical assistance. Likewise, only 33.3% of UPAs use irrigation (this percentage fell to 11.5% among ethnic groups), although 87% have access to water sources. In addition, 84% do not use machinery or have infrastructure for agricultural production, and only 11% have applied for agricultural credit (DANE, 2016).

Resident farmers (those who live in their farms) account for 27% of the rural inhabitants with UPA, and only one third are women. At least 17% are illiterate, which is consistent with the 19% who declared they did not have any school degree, while 57% barely achieved elementary education, and less than 2% have a history of technical studies (DANE, 2016).

On the other hand, although health care coverage has reached significant levels (96% of residents), only 13% belong to the contributory regime, which means that farmers have limited access to the health system through formal jobs or with a paid enrollment (DANE, 2016). According to Fedesarrollo (2017), the predominant form of employment in rurality is the self-employed worker, which represents about half of employment in these areas (about 51%), followed by the positions of private employee (18%) and day laborer (12%). The CNA2014 also revealed that 74% of resident farmers in rural areas do not belong to any type of organization (association, union, cooperative, etc.), just 13% are linked to farmers' association or cooperatives, and 12% pertain to ethnic, women, elderly or young organizations (DANE, 2016).

Moreover, the social situation in rural areas suggests that many challenges still need to be addressed in Colombia. Poverty remains high despite improvements in the last decade. According to DANE (2019d), the Multidimensional Poverty Index (MPI) in 2018 increased to 39.9% after having declined 15.5% between 2010 and 2016. This rate is significantly higher than the urban rate of 13.8%. The gauges recording the greatest lags in rural areas are labor informality, low educational performance, lack of clean water, household members with educational backwardness, and lack of sanitation systems (28%) (DANE, 2019d). These precarious conditions have a higher incidence in rural households run by women with 45% of them experiencing multidimensional poverty, while only 38 % of man-headed households experiences the same (DANE, 2019d).

Likewise, monetary poverty and extreme poverty in rural areas (36% and 15% respectively) widely exceeded their levels compared to the urban context (24% and 5%) in 2018 (DANE, 2019c). Like in multidimensional poverty, their incidence is highly-concentrated in female-headed households (41% are poor compared to 35% of male-headed households).

Scholars have recognized poorly defined land markets and the unequal distribution of land as another hurdle to agricultural development. According to the *Misión para la Transformación del Campo (MTC)* carried out by the government in 2015, "Land tenure in Colombia is informal, disorderly and insecure, and its use lacks an order that administrates and assigns the rights of use and property in a functional way to the social and productive context of each territory" (Varón, Torres, & Baron, 2015, p. 60). In that sense, land ownership in Colombia faces several structural problems in rural areas: 1) high informality, 2) insufficient access to the land needed to generate adequate livelihoods, 3) high concentration due to land grabbing through violent practices, and 4) deficiencies in the administration of public lands (Centro Nacional de Memoria Histórica, 2016; Ibáñez & Querubín, 2004; Ibañez & Velásquez, 2018; Restrepo & Bernal, 2014; Varón et al., 2015). Also, the land rental market is almost non-existent, limiting alternative mechanisms for its access and use (Centro Nacional de Memoria Histórica, 2016; Ibañez & Velásquez, 2018).

Using the Quality of Life Survey, Gáfaro, Ibañez & Zarruk (2012) calculated that only 42% of rural households had access to land in 2011. By 2014, the CNA reported that 72% of UPAs declared ownership of land; however, several studies conclude that there are high rates of informality in land property (DNP, 2015b; Gáfaro et al., 2012; Garay, 2013; Ibáñez & Muñoz, 2011; Ibañez & Velásquez, 2018; PNUD, 2011; Suescún, 2013; UPRA, 2016b; UPRA & MADR, 2017). According to UPRA (2016b), the degree to which this problem affects municipalities is variable. By 2016, the DNP estimated that 64% of the national territory (722 municipalities) had outdated cadastral information and 28% did not even have a cadaster (DNP, 2016).

Besides, a significant number of rural households had to abandon their lands due to the conflict (ILSA, 2015; PNUD, 2011). Although there are different sources of data about this problem, the *Registro Único de Población Desplazada (RUPD)* administered by the government had reports of 6.5 million hectares abandoned or dispossessed between 1999 and 2010, with the most critical period being between 2002 and 2008 (Centro Nacional de Memoria Histórica, 2016). For Ibañez & Velasquez (2018), this figure reached 7.3 million hectares in 2017, from which 4.6 million lacked titles; therefore, chances of property recovered are minimal.

Important constraints like poor conditions in road infrastructure, inadequate transportation services, and scarce information systems also affect the performance of agriculture and social wellbeing of rural communities (Fedesarrollo, DNP, & IGCVC, 2017; Lozano & Restrepo, 2016; Narvaez, 2017). In the same way, low-level agricultural R&D funding (Corpoica, 2015b), and the increasing vulnerability to climate change (Fedesarrollo et al., 2017) have limited the sector.

According to Lozano & Restrepo (2016), in Colombia, 9% of the roads correspond to the primary road network and 5% to the secondary network, while the remaining 86% make up the tertiary network, which is the one that connects scattered areas with populated centers and municipalities. Only about 19% of the last group has suitable conditions to be utilized throughout the year. The other 81% can only be used during the dry season, and 51% do not even have the surface graded, limiting the availability of transportation options and increasing costs of mobilization (Lozano & Restrepo, 2016). The responsibility for the maintenance of tertiary roads lies mainly (72%) with the municipalities (Narvaez, 2017).

Regarding agricultural R&D, diagnoses indicate that the Agricultural Science and Technology System operation is affected by a limited articulation between public and private institutions and insufficient capabilities to advance in a timely research agenda for the sector (Corpoica, 2015a; DNP, 2015a; Nupia, 2020). Figures of public investment in such activities barely reached an average of 0.71% of sector GDP between 2000 and 2013 and average annual growth of 2.27%, localizing Colombian public expenditure well below the percentage invested in other countries with an agricultural vocation in the region (Brazil, Chile, and Argentina) (Corpoica, 2015b).

Conflict, Peace Agreement, and Illicit Crops

From the 1960s, Colombia has experienced one of the most prolonged and complicated armed conflicts in the contemporary world (GMH, 2013; Gobierno Nacional de Colombia et al., 2016). Although it is rooted in peasant struggles that advocated for agrarian reform, poverty and inequality reduction, and counteraction of regional hegemonic powers (Bushnell, 2013; GMH, 2013; Salgado, 2000), these objectives gradually were coupled with new social and economic phenomena that increased the complexity of these problems (Diaz & Sanchez, 2004; GMH, 2013; Ramirez, 2011). In the words of the researchers of *Grupo de Memoria Histórica* created in 2011 to unveil the historical facts of the Colombian conflict:

> All the reports illustrate the gradual convergence between the war and the agrarian problem (violent dispossession, the idle concentration of land, inadequate uses, colonization, and failed issuance of property titles). But to the old problems are added new ones, which show the dynamics inaugurated by drug trafficking, mining, and energy exploitation, agro-industrial models, and criminal alliances between paramilitaries, politicians, public servants, local economics, and business elites, and drug traffickers...(GMH, 2013, p. 21).

In that way, in addition to the rural guerrillas (*Fuerzas Armadas Revolucionarias de Colombia - FARC-EP, Ejército Popular de Liberación - EPL y Ejército de Liberación Nacional - ELN*), and the urban insurgency of *Movimiento 19 de Abril (M-19)*, other actors found the opportunity to take advantage of this context to carry out illegal actions to benefit their interests (Borrero, 2004; Diaz & Sanchez, 2004; DNP, 2015b).

Thus, paramilitary groups (illegal armed groups of the extreme right) emerged as actors linked to local political and economic powers that, in absence or complicity to state agents, executed their counterinsurgency strategy attacking civil population (Borrero, 2004; GMH, 2013). On the other side, drug traffickers who organized the production and commercialization of illicit drugs stimulated the establishment of illegal crops, especially in the most poor, rural areas (Diaz & Sanchez, 2004; DNP, 2015b). Finally, the cartels of other illicit activities (illegal mining, smuggling, and micro-trafficking) took advantage of this scenario to exploit natural resources indiscriminately and generate a speculative economic bubble (Ramirez, 2011).

With time, all these actors developed alliances and coalitions to operate in the territory collaboratively and violently (Diaz & Sanchez, 2004; GMH, 2013; Ramirez, 2011). Among them, the partnership between the guerrilla and drug traffickers stands out in most of the regions where they overlapped. Through their arrangements, the guerrillas initially dedicated to collecting taxes from coca commercialization rapidly became guardians and operators of plantations, cocaine labs, and infrastructure, inputs smuggling (e.g., gasoline and chemicals), and commercialization routes (Borrero, 2004; Diaz & Sanchez, 2004; GMH, 2013). Likewise, alliances between paramilitaries and politicians, local elites or drug traffickers have also been mechanisms for drug business, land grabbing, rural population displacement, corruption and crime (GMH, 2013; Salgado, 2000).

Although urban areas experienced some of the conflict, peasant, indigenous, and black communities, as well as rural landscapes and natural resources, suffered the most intense effects (GMH, 2013; PNUD, 2011). In the words of researchers of *Grupo de Memoria Histórica*:

> Seeing violence from the perspective of the land and the territories reveals another distinctive feature of its history: the war has been fought mostly in the Colombian countryside, in the hamlets, villages and municipalities, far from the central country or large cities (GMH, 2013, p. 22).

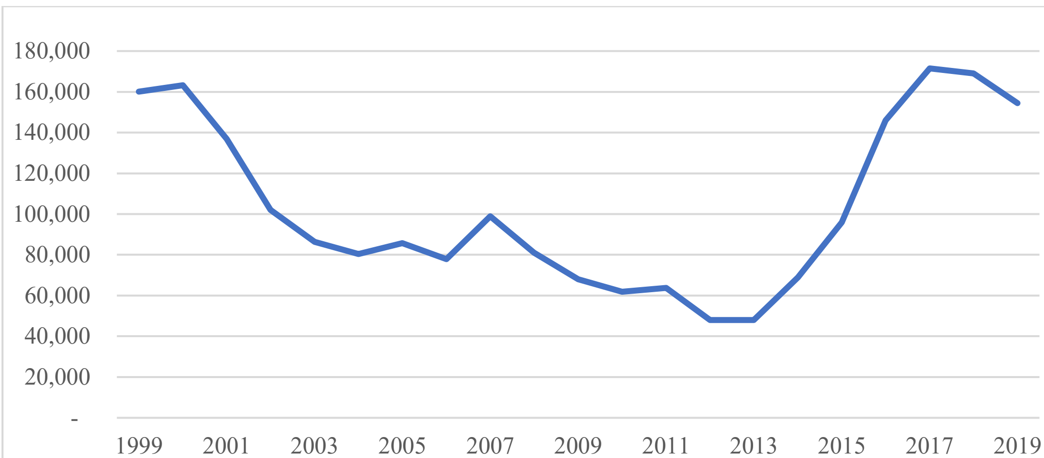
The official figures indicate that the nation has recognized 8.9 million victims of actions occurring between 1985 and 2018. Of these, 7.6 million people were violently displaced, 1.2 million were killed or forcibly disappeared, and at least 37,000 kidnapped (Unidad Nacional de Victimas, 2019). Between 2016 and 2018, 437 homicides of social leaders were reported, 85% of them belonged to peasant or community movements engaged with the defense of land rights and environment. Moreover, those advocating for finding substitutes for illicit crops have also been killed (Defensoría del Pueblo, 2019; OEA & CIDH, 2019).

As was indicated previously, illicit crops and conflict have coexisted and reinforced each other in Colombia since the 1960s (Diaz & Sanchez, 2004). After the decline of marijuana areas in the north of the country (Sierra Nevada de Santa Marta and Serranía del Perijá), coca became the dominant illicit crop since the middle 1980s. The 1990s were characterized by its expansion, from 37,500 hectares in 1992 to 163,000 in 2000 (Borrero, 2004; Diaz & Sanchez, 2004). After 2001, areas involved in coca production steadily decreased until 2012 when 47,790 hectares were under production, largely because of the eradication and alternative development programs (Plan Colombia) implemented by the government and international agencies (Uribe, 2019). However, with the suspension of aerial spraying, coca production grew progressively again to 171,000 hectares in 2017, creating a W-shape line when graphing the period (UNODC, 2005, 2013, 2017, 2019b; Uribe, 2019). Since then, coca areas have shown a steady trend downward (Figure 1).

In that context, the Peace Agreement between the government and FARC-EP was negotiated and signed. This process not only allowed that 14,900 guerrillas begin their process to reincorporate into the civil society (CINEP & CERAC, 2018a), but was also understood as the most concrete commitment to address the structural problems in rural areas.

Within the agreement, both the *National Comprehensive Plan for Substitution of Illicit Crops* (PNIS) and the *Rural Development Plans with a Territorial Approach* (PDET) are the actions generating most expectations in rural population. Regarding PNIS, about 130,000 households signed agreements for illicit crops substitution by the end of Santos' presidency [8] (Garzón et al., 2019). Out of them, 99,097 were fully enrolled in the program by the end of 2018, and 34,767 hectares of 52,000 had been voluntarily eradicated with government assistance[9] (Garzón et al., 2019; Kroc Institute, 2019a; UNODC, 2019a).

Figure 1. Coca Area in Colombia 2001-2019 (Hectares)



Source: UNODC – SIMCI (2004 – 2018)

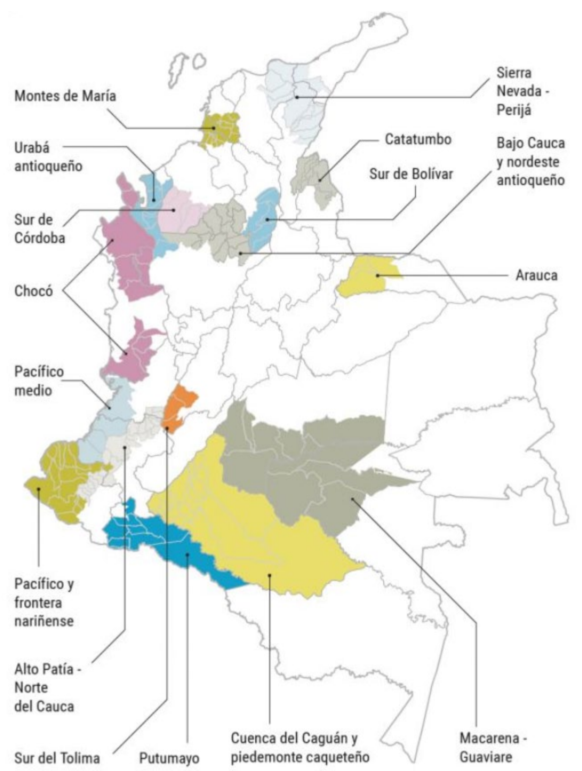
By October of 2019, the PNIS had prepared 8,838 investment plans for projects oriented to foster activities like silvopastoral systems, cacao, coffee, hog farming, plantain, fruits, and fish farming (Garzón et al., 2019; Kroc Institute, 2019b; UNODC, 2019a). If the distribution of such activities maintained the farmers preferences in 2018, at least 10% would correspond to cacao plantations, though some municipalities could show more interest in this crop; this is the case for

[8] Juan Manuel Santos finished his second presidency period in August 2018.
[9] Households received 1 million pesos monthly (US$338 based on 2018 average exchange rate) for working in substitution or community infrastructure. Besides, for one time, they receive 1.8 million pesos (US$ 609) for food security, and 9 million pesos (US$ 3,045) for quick-revenue projects. In the second year, the government invest up to 10 million pesos (US$ 3,383) per family for projects and 3.2 million pesos (US$ 1,083) for technical assistance (Alta Consejería para el Posconflicto, 2017; UNODC, 2019a).

Cumaribo (Vichada) in the east of Colombia, where 56% of projects focused on cacao plantation and 11% on cacao maintenance (UNODC, 2020). By March of 2020, the government reported that voluntary eradication had reached 36,139 hectares and food assistance reached 59,940 households (UNODC, 2020).

On the other hand, for PDET, the government enacted Decree 893/2017 to define the program configuration prioritizing 170 municipalities grouped into 16 regions (Figure 2) where the extra effort to balance the development among territories would be undertaken (Gobierno Nacional de Colombia et al., 2016). Besides, a "community-based planning process" yielded a 10-year Action Plan for Regional Transformation (PATR) in each region (Kroc Institute, 2019a).

Figure 2. PDET Regions Defined by the Decree 893 of 2017.



Source: El Espectador, 2019

PDET regions are spread across the country and cover 6.6 million people (including 24% of the rural population and 57% people with multidimensional poverty); 2.5 million of them are recognized victims. Likewise, these regions correspond to the 36% of the national territory and include 45% of Natural National Parks (Agencia de Renovación del Territorio, 2020b). Among them are the two regions covered in this research, 1) Montes de María and 2) Sierra Nevada de Santa Marta-Perijá, both located in the north of the country. They were selected according to the goals of the *Cacao for Peace* project, where this research is framed. In addition, they represent previous efforts of fostering cacao to assist communities affected by the conflict. Therefore, those places are relevant cases to get in-depth on the questions proposed in this research.

According to the ART (2019a), PATR construction finished in February of 2019 involved about 200,000 people from local governments, victims groups, communities, and stakeholders (Kroc Institute, 2019; Ramos, 2019) belonging to 11,000 villages (Agencia de Renovación del Territorio, 2019c, 2020a). Using a methodology based on an eight-pillar framework (Figure 3), local communities focused on rethinking the development goals for those regions. Thus, participants discussed and agreed 32,808 municipal and 1,178 regional initiatives in the 16 PATR (Agencia de Renovación del Territorio, 2017, 2020b; CINEP & CERAC, 2019).

Figure 3. Eight-pillar Framework for PATR Planning

Source: Agencia de Renovación del Territorio, 2017b

Within the group of municipal proposals, participants prioritized initiatives focused on responding to *Early Childhood and Rural Education* issues (21%), followed by those of *Economic*

Revival and Agricultural Production (18%), and *Reconciliation, Coexistence and Peace* (14%) (Agencia de Renovación del Territorio, 2020b).

Exploring in detail the pillar *Economic Revival and Agricultural Production* (6,065 initiatives), the results show that 27% of initiatives refer to productive projects; among them, 56% are related to crop production, 37% to livestock, and 8% to forestry (Agencia de Renovación del Territorio, 2019b). Secondly, 20%, refer to initiatives oriented to assets and infrastructure, followed by those on agricultural extension, and off-farm activities (14% each) (Agencia de Renovación del Territorio, 2019b). Initiatives regarding economic development and communities' livelihoods declared interest in coffee, cacao, livestock, beans, pisciculture, rice, dairy, maize, fruits, cassava, brown sugar, pepper, plantain, and beekeeping, among others (Agencia de Renovación del Territorio, 2020a; Kroc Institute, 2019b; Zambrano, 2019).

Particularly in Montes de María, communities of San Jacinto, San Juan Nepomuceno and El Carmen placed priority on pillars of *Reconciliation, Coexistence, and Peacebuilding*, and *Social Management of the Rural Property and Land Use* like those aimed at land adjudication, restitution, and titling (ART & Alcaldía El Carmen de Bolívar, 2018; ART & Alcaldía San Jacinto, 2018; ART & Alcaldía San Juan Nepomuceno, 2018; ART & Municipios Montes de María, 2018; CINEP & CERAC, 2019). On the other hand, in the *Economic Revival and Agricultural Production* pillar, the three municipalities included cross-cutting projects such as construction of collection centers, supply of machinery, irrigation infrastructure, and training and extension. Economic activities prioritized by local participants were diverse, both agricultural and non-agricultural, and highlighted cacao and their complementary crops (plantain, maize, and forestry) (ART & Alcaldía El Carmen de Bolívar, 2018; ART & Alcaldía San Jacinto, 2018; ART & Alcaldía San Juan Nepomuceno, 2018).

Similarly, the Sierra Nevada region, and particularly Santa Marta, Ciénaga, and Dibulla focused on pillars of *Reconciliation, Coexistence, and Peacebuilding*, *Social Management of the Rural Property and Land Use*, and *Economic Revival and Agricultural Production*. The last two included initiatives to improve land access and cadastral management, as well as projects aimed at increase the production of crops already incorporated into local dynamics (ART & Municipios Sierra Nevada - Perijá, 2018). Municipalities also validated initiatives to provide extension services, strengthening agricultural organizations, access to credit, infrastructure, and machinery and equipment provision. In Santa Marta, Ciénaga, and Dibulla, the communities' interest in improving cacao production as an economic development strategy was made explicit by singular initiatives (ART & Alcaldía Ciénaga, 2018; ART & Alcaldía Dibulla, 2018; ART & Alcaldía Santa Marta, 2018). The agricultural proposals prioritized in the six municipalities covered by this research and the specific projects for cacao are summarized in Table 1:

Table 1. Productive proposal in Montes de María and Sierra Nevada by PATR – PDET

Region	Specific Initiatives in Cacao or Cross-cutting
San Jacinto	Building and provide machinery, tools, and supplies for the agro-industrialization of **cacao**, to serve families in the villages of Paraiso, Arroyo, Brasilar, other.
El Carmen de Bolívar	Promoting technological innovation for agricultural production in the upper, middle, and lower areas of the municipality.
	Providing manual and mechanical tools for the agricultural production of the communities of the high, medium, and low zone of the municipality.
	Transferring and update of the agricultural technological offer for farmers and technicians in the upper, middle, and lower areas of the municipality.
San Juan Nepomuceno	Building a nursery for grafted seedlings of forest species, fruit trees, and native.
	Implementing a research and transformation program for traditional agricultural products to generate added value for rural families.
PATR Montes de María	Strengthen the production, research, transformation, certification, and commercialization for value chains of rice, avocado, maize, sesame, sweet potato, fruits, cassava, yam, and cacao.
Santa Marta	Formulate and present comprehensive productive projects of coffee, mango, cacao, avocado and musaceae in the municipality.
Ciénaga	Strengthen the productive systems of coffee, cacao, mango, blackberry, lulo and banana in the municipality.
Dibulla	Strengthen the plantain, cacao and cassava systems to increase the production of small and medium producers in the municipality.

PATR Sierra Nevada y Perijá	Strengthen production systems through comprehensive production projects for cacao, coffee, guava, passion fruit, melon, and fique. As well as promising plantain and avocado products in the municipalities of Dibulla, Fonseca, Juan del Cesar (La Guajira) in a way that allows the production of small and medium producers to increase.
	Strengthen production systems through comprehensive production projects for cacao, coffee, banana, plantain, and mango, as well as promising avocado and 'fique' crops in the municipalities of Aracataca, Ciénaga, Fundación, Santa Marta (Magdalena) that allows for increasing the production of small and medium producers.

Source: PATRs Montes de María and Sierra Nevada -Perijá. PMTRs El Carmen de Bolívar, San Jacinto, San Juan Nepomuceno, Santa Marta, Ciénaga y Dibulla.

The Context in Montes de María

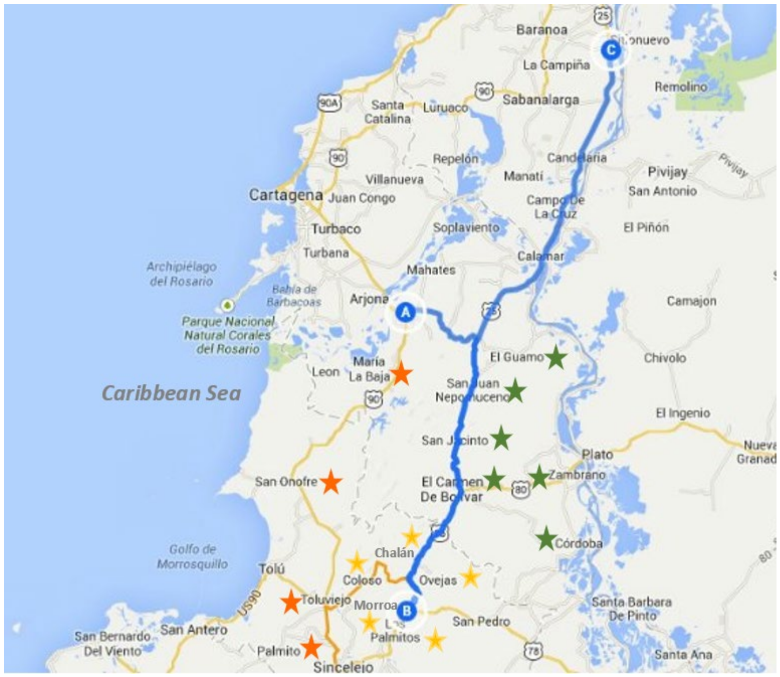
Montes de Maria is a region of national strategic importance, located in the north of the country, close to the urban-regional corridor made up by Cartagena, Barranquilla, and Santa Marta, whose seaports are used to export merchandise through the Caribbean (Aguilera-Díaz, 2013; ART & Municipios Montes de María, 2018). The region encompasses 6,466 km2 and comprises 15 municipalities belonging to three areas between Bolivar and Sucre (Figure 4) (Bocchi, 2011).

1) *Mountain area.* With an area that reaches 1,053 km², it is composed of the municipalities of Chalán, Colosó, Morroa, Ovejas and Los Palmitos. It is located between the Troncal del Caribe and the Troncal de Occidente.

2) *Troncal del Magdalena area***.** Located to the northeast of Montes de María, in the department of Bolívar. The municipalities of Córdoba, El Carmen de Bolívar, El Guamo, San Jacinto, San Juan Nepomuceno and Zambrano are part of it in an area of 3,262 km².

3) *Western foot land area.* Located between the Caribbean Transverse Highway and the sea. It includes María La Baja, San Antonio de Palmito, Tolúviejo and San Onofre with an area of 2,151 km² (Bocchi, 2011; PNUD, 2010).

Figure 4. Map of PDET Montes de María



Source: Agencia Nacional de Infraestructura – ANI

Montes de María has been historically recognized for its abundance and agricultural diversity, which is why some authors refer to it as “the pantry of the Caribbean” (PNUD, 2010). Its economy is based on farming, with a tradition in cattle raising and peasant crops of cassava, yams, corn, rice, plantain, tobacco, coffee, and avocado. During the first decade of this century, commercial crops of oil palm, cacao, and black pepper have increased (Aguilera-Díaz, 2013). Except for livestock areas, the Magdalena River valley is populated mainly by peasants and Afro-Colombian communities that settled there after the period of slavery (PNUD, 2010).

Production has been led by organized smallholders and landless peasants, since the beginning of the 20th century, to claim their right to land. By the end of the 1960s, grassroots movements led the government to carry out a process of land redistribution through the purchase of land from large landowners and their subsequent sale to peasants via agrarian loans[10]; moreover, the government provided technical assistance, and commercialization of crops through public purchases (Aguilera-Díaz, 2013; PNUD, 2010). During the 1980s, farmers experienced difficulties after adopting chemical fertilizers and “improved seeds” that they could not pay for with the harvests, further increasing their debt for land acquisition (Centro Nacional de Memoria Histórica, 2016; PNUD, 2010). Numerous peasant sold their land for subsistence (PNUD, 2010).

In this scenario, monocultures such as oil palm were introduced using farmers as suppliers or workers for large-scale plantations. Thus, the needs and preferences of those who did not join this development model supported by the government were excluded, (Centro Nacional de Memoria Histórica, 2016; PNUD, 2010). This situation was complicated by the guerrilla groups who found refuge in the Serranía de San Jacinto - an area with difficult access. There were groups from the FARC (Revolutionary Armed Forces of Colombia), the ERP (People's Revolutionary Army) and the ELN (National Liberation Army), who used cattle theft and ranchers’ kidnapping as source of funding. Their harassment of landowners and police was interpreted as a strategy articulated to peasants' occupation of land increasing the pressure on class conflict (PNUD, 2010, 2011).

In the 1990s, the problems were exacerbated by paramilitary groups and drug traffickers. Paramilitarism was linked to landowners who wanted to recover land purchased by the government, prompted the violence against peasants and indigenous people (Centro Nacional de Memoria Histórica, 2016; GMH, 2013; PNUD, 2010, 2011). Between 1997 and 2003, paramilitaries displaced around 100,000 people and committed at least 115 massacres in Montes de Maria (Centro Nacional de Memoria Histórica, 2016; GMH, 2013; PNUD, 2010).

[10] Between 1963 and 2007, Incoder (Colombian Institute of Rural Development) delivered 134,230 hectares to 10,736 rural households in Montes. In Bolívar, 59% of hectares were in El Carmen, María La Baja and San Jacinto (Aguilera-Díaz, 2013).

On the other hand, traffickers utilized the region as a route to move coca paste and cocaine from the south of Bolivar to the Gulf of Morrosquillo (over the Atlantic), where it is transported to Caribbean countries (Bocchi, 2011; PNUD, 2010). This step generated an illegal local economy that directly impacted the social fabric of the municipalities in the region (PNUD, 2010). Finally, violence and power disputes in Montes de María have been also related to smuggling, clientelism, corruption, and election fraud (PNUD, 2010).

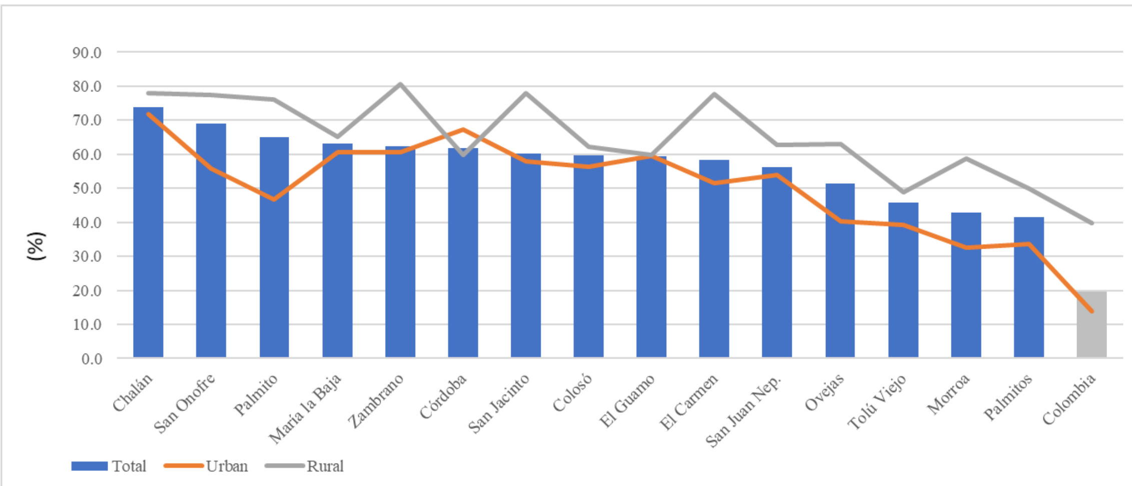
In summary, four structural causes are important for understanding conflict context in the region: tensions in land tenure, rural development model influenced by power unbalances, political and class conflict, and corruption (Aguilera-Díaz, 2013; PNUD, 2010, 2011).

According to the National Census of Population and Housing 2018, the region has 372,504 inhabitants, 56% of which are in urban areas and 44% in rural areas, representing almost double the proportion of national rural population. Regionally, the percentage of women and men is 49% and 51% respectively (DANE, 2019b). Particularly, the municipalities of this research concentrate 48% of urban and 19% of rural population in Montes de Maria; However, the individually rural population is 13% in San Jacinto, 26% in San Juan Nepomuceno, and 28% in El Carmen.

Regarding social conditions, Census 2018 indicates that the average Multidimensional Poverty (MP) of Montes de María municipalities was 58%; however, in rural areas the average reached 66%. The municipality with the highest rural MP was Zambrano (81%) followed by San Jacinto and Chalán (78% each), and El Carmen and San Onofre (77%). Although the other municipalities were below those percentages, the region shows a poverty incidence notably above the 39.9% of national rural average (Figure 5) (DANE, 2020).

Within MP, some dimensions reached important levels. Labor informality occurs, on average, in 95% of rural households. This is followed by low educational achievement, with 79%, but it rises to 84% in municipalities such as El Carmen and San Jacinto (DANE, 2020). They are followed by four dimensions whose incidences are between 42% and 48%, including lack of a sanitation, no access to clean water, a high dependency ratio (three or more people depending per employed person), and inappropriate material for the household floor. Illiteracy levels are over 39% in the rural population, compared to 29% in the urban population of Montes de María or 22% of rural population at a national level (DANE, 2020).

Figure 5. Multidimensional Poverty in Montes de María. 2018



Source: DANE, Census 2018

Regarding to land suitability, the region surface ought be utilized 48% for forest, 41% for agriculture and only 5% for livestock (ART & Municipios Montes de María, 2018). However, in Montes de María two types of landscapes of appropriation and land use occur: "One of them comprises the high parts of the mountains, where the land use has been especially agricultural. The other type of landscape is the valleys, where cattle ranching and intensive tobacco cultivation were traditionally established" (Bocchi, 2011, p. 24).

According to Ministerio de Agricultura y Desarrollo Rural (MADR), between 2008 and 2018, planted area decreased by 10% from 102,705 to 92,880 hectares, while production increased by 1%. By 2018, Montes de María generated 601,504 tons of cassava, maize, yam, oil palm, plantain, avocado, tobacco, rice, and cacao, among others, showing an expansion of long-term cash crops and reduction of annual crops (Bocchi, 2011). Thus, traditional products like cotton practically disappeared, and maize area declined 50%, rice 32%, and yam 6% (MADR, 2019).

On the other hand, avocado, recognized as traditional crop shows area reduction of 23% due to plant health issues experienced in the last few years (ICA, 2019b). Other crops like oil palm, plantain, and cacao increased by 301%, 176%, and 125%, respectively. Particularly San Jacinto, San Juan Nepomuceno, and El Carmen concentrate 41% of area and 50% of production in Montes de María; however, this percentage is higher in certain crops, such as in cacao (Table 2) (MADR, 2019).

Table 2. Agricultural Production in San Jacinto, San Juan Nepomuceno and El Carmen (2018)

	Planted Area (Ha)	Harvested Area (Ha)	Production (Ton)	% of Montes de Maria region		
				Planted	**Harvested**	**Production**
Cassava	11,125	11,125	116,250	47%	47%	51%
Maize	10,300	10,300	17,375	48%	48%	45%
Yam	10,160	10,160	128,570	62%	62%	66%
Plantain	1,550	1,340	13,080	35%	33%	33%
Avocado	2,670	2,230	24,050	77%	76%	82%
Tobacco	600	600	990	24%	24%	15%
Rice	620	620	740	25%	25%	9%
Cacao	858	426	136	68%	66%	56%
Beans	255	255	204	20%	20%	16%
Other	230	172	1,129	11%	9%	9%
Total	**38,368**	**37,228**	**302,524**	**41%**	**42%**	**50%**

Source: Evaluaciones Agropecuarias Municipales, MADR, 2019

Regarding the livestock sector, by 2018, the cattle inventory of Montes de María was 411,086 heads in 7,685 farms representing 18% of the total inventory and 23% of farms with cattle in Bolivar and Sucre (ICA, 2019a). Between 2008 and 2018, the number of heads increased in the region by 19%, although this situation occurred differently among municipalities.

Since the 1960s, calculations of the Gini coefficient for rural property indicate that Colombia always had an indicator above 0.8. In fact, in 2009 Gini reached 0.885, which means a very high concentration of land ownership (UPRA, 2016a). In the case of Montes de María, figures indicate that the highest concentration is found in Palmito and the lowest in Ovejas (UPRA, 2016a) (Table 3).

Table 3. Gini Index of Land Property in Montes de María

<table>
<tr><th>Municipality</th><th>Gini Index</th><th>Year</th><th>Municipality</th><th>Gini Index</th><th>Year</th></tr>
<tr><td>Palmito</td><td>0.8176</td><td>2011</td><td>El Guamo</td><td>0.6595</td><td>2010</td></tr>
<tr><td>Tolúviejo</td><td>0.8171</td><td>2011</td><td>El Carmen</td><td>0.6578</td><td>2011</td></tr>
<tr><td>Zambrano</td><td>0.7730</td><td>2013</td><td>Palmitos</td><td>0.6532</td><td>2009</td></tr>
<tr><td>San Onofre</td><td>0.7453</td><td>2012</td><td rowspan="2">San Juan Nepomuceno</td><td rowspan="2">0.6512</td><td rowspan="2">2014</td></tr>
<tr><td>Córdoba</td><td>0.7364</td><td>2011</td></tr>
<tr><td>María la Baja</td><td>0.7172</td><td>2013</td><td>Colosó</td><td>0.6442</td><td>2011</td></tr>
<tr><td>Morroa</td><td>0.6974</td><td>2011</td><td>Chalán</td><td>0.6219</td><td>2011</td></tr>
<tr><td>San Jacinto</td><td>0.6751</td><td>2008</td><td>Ovejas</td><td>0.6055</td><td>2014</td></tr>
</table>

Source: UPRA (2016a).

According to UPRA (2016a), Bolívar has 65,534 rural properties identified in the cadastre; 75% of them correspond to private properties with agricultural use 85% of which have 50 hectares or fewer, and even about 12% barely reach 1 hectare or fewer. In 2017, the IGAC reported that 53% of the municipalities in the Montes de María region did not have an updated cadastre (ART & Municipios Montes de María, 2018). As a result of the large-scale dispossession and forced abandonment of land, the displaced population in the region reaches 255,536 people; 14 of the 15 municipalities (including those covered by this study) have filed 7,036 requests for land restitution of which only 31% have received a sentence (ART & Municipios Montes de María, 2018).

Among other problems, despite the presence of lagoons and groundwater, 67% of the UPAs in rural areas have difficulties in accessing water. Besides, 85% of them do not have technical assistance and 93% do not have any type of machinery (DANE, 2016).

Finally, the region presents significant deficiencies in road infrastructure (Lozano & Restrepo, 2016). Bolívar has 3,099 km of primary, secondary, and tertiary roads. 29% of them belong to the Montes de María region (Aguilera-Díaz, 2013; ART & Municipios Montes de María, 2018). Some analyses indicate that this infrastructure barely represents the availability of 36 km of roads per 100 km2, ranking 5th in the departments with the smallest road network in Colombia. Likewise, approximately 90% of roads correspond to the unpaved tertiary network which is only available for use in the dry season (Lozano & Restrepo, 2016).

The Context in Sierra Nevada -Perijá – Zona Bananera

The PATR named Sierra Nevada[11]– Perijá – Zona Bananera includes 15 municipalities of three departments. Aracataca, Ciénaga, Fundación, and Santa Marta in Magdalena; Dibulla, Fonseca and San Juan del Cesar in La Guajira; and Agustín Codazzi, Becerril, La Jagua de Ibirico, La Paz, Manaure, Pueblo Bello, San Diego, and Valledupar in Cesar (ART & Municipios Sierra Nevada - Perijá, 2018) (Figure 6).

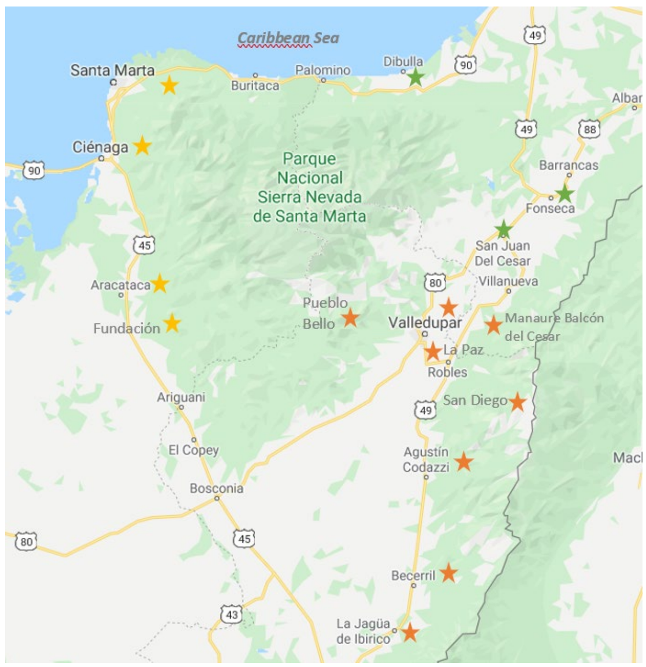
Although these municipalities share important socioeconomic and environmental dynamics, compared to Montes de María they have typically been part of a less-integrated region. Instead to address the contextualization of the entire PDET area, this section will concentrate on the Sierra Nevada of Santa Marta area, since it includes the municipalities of interest in this research (Santa Marta, Ciénaga and Dibulla), and represents the area where the conclusions and recommendations could be applicable.

Sierra Nevada de Santa Marta is a mountainous massif located at 46 km of the north coast of Colombia. Its pyramid shape has an approximate area of 17,000 km2 (1.7 million of hectares) and its highest peak reaches 5,775 meters above sea level (Balaguera et al., 2005). In the north, the

[11] According to the political division, Sierra Nevada de Santa Marta comprises 18 municipalities belonging to La Guajira, Magdalena, and Cesar (Fundación Prosierra, 2018). From them just nine coincide with those included in the sub regional PATR.

Sierra Nevada borders the Caribbean Sea from the south of La Guajira to Santa Marta city (Fundación Prosierra, 2018). The westward side is surrounded by an extensive swamp named Ciénaga Grande and the alluvial plain of Magdalena River. The eastern side is framed by the valleys of Cesar and Ranchería rivers and the Serrania del Perijá (Balaguera et al., 2005). Because of its altitudinal variability it contains a mosaic of biomes that provide a diversity of temperatures and precipitation (Fundación Prosierra, 2018; Viloria, 2005).

Figure 6. Map of PDET Sierra Nevada - Perijá - Zona Bananera



Source: Agencia Nacional de Infraestructura – ANI

Several studies highlight the importance of Sierra Nevada since at least 35 rivers go down from the massif, supplying aqueducts for 1.5 million people in the surrounding municipalities (Benjumea, 2015; Fundación ERIGAIE & CEEP, 2015; Fundación Prosierra, 2018). Furthermore, these rivers enable areas for agriculture, livestock, and mining (Balaguera et al., 2005; Benjumea, 2015; Fundación ERIGAIE & CEEP, 2015) that are not properly supplied due to the lack of infrastructure and normalization of flows (Viloria, 2005).

As mentioned above, climate change generates significant challenges to the availability of water in Sierra Nevada because the region lacks adequate basins management (Benjumea, 2015; Fundación ERIGAIE & CEEP, 2015; Martinez et al., 2018) and scarce infrastructure (Viloria, 2005); Besides, extreme temperatures and rainfall - both excess and deficit - makes the availability of water variable, increasing the vulnerability of agricultural activities (MADS, 2019). In general, Fundación ERIGAIE & CEEP (2015) indicated that water use conflicts are driven by: 1) unequal access, 2) irregular use of collection infrastructure and deviation of rivers, and 3) lack of institutional capacity to enforce the appropriate uses of concessioned water.

Regarding its inhabitants, Sierra Nevada is a multi-ethnic conglomerate with diverse origins, traditions, and customs. According to Balaguera (2005), the region brings together indigenous people, amphibious culture communities, coastal peasants, Afro-Caribbeans, and migrants from the center of the country (*Cachacos*). Particularly, tradition speaks of four indigenous groups of Sierra Nevada: Koguis, Arhuacos, Wiwas (Malayos), and Kankuamos, currently located in three recognized reserves which concentrate 602,251 hectares (Viloria, 2005). However, Wayúu people displaced from the middle and upper Guajira due internal tensions are settled on the eastern slope (Balaguera et al., 2005; Hettler & Gil, 2019; Viloria, 2005).

By 2018, the indigenous reservations in the Sierra Nevada had a population of 49,961 people (DANE, 2019a). Although another 97,889 members of these communities were registered outside the reservations, in Valledupar, Dibulla, Riohacha, San Juan Del Cesar, Aracataca, Ciénaga, Santa Marta, Pueblo Bello and Fundación (DANE, 2019a). Regarding Wayuu people, the 2018 Census reported five reservations (Zahino-Guayabito-Muriaytuy, San Francisco, Mayabangloma, Provincial, and Trupigancho-La Meseta,) in which 4,182 indigenous people live, additionally another 17,215 people were located out of them (DANE, 2019a).

On the other hand, peasant populations settled several decades ago in the region migrated mainly from the center of Colombia and configured different territorial divisions in the Sierra Nevada coffee belt between 600 and 1.500 masl (Balaguera et al., 2005; Viloria, 2005). Some of them have even located above and below this level up to 2,000 masl (Fundación ERIGAIE & CEEP, 2015). On the north slope, peasants from Santander arrived to work on coffee plantations introduced in the region in the 18th century, while migrants from Tolima settled on the western side to carry out similar work. Peasants from Antioquia, Boyacá, and Caldas also came to the Sierra Nevada (Viloria, 2005).

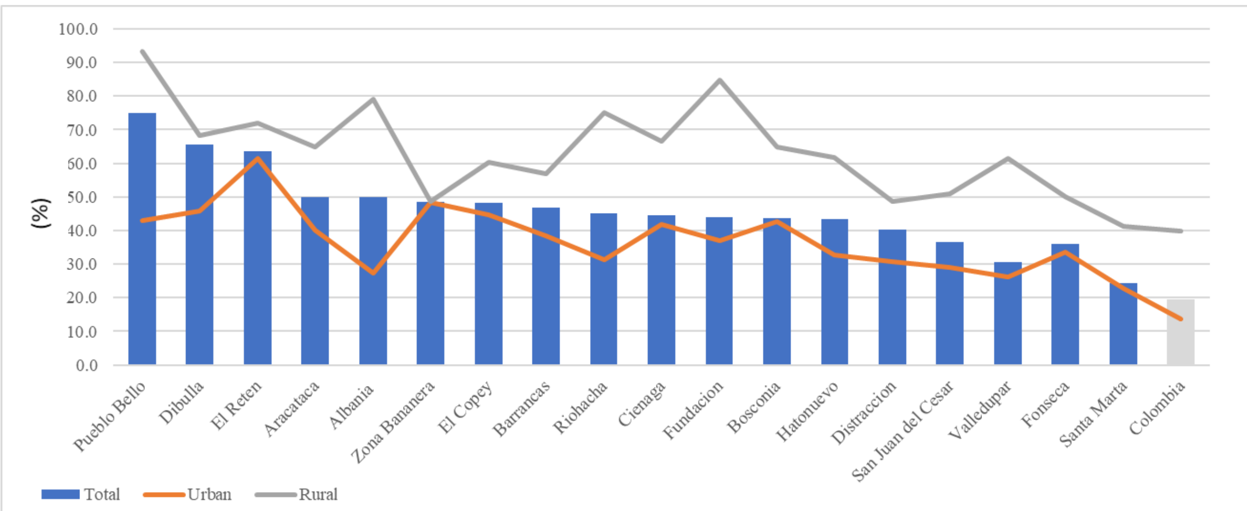
Finally, settlers of lower massif and planes are mostly of coastal origin (Viloria, 2005) and Afro-descendants (Fundación ERIGAIE & CEEP, 2015). As in other territories, Afro-descendant communities have formed Community Councils[12] requesting land titling as collective territories (Observatorio de Territorios Etnicos y Campesinos, 2019). Ministerio del Interior has a record of 21 of them in the region[13] (ART & Municipios Sierra Nevada - Perijá, 2018; MinInterior, 2019).

All this diversity has been compiled by the Census 2018. Its reports show that the 18 municipalities of the region have 1,814,665 inhabitants, 78% of which are in the municipal heads and 22% in rural areas, showing an urban-rural distribution similar to that registered for the national population (DANE, 2019b). Regionally, the percentage of women and men is 51% and 49% respectively. Particularly Santa Marta, Ciénaga, and Dibulla concentrate about 36% of the Sierra population; however, it is mainly located in urban areas. For Santa Marta and Ciénaga, 91% and 87% of the population live in the municipal head. In contrast, the urban proportion in Dibulla reaches 13%, which makes it a primarily rural municipality (DANE, 2019b).

[12] According to Decree 1745 of 1995, a black community is a group of Afro-Colombian families who share a history and have their own culture, traditions and customs that distinguish them from other ethnic groups. Such community may organize themselves as a Community Council, a legal figure through which exercise the highest administrative authority within its lands (MADR, 1995).
[13] Community Councils in the Sierra Nevada: Riohacha (8), Valledupar (6), Zona Bananera (3), El Retén (2), Aracataca (1) and Fundación (1). Detailed information in http://sidacn.mininterior.gov.co/DACN/Consultas/ConsultaCertificadosOrgConsejoPublic

Regarding social conditions, Census 2018 indicates that the average Multidimensional Poverty (MP) of the region was 46%; however, in rural areas the average reached 64% (DANE, 2020). The municipality with the highest MP in rural areas was Pueblo Bello (93%) followed by Fundación (85%), Albania (79%) and Riohacha (75%). In general, the region showed a poverty incidence notably above the 39.9% registered for the national rural areas (Figure 7) (DANE, 2020).

Figure 7. Multidimensional Poverty in Sierra Nevada de Santa Marta. 2018



Source: DANE, Census 2018

Among MP dimensions, those generating the biggest concerns are labor informality, which affects 91% of rural households in the region. This is followed by low educational achievement with an average of 72%, and three dimensions whose average incidences are between 42% and 55%, lack of sanitation systems, inadequate floor material, and dependency ratio. Lack of access to clean water affects 38% of rural population compared to 9.5% in urban areas (DANE, 2020).

Different from Montes de María where municipalities showed similar poverty conditions, in Sierra Nevada de Santa Marta municipalities showed significant differences among them. On one side, Santa Marta as the capital city of Magdalena exhibited lower poverty rates than Dibulla and Ciénaga, and its rural poverty rate remained close to the national rate. On the other hand, rural poverty rates in Ciénaga and Dibulla were 66.7% and 68.4% respectively (DANE, 2020).

Balaguera (2005) asserts that the massif constitutes a natural unit rich in ecological diversity surpassing other areas of Colombia, since it contains optimal lands for agriculture in the country and has a high potential for mining. In that sense, researchers agree that in the Sierra Nevada diverse forms of natural resources use coexist; there is the traditional indigenous livelihoods, small and medium-sized agriculture of settlers and peasants, agro-industrial plantations for export, forestry, logging and mining (e.g., coal, gas, and lime), various forms of tourism, hunting and artisanal fishing (Balaguera et al., 2005; Fundación ERIGAIE & CEEP, 2015; Martinez et al., 2018; Vilardy-Quiroga & González, 2011; Viloria, 2005). In addition, for several decades, illicit crops such as marijuana, coca, and poppy were also part of the local livelihoods (Viloria, 2005).

Several documents describe the history of agricultural development in Sierra Nevada linked to coffee plantations established in the middle lands since the end of the 18th century (Viloria, 2005, 2018, 2019). Coffee was utilized to "colonize" the massif and open the native forest for other crops. Although during the 19th and 20th century, peasants were brought to the region as labor for the *haciendas* owned by foreigners. Later they planted their own plots of coffee, timber, and subsistence crops (Viloria, 2018).

According to Viloria (2018), cacao was also planted to a large extent by foreigners involved in the Sierra Nevada colonization who were benefited from public land granted to them in Santa Marta, Ciénaga and Dibulla. Although their main activity was coffee, they also established cacao plantations in the lowland and formed a commercial society to export it to Germany in the 19th century. The expansion of cacao, tobacco, and bananas for export pressured the construction of infrastructure such as the Santa Marta railway, and the granting of permits for irrigation channels development (Viloria, 2018). By the end of the 20th century, cacao was promoted for income diversification in the coffee area (Viloria, 2005), and for substitution of illicit crops (Comunidad Andina de Naciones, 2011; EFE Agency, 2017; Lale-demoz & Demoz, 2011). Currently, cacao is located in the North and Southeast of the Sierra Nevada where it is grown primarily by peasants and indigenous people (Fundación ERIGAIE & CEEP, 2015).

At the same time, indigenous communities preserved their practices of gathering and using native species for their livelihood (malanga, maize, taro, potato, yam, pigeon pea, agave and others), while other more commercial developments, such as plantain and banana for export, oil palm, irrigated rice, cotton, corn, and cassava were planted in the middle-low zone. In the low coffee border, cacao, avocado, orange, mango, soursop, guava, papaya, squash, beans, beekeeping, and livestock were also developed (Balaguera et al., 2005; Fundación ERIGAIE & CEEP, 2015; Viloria, 2005). Finally, in plain areas is common to find subsistence agriculture and artisanal fishing, aquaculture, and commercialization of raw materials (Balaguera et al., 2005).

For Balaguera (2005), in the previous decades agricultural production in Sierra Nevada has been characterized by low social remuneration, scarce application of technology according to the biophysical conditions, inadequate labor, limited working capital, concentration of property, negative impact on soil and water resources, and institutional and infrastructure deficiencies.

According to MADR, between 2008 and 2018, the total planted area in Sierra Nevada increased by 4%, from 150,237 to 156,501 hectares. By 2018, the region harvested 140,837 hectares and 957,986 tons from oil palm, coffee, banana, maize, cassava, plantain, cacao, mango, and coconut among others (MADR, 2019). As in Montes de Maria, Sierra Nevada also experienced an expansion of long-term cash crops and reduction of traditional annual crops (Bocchi, 2011). In that sense relevant crop area for products of local food security like maize, rice, beans, and cassava was reduced by 56%, 52%, 41% and 33% respectively. This situation represented almost 55,000 tons fewer in 2018 compared to 2008. By contrast, mango, oil palm, plantain, banana, and cacao experienced growth in planted area ranging between 17% (cacao) and 61% (oil palm). Such an increase does not necessarily represents higher volumes, since cacao reported a reduction of 19% in production in the same period (MADR, 2019).

Particularly in Santa Marta, Ciénaga and Dibulla data indicates they concentrate 24% of total planted area and 21% of total production in the region (Table 4) (MADR, 2019).

Table 4. Agricultural Area and Production in Santa Marta, Ciénaga y Dibulla (2018)

	Planted Area (Ha)	Harvested Area (Ha)	Production (Ton)	% of Sierra Nevada region		
				Planted	Harvested	Production
Coffee	16,110	14,376	8,937	53%	53%	53%
Plantain	3,121	2,994	14,151	51%	56%	49%
Banana	2,966	2,926	100,454	18%	19%	19%
Mango	2,495	1,935	15,635	69%	68%	67%
Maize	1,995	1,995	2,686	21%	21%	17%
Cacao	1,903	1,313	640	34%	34%	34%
Coconut	1,388	1,333	7,146	96%	95%	95%
Cassava	1,227	1,227	11,624	19%	19%	19%
Oil Palm	1,188	828	2,941	2%	2%	2%
Others	4,949	4,559	36,577	39%	41%	41%
Total	**37,342**	**33,486**	**200,791**	24%	24%	21%

Source: Evaluaciones Agropecuarias Municipales, MADR, 2019

Regarding the livestock sector, by 2018, the cattle inventory of municipalities in Sierra Nevada included 620,581 heads on 6,618 farms representing 20% of total cattle inventory in Magdalena, Cesar, and La Guajira (ICA, 2019a). The municipalities of interest in this study concentrate just 6% of the regional inventory of cattle (ICA, 2019a).

As mentioned above, land tenure in Colombia is markedly uneven, and the Sierra Nevada is not an exception, mainly peasants and indigenous people, lack clear property rights on the land. This is due to multiple factors: among them stand out the realignment of indigenous reservations

(Fundación ERIGAIE & CEEP, 2015; Viloria, 2005) and Tayrona and Sierra Nevada National Natural Parks (Balaguera et al., 2005), as well as the outdated rural cadaster and the low capacity of the government to meet the titling requests (Fundación ERIGAIE & CEEP, 2015; Martinez et al., 2018). For instance, the Fundación ERICAIE & CEEP (2015) found that some peasant properties located within the Arhuaco reservation cannot obtain titles until they prove their families arrived before the formation of the reservation (1984). Likewise, many peasants only have purchase letters instead of titles which limit them to access public goods and services (e.g., credit, insurance, irrigation, etc.) (Fundación ERIGAIE & CEEP, 2015).

Besides, historic land grabbing (Balaguera et al., 2005; Martinez et al., 2018; Viloria, 2005) and harassment by illegal groups have generated a land dispossession scenario affecting local communities (Viloria, 2005). Finally, the relocation of indigenous houses from the paramo to lower areas, and the increase of tourist activity (hotels and services) put pressure on the land market, distorting prices and causing them to rise (Fundación ERIGAIE & CEEP, 2015).

Data of land distribution show that in Sierra Nevada, Zona Bananera has the highest property concentration with an index of 0.789, and Pueblo Bello the lowest with 0.535 (Table 5) (UPRA, 2016a).

Table 5. Gini Index of Land Property in Municipalities of Sierra Nevada de Santa Marta

Municipio	**Gini Index**	**Year**	**Municipio**	**Gini Index**	**Year**
Zona Bananera	0.789	2011	San Juan del Cesar	0,676	2008
El Retén	0.758	2009	Fonseca	0,671	2005
Santa Marta	0.736	2014	Fundación	0.667	1994
Distracción	0.712	2008	Riohacha	0.645	2014
Hatonuevo	0,710	2014	Albania	0,639	2014
Dibulla	0.710	2014	Ciénaga	0.635	2010
Bosconia	0.698	2008	El Copey	0.631	2011
Valledupar	0.697	2014	Aracataca	0.622	1994
Barrancas	0.683	2003	Pueblo Bello	0.535	2011

Source: UPRA, 2016a

Regarding property informality, UPRA calculated an index registering what percentage of total properties have an anomaly or insufficiency of documentation to support ownership (UPRA, 2016b). At the national level, the index was 54.3%; however, 15 of 18 municipalities of Sierra Nevada surpassed that index and ranged between 57% and 95%. Dibulla and Ciénaga stand out with 72% and 95%, respectively.

Sierra Nevada also presents significant deficiencies in its road infrastructure (Lozano & Restrepo, 2016; Viloria, 2005), irrigation (Fundación ERIGAIE & CEEP, 2015), agricultural extension, and other services for rural population (ART & Municipios Sierra Nevada - Perijá, 2018). According to Viloria (2005), cultivation of coffee and marijuana drove the settlement of new populations; therefore, new roads, mostly unpaved, usable only in the dry season emerged to connect towns with Troncal del Caribe between Santa Marta and Riohacha since 1971[14].

The Magdalena Development Plan 2020-2023 (Gobernación del Magdalena, 2020) reports that the department has 6,653.6 km of roads, 11% of which correspond to primary, 17% to secondary, and 72% to the tertiary network that connects villages with municipal heads. Most of the road network (91%) in the charge of the Magdalena government, has paving quality problems or do not have pavement. During rainy seasons, transit through these road is difficult due to floods, affecting costs of inputs and harvest transportation while hindering access to health services and education (ADR, FAO, & Gobernación del Magdalena, 2019; Gobernación del Magdalena, 2020).

Similarly in La Guajira, the Development Plan 2017-2019 indicates that the primary network has 377.64 km, 82% of which is paved although only 50% of them are in good condition, and 20% are in critical condition (Gobernación de La Guajira, 2017). The secondary network has 497 km, 67% of which is paved, 23% is unpaved, and 10% does not even have the surface graded.

[14] The principal roads are Dibulla-San Antonio, Santa Marta-Minca-El Campano-La Tagua, Minca-Cerro San Lorenzo, La Gran Vía-San Pedro de la Sierra, La Bodega-Palmor, Valencia de Jesús-Pueblo Bello-Nabusímake and Valledupar-Atánquez.

Barely 24% of the paved roads are in good condition. Regarding tertiary roads, official information reports 1,299 km managed by INVIAS and the department; however, there is no diagnosis of the network administered by the municipalities. It is estimated that tertiary roads correspond to 60% of the total network in the department (Gobernación de La Guajira, 2017).

Despite of having 36 rivers, the available irrigation is insufficient for the agricultural potential of the region. This situation corresponds to the high number of irrigation districts out of operation (Balaguera et al., 2005; Fundación ERIGAIE & CEEP, 2015; Lozano & Restrepo, 2016).

For Lozano & Restrepo (2016), available irrigation is insufficient for the regional economic potential. By 2007, Magdalena had 14 irrigation districts, 43% of which operated serving 2,099 users and 30,370 hectares; this represents only 9% of the susceptible area to irrigation. They also indicated that the land used in agricultural activities by 2014, was only 27.5% of the Magdalena area while 83% have such capacity. On the other hand, La Guajira had 13 irrigation districts, all out of operation despite 45% of its area being suitable for agriculture (Lozano & Restrepo, 2016).

Thus, although Sierra Nevada has four of the 18 large-scale districts in Colombia (IGAC, 2017), water supply is limited to certain crops in the area. According to Fundación ERIGAIE & CEEP (2015), regional irrigation is concentrated in oil palm and banana in the western massif. Several decades ago, the United Fruit Company intervened so that the Frío, Sevilla, Tucurinca, and Aracataca rivers could irrigate export bananas.

Later the infrastructure was bought and transformed into irrigation districts by the government (INCORA), and finally, they were handed (Asosevilla) or delegated for administration to user associations (AsoRiofrío, Asotucurinca, and Usoaracataca) at the beginning of this century (Fundación ERIGAIE & CEEP, 2015).

According to ADR, in charge of agricultural irrigation and drainage, eight districts are currently registered in this region (Table 6).

Table 6. Irrigation Districts in Sierra Nevada de Santa Marta.

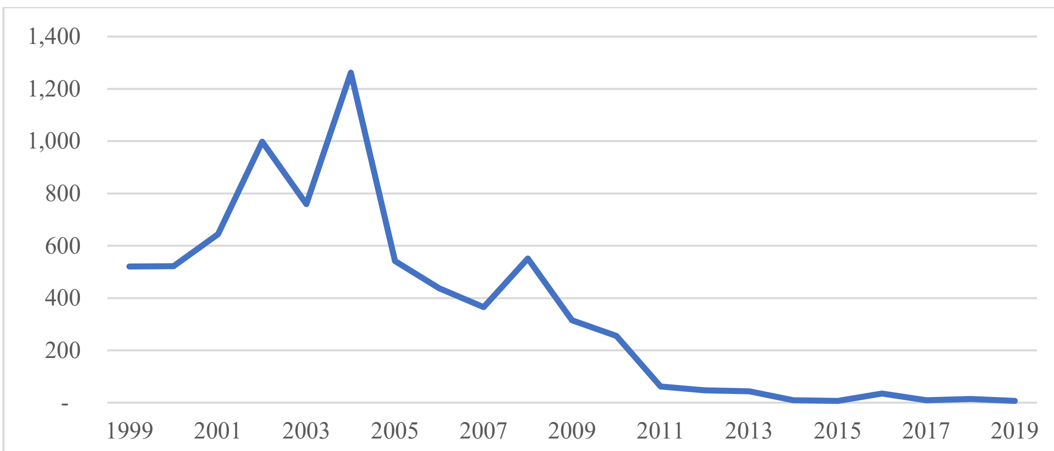
District name	Scale	Type	Department	Municipalities	Water source	Area (Hectares)	Activities	Association	Families
Rio Frio	Big	Irrigation and drainage	Magdalena	Ciénaga and Zona Bananera	Frio river	5,672	Banana, oil palm, fruits, and subsistance.	AsoRiofrio	630
Sevilla	Big	Irrigation and drainage	Magdalena	Zona Bananera	Sevilla river	7,100	Banana, oil palm, fruits, mango, lemon, and subsistance	AsoSevilla	418
Tucurinca	Big	Irrigation and drainage	Magdalena	Zona Bananera	Tucurinca river	8,304	Oil palm, banana, rice, fruits, subsistance, pastures and livestock	AsoTucurinca	352
Aracataca	Big	Irrigation and drainage	Magdalena	Aracataca and El Retén	Aracataca and Fundación rivers	10,434	oil palm, pastures, banana, rice, subsistance, cassava, citrics, fruits and cacao	UsoAracataca	444
El Porvenir*	Small	Irrigation	Magdalena	Santa Marta	El Silencio, Tigrera and Tamacá creeks	46	Coconut, mango, lemon, orange, avocado, pastures, papaya, vegetables, cassava, maize, tomato, and soursop.	AsoPorvenir	22
San Pedro	Small	Irrigation	La Guajira	Barrancas	Palomino and Manantial rivers	100	Subsistance, vegetable, livestock, minor species, potato, cassava, maize, sweet pepper, guineo, plantain, pastures.	Asociación de Usuarios	50
Los Haticos*	Small	Irrigation	La Guajira	San Juan del Cesar	n.d.	60	Cassava, maize, beans, tomato, watermelon, melon, pastures, and plantain.	AsoHaticos	50
Tabaco Rubio	Small	Irrigation	La Guajira	Riohacha	Tapias river and Robles canal	350	Oil palm, banana, livestock, minor species, tobacco, fruits, subsistance, pastures, tomato, maize and cotton.	AsoTabaco Rubio	34

Source: ADR, 2019

Like Montes de Maria, Sierra Nevada is recognized as one of the regions where the armed conflict and drug business took advantage of territorial complexity, negatively affecting the population, primarily rural inhabitants (Balaguera et al., 2005; Defensoría del Pueblo, 2003; GMH, 2013; Vilardy-Quiroga & González, 2011). In the early 1970s, the first marijuana crop appeared in the region and spread around the massif perimeter until the middle of the 1980s (Viloria, 2005). In that way, an illegal economy emerged. Later it was intensified by the bonanza of coca (Figure 8) and the arrival of criminal groups (drug traffickers, paramilitaries, and guerrilla) disputing control of drug businesses (Borrero, 2004; Diaz & Sanchez, 2004; GMH, 2013).

The advancement of guerrilla groups in the 1980s was experienced mainly in Santa Marta, Dibulla, Ciénaga, Fundación, San Juan del Cesar, and Pueblo Bello (Balaguera et al., 2005; Defensoría del Pueblo, 2003; Viloria, 2005). The groups even occupied areas of National Park and indigenous reserves (Fundación ERIGAIE & CEEP, 2015). Nourished by illegal income, there were at least five guerrilla fronts (FARC-EP, ELN and EPL) in the 1990s (Viloria, 2005). Almost in parallel, the paramilitary model arrived into the region supported by drug traffickers to eliminate the guerrillas dedicated to collect "taxes" of illicit crops (Defensoría del Pueblo, 2003).

Figure 8. Coca Area in Sierra Nevada de Santa Marta Municipalities (Hectares)[15]



Source: Observatorio de Drogas de Colombia – Min Justicia

The confrontation among groups for territory and coca business generated a scenario of violence against public and private properties, violation of human rights, and overexploitation of natural resources (Diaz & Sanchez, 2004; Fundación ERIGAIE & CEEP, 2015; Viloria, 2005). The government reports 586,436 victims between 1985 and 2019: the most affected have been Santa Marta, Valledupar, Cienaga, Fundacion, Zona Bananera and Riohacha, concentrating 71% of them (Unidad Nacional de Victimas, 2019).

[15] According to Observatorio de Drogas de Colombia (http://www.odc.gov.co/sidco), the municipalities have had coca crops, in different years, between 1999 and 2019 are Dibulla, Riohacha, San Juan del Cesar, Aracataca, Ciénaga, Fundacion, y Santa Marta.

Alternative Development approach in Colombia

According to Zorro (2005) the factors that contribute to the dynamics of illicit crops are not just the appropriate agro-ecological conditions for planting, but also the precarious economic situation of farmers, social exclusion of producing regions, and the political and institutional instability. However, the control of illicit crops in Colombia has been primarily based on plants eradication and their substitution by other crops or activities through programs guided by the so-called Alternative Development approach (Diaz & Sanchez, 2004; Vargas, 2010; Zorro, 2005)[16].

For the government, eradication refers to official actions aimed to eliminate the plantations, either by manual removal or chemical spraying, in a forced or voluntary manner, exerting direct control over the availability of raw material for drug production (Minjusticia, 2007). On the other hand, Alternative Development (AD) is the set of actions aimed to improve the socio-economic and environmental conditions of people involved with such plantations to reduce their incentives to sow illegal crops and facilitate their transition to licit activities (DNP, 1994, 1995; Minjusticia, 2007). Officially, AD is conceived as a complementary action to eradication, which is applied exclusively to small growers (less than 3 hectares) and indigenous people who obtain their livelihoods from illicit crops (DNP, 1995; Vargas, 2010; Zorro, 2005).

In Colombia, substitution began with projects for crop replacements in the Amazon region funded by the United Nations Program for International Drug Control-UNDCP between 1985 and 1989 (Minjusticia, 2007; Zorro, 2005). Later, the first Alternative Development Program (PDA) was implemented in 1992 and PLANTE program began in 1994 (Caicedo, 2012; Zorro, 2005).

[16] According to UNODC (2010, p. 4), the UN General Assembly in 1998 defined Alternative Development as "a process to prevent and eliminate illicit crops of plants that contain narcotic drugs and psychotropic substances, through rural development measures specifically designated in the context of national economic growth, and sustainable development efforts in the countries that undertake initiatives against drugs, taking into account the socio-cultural characteristics of the groups and communities in question, within the framework of a global and permanent solution to the problem of illicit drugs ".

PLANTE was created through the CONPES[17] 2734 aimed at offering legal alternatives for income generation, improvement of quality of life, environmental conservation, and promotion of peace (DNP, 1995). Its funding came from national budget and international aid, and it was primarily focused on the southern part of the country (DNP, 1994, 1995; Minjusticia, 2007). Its actions included credit, housing, roads, resettlement and promotion of plantain, rubber, cacao, and oil palm to substitute coca and marijuana crops (Caicedo, 2012). However, scholars indicate that many projects could not generate stable income, especially those depending on monoculture while others reproduce deforestation and inadequate management of natural resources (Vargas, 2010).

By 1998, the Pastrana government set an anti-drug policy framed in a broader strategy called *Plan Colombia*, primarily supported by the United States government[18]. Actions were organized into four components: 1) negotiation with illegal groups, 2) the fight against drug trafficking, 3) economic and social recovery, and 4) institutional development (Salazar et al., 2017; Vargas, 2010). In this period, criticism focused on the priority given to the eradication, leaving behind the protection of households livelihoods' and food security (Caicedo, 2012; Vargas, 2010).

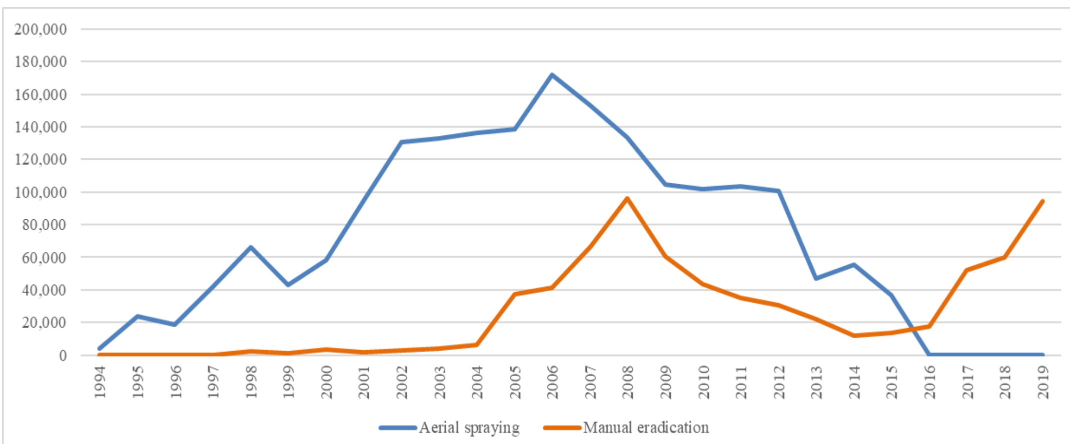
The weaknesses of projects were associated with at limited knowledge of the quality of soils and agricultural suitability of alternatives in each region. Besides, poor actions to improve commercialization affected the stability of income generation and motivated the abandonment of projects (Vargas, 2010). In this period rubber, plantain, peach-palm heart, livestock, minor species, borwn sugar, and pepper were promoted (Vargas, 2010).

According to Uribe (2019), since coca eradication registration began in 1987, few hectares were eradicated until 1998 when it showed an increment of 66,000 hectares sprayed and 2,036 hectares manually eradicated. Later, although manual eradication remained at a similar level in

[17] CONPES are documents of public policy recommendation issued by the National Council for Economic and Social Policy.
[18] Between 2000 and 2005, USAID contributed US $557 million to Plan Colombia supporting AD among others (Vargas, 2010).

2002, aerial intervention sprayed almost twice the area done in 1998, reaching 130,364 hectares in 2002 (ODC & Minjusticia, 2019) (Figure 9). At this point, after 15 years of AD implementation, coca had grown to 163,000 hectares in 2000 (ODC & Minjusticia, 2019; Uribe, 2019). Results showed failures to effectively provide mechanisms to encourage producers in dropping the illicit cultivation. Certain interventions made peasants and indigenous people's conditions more precarious, allowing guerrillas and paramilitaries to take advantage of them (Caicedo, 2012; Palau & Arias, 2011; Vargas, 2010).

Figure 9. Coca Eradication in Colombia 1994-2019 (hectares)



Source: Observatorio de Drogas de Colombia, Minjusticia, 2019

In 2003, CONPES 3218 set the new guidelines for AD now framed in the *Politica de Seguridad Democrática* of first Uribe government (2002-2006). Like previously, this policy focused on drug production and trafficking dismantling (DNP, 2003; Minjusticia, 2007). Eradication increased significantly, particularly the manual eradication which consisted in pulling the coca plant directly (Salazar et al., 2017; Vargas, 2010). Between 2003 and 2006, the annual spraying area rose from 132,817 to 172,025 hectares, while manual eradication increased from 4,220 to 41,346 hectares (ODC & Minjusticia, 2019). On the other hand, AD was integrated into the agricultural and environmental policy by three components: 1) Productive and income-generating projects (PPP in Spanish); 2) Forest Ranger Family Program (PFGB in Spanish); and 3) Institutional strengthening, social development, and monitoring (Caicedo, 2012; Vargas, 2010).

The PPP were oriented to food security and cash flow generation in medium and long-term interventions. In this version, projects focused on what the government considered competitive value chains and high value-added products. Crops such as cacao, rubber, oil palm, coffee, timber and non-timber forests were expected to meet this requirement (DNP, 2003; Vargas, 2010). For that, peasants' integration with "modern" businesses was fostered through alliances co-financed with a public budget that covered the contribution of peasants (Vargas, 2010; Zorro, 2005).

On the other hand, the Forest Ranger Families Program (PFGB) corresponded to a direct monetary transfer to families organized in clusters within targeted areas; these transfers operated as compensation for elimination and not replanting of illicit crops, favoring the regeneration and conservation of natural ecosystems (Caicedo, 2012; DNP, 2003). This program operated by four actions: 1) technical support and training about conservation and forest recovery, 2) income-generation projects, 3) social component aimed at improving relations between relatives or neighbors, and strengthening organizations, and finally, 4) citizen participation (Meneses, 2016; PNUD, 2011; UNODC, 2010).

Sierra Nevada was one of the principal regions involved in the Ranger Families Program, while Montes de Maria was served mainly by productive projects in PPP.

The main criticisms pointed out that projects continued as trial and error process that relapsed into the same problems of PLANTE (Caicedo, 2012; Vargas, 2010; Zorro, 2005). Also, the imposition of standardized alliances between peasant and agribusiness sectors was not feasible for all contexts and communities. Likewise, the complete eradication of coca before the intervention was questioned (Vargas, 2010; Zorro, 2005). Finally, the financial scheme for ranger families turned unsustainable (UNODC, 2010); by the end of the period, 72% of them did not have any productive project (Vargas, 2010).

The second government of Uribe (2006-2010) implemented the *Política Nacional de Consolidación Territorial* (PNCT) continuing with aerial and manual eradication, PFGB, PPP, and zero illicit protocol (Bocchi, 2011; Palau & Arias, 2011; Salazar et al., 2017). The novelty was the inclusion of new regions into the programs[19]. Thus, areas affected by the drugs' economic system and conflict despite not having coca plantations were covered.

In that way, plans included Montes de María and Sierra Nevada de Santa Marta, where the area was minimum although the violence great (DNP, 2010a; Palau & Arias, 2011; Presidencia de la República de Colombia, 2009). Thus, the strategy aimed at creating an environmental buffer area combined to indigenous resettlement eliminating the small areas of coca in Sierra Nevada, while in Montes de María, the project was associated with the return and rights restitution for displaced population and the reinsertion of paramilitaries (Bocchi, 2011; Vargas, 2010).

By the end of this period, five phases of the PFGB had already been completed with 126 municipalities and 113,779 families involved (Meneses, 2016). On the other hand, the PPP between 2003 and 2010 had provided resources to 94,093 hectares planted in projects of 43,938 families (DNP, 2010b). In this opportunity, problems were around the weak integration of local governments in the implementation of *Plan de Consolidación* in regard to road infrastructure, market development, and logistics (DNP, 2010b; Palau & Arias, 2011; Zorro, 2005).

[19] According to Palau & Arias (2011), the criteria for areas selection were: 1) terrorism and drug trafficking, 2) violation of Human Rights, 3) weak institutional presence, and 4) illicit crops. In 2009, the government announced the categories as 1. Areas to begin consolidation, 2. Areas of transition towards consolidation (where Montes de María and the Sierra Nevada were included), and 3. Complementary Areas.

In 2010, CONPES 3669 of *Política Nacional de Erradicación Manual de Cultivos Ilícitos y Desarrollo Alternativo para la Consolidación Territorial* focused on four objectives to address the low sustainability of eradication and AD: 1) strengthening inter-agency coordination; 2) improving statistics of illegal crops; 3) promoting a culture of legality, and 4) comprehensive and sustainable attention for communities (DNP, 2010b). In the latter, the government proposed tackling the AD constraints improving synchrony in goods and services supply. In this way, the PFGB and PPP programs had to act over factors like land tenure, infrastructure provision, basic sanitation, housing, education, health, and financial inclusion (DNP, 2010b).

Moreover, the policy recognized the inconvenience of limiting projects to the export agenda (cacao, rubber, coffee, palm, and forestry) despite their hypothetical higher chances of success. The experience showed that participants had not the conditions or expertise required for the implementation of those alternatives (DNP, 2010b). Finally, the policy acknowledged the problem of eradicating before providing attention since communities relapse into coca while they were able to get involved in the AD programs (DNP, 2010b).

In 2012, the government added the post-eradication and containment strategy through which food assistance and provision of seed capital, inputs, training, and tools were offered, while coca eradication was executed (Unidad de Consolidación Territorial, 2014, 2015). On the other hand, the government included the graduation strategy focused on business strengthening; under this modality, organizations began their exports of cacao, rubber, honey, and fishery products (Unidad de Consolidación Territorial, 2015).

Between 2012 and 2014, *Plan de Consolidación* served about 81,000 households and 1.4 million hectares kept free of illicit crops. Based on the 2014 evaluation, 81% of families indicated having improved their production techniques, 39% declared improvements in their environmental protection activities, 37% in income generation, and 22% in job sources (Unidad de Consolidación Territorial, 2015).

In the next period (2014 -2018), efforts focused on concluding the negotiation of Peace Agreement with FARC-EP while closing the *Plan de Consolidación*. According to the 2018 report of Agencia de Renovacion del Territorio (ART)[20], which assumed the responsibilities of Unidad de Consolidación Territorial since 2015, the final actions of AD resulted in 17,729 households served by the post-eradication program, and seven organizations (1,240 families) in the strategy of graduation with projects of cacao, coffee, fruits, and fish farming (ART, 2018).

Finally, the new approach for AD in the Peace Agreement took shape primarily by the *National Comprehensive Plan for Substitution of Illicit Crops (PNIS*) undertaken directly by Alta *Consejería para la Estabilización y la Consolidación in Presidencia de la República* and ART (Gobierno Nacional de Colombia et al., 2016; Salazar et al., 2017). Both PDET and PNIS strategies were widely explained above in this section.

Cacao as an Alternative for Illicit Crop Substitution and Rural Development

Cacao in Colombia

Theobroma cacao L. excels as one of the most relevant species by its economic importance in industries like confectionery, cosmetics, and pharmaceutical worldwide; within it, two subspecies have been described as *Theobroma Cacao ssp.* and *Theobroma Cacao ssp. Sphaerocarpum* (MADR, FNC, & FEDECACAO, 2013; Villamil, Martinez, Aranzazu, & Cadena, 2013). The first, generally known as *Criollo* variety is characterized by elongated fruits with pronounced furrows and white beans; the second is known as *Forastero* and presents rounded fruits with shallow furrows and purple seeds. The artificial crossing of these two types of cacao

[20] By 2015 the government created Agencia de Renovación del Territorio (ART) attached to the MADR and responsible of coordinating the intervention of national and local entities in post-conflict areas mainly through the 16 PDET (ART, 2018).

gave rise to a third type called a *Hybrid* or *Trinitarian* cacao characterized by a wide variability of shapes, sizes, and performance, this is the most widespread in Colombia (MADR et al., 2013).

As a perennial crop, cacao begins to produce fruit after 24 or 36 months, although some do it up to 48 months depending on the variety (Avila, Campos, Guharady, & Camacho, 2020; FEDECACAO, 2015b). Later, with proper agronomic management, the production increases regularly until the plant reaches maturity and stabilization about the seventh year, then production can remain at acceptable levels up to year 20. After that, trees reach very low productivities; therefore, it is recommended the trees are replaced to renovate the cultivation. Grafted cacao trees begin production in the third year, reaching maturity and stable production after the seventh year. When hybrid cacao is planted, production begins between 7–8 years of age (FEDECACAO, 2015b; MADR et al., 2013; Zamora & Ochoa, 2020).

For its development, cacao cultivation needs to be located between 0 and 1,200 meters above sea level, in places where temperature range between 22°C and 30°C. The water requirement for cacao is between 1,500 and 2,500 mm well distributed during the year; therefore, crops' rainfall dependents should have alternatives for irrigation to accomplish appropriate water needs in low precipitation seasons. The optimum relative humidity for the cultivation is 80%, although it could be managed between 70% and 90% (FEDECACAO, 2015b; MADR et al., 2013).

Implementing pest and disease control during the entire crop cycle is important since they can cause considerable damage. However, diseases generate the greatest economic losses for farmers in Colombia (FEDECACAO, 2015b; MADR et al., 2013). According to Jaimes & Aranzazu (2013), those with the greatest potential for damage are *Moniliophthora roreri* (Moniliasis or Frosty Pod Rot) and *Moniliophthora perniciosa* (Witches' broom). Moniliasis is the most widespread: the causative fungus attacks the fruit (pods) at any stage of development but its intensity varies according to climatic conditions, particularly high temperature and relative humidity (Correa, Castro, & Coy, 2014a; Pinto & Saurith, 2006). Excessive shade or competition for nutrition and water also contribute to generating the disease (FEDECACAO, 2015b).

In addition to disease control, other activities are key for cacao cultivation. In the sowing phase, both genetic material selection and its propagation strategy (usually by grafting[21]) influence final production; during maintenance, appropriate techniques and opportuneness in pruning are critical. Finally, in post-harvest, adequate timing, implements, and techniques for cacao fermentation and drying is essential (Arvelo, González, Maroto, Delgado, & Montoya, 2017; Barón, 2016; FEDECACAO, 2015b).

Cacao cultivation can be carried out as monoculture in open areas; however, increasingly technical recommendations turn to planting cacao in Agroforestry Systems (AFS) since they provide a wide range of economic, sociocultural, and environmental benefits (Armengot, Barbieri, Andres, Milz, & Schneider, 2016; MADR et al., 2013; Sanial, Fountain, Hoefsloot, & Jezzer, 2020). Although a unified definition of cacao AFS is lacking (Sanial et al., 2020), it can be defined as the multifunctional association of cacao with other trees, crops, and/or animals in the same area, characterized by ecological and economic interactions between them. It may include existing native plants and forests established by farmers (Cacao Forest, 2016; FEDECACAO, 2015b; MADR et al., 2013; Sanial et al., 2020; Somarriba & Lopez-Sampson, 2018). In such arrangements, cacao grows in the lower strata where humidity is high and shade species filter the light ameliorating the microclimate while storing water and nutrients (Cacao Forest, 2016).

Besides, AFS represent an alternative highly adaptable to different types of farmers, from those with subsistence crops using rudimentary systems to those who manage full-design and

[21] According to Torres & Rodriguez (2015), vegetative propagation is the most time and cost-efficient method to increase quality productivity, and disease tolerance in cacao. Clones grafted on hybrid rootstocks begins early and yields can exceed 2,500 Kg.

technified plantations (Cacao Forest, 2016; RESET.org, 2018). Especially the smallest ones find in crops diversity greater opportunities to generate their own food while improving their income and living conditions (Somarriba & Lopez-Sampson, 2018). In Colombia cacao areas managed as AFS are estimated between 75% (Somarriba & Lopez-Sampson, 2018) and 95% (interview with Fedecacao, 2018) of the total area.

On the other hand, cacao cultivation requires a significant amount of labor supplied mainly by household members and eventually with wage workers for specialized tasks (e.g., grafting, pruning, and post-harvesting) (Abbott et al., 2018; Castellanos et al., 2007; Torres & Rodriguez, 2015). In the maintenance stage, every three cacao hectares generates one job in Colombia, so it is recognized as a sector of employment generation (MADR et al., 2013). Between 2011 and 2019, sectoral employment increased by 3.2% on average each year, reaching about 91,500 direct and 73,000 indirect jobs in 2019 (MADR, 2020).

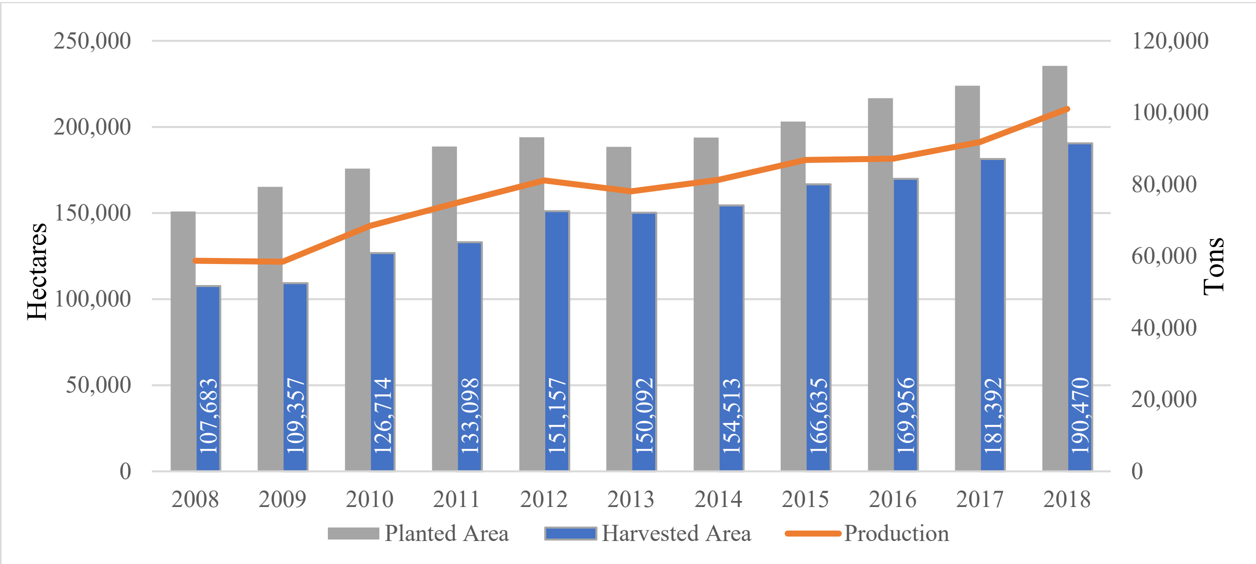
According to MADR (2019), cacao production increased by 5.4%, on average, between 2008 and 2018, reaching 101,020 tons. Despite this growth, such performance is mainly due to the increase in harvested area (5.6% annual average) that achieved 190,470 hectares; in contrast, yields decreased by 0.2% (Figure 10).

During this period, yields fluctuated between 506 and 563 Kg/hectare/year, ending by 2018 with barely 530 Kg, this contrasts with its potential calculated between 1,500 and 2,000 Kg/hectare/year [22] (FEDECACAO, 2015b; MADR, 2019). Regionally, departments exhibit varied productivity results. Among the principal producers in 2018, Antioquia, Arauca, and Norte de Santander were above the national average with 624, 600, and 571 Kg/hectare/year respectively. Regarding the departments in this study, La Guajira (585Kg) showed a figure over the national

[22] Plantations implemented from agroforestry systems using high-performance cacao clones, with 1,000 to 1,100 trees per hectare can produce 1,500 kilograms/hectare/year. Cacao planted without shade can produce up to 2.000 Kg (FEDECACAO, 2015b).

average while Magdalena (488 Kg) and Bolivar (414 Kg) were far below it (MADR, 2019).

Figure 10. Cacao Area and Production in Colombia (2008-2018)



Source: MADR, Evaluaciones Agropecuarias 2019

From the perspective of post-conflict regions, 149 of 170 PDET municipalities concentrated 46% of the harvested area and 43% of cacao production by 2018. The average yield of this group barely reached 492 Kg/hectare/year, which is 14.5% less compared to the 563 achieved by non-PDET municipalities. Regarding municipalities with coca, data shows that 167 (88%) of them also have cacao plantations; like in PDET municipalities, cacao area and production correspond to 47% and 42% respectively. However, its performance is even more limited. In municipalities with coca, the average yield barely reached 473 Kg/hectare/year in 2018 while those without coca areas obtained 581 Kg (MADR, 2017, 2019; ODC & Minjusticia, 2019).

According to official figures, about 65,300 households were dedicated to cacao cultivation by 2019 (MADR, 2020); in contrast, FNC-FEDECACAO reported 52,569 households by 2018. From them, about 30,600 growers were characterized through a survey undertaken by FEDECACAO in 365 municipalities between 2016 and 2018. Preliminary results showed that 77%

of farmers were men and 23% women; however, women's participation ranged between 8% to 42% at the department level. Other results indicated that the age of cacao farmers was, on average, 53 years old; regarding education level, 65% of farmers barely reached elementary, 27% secondary education, and only 6% had a technical or undergraduate level (FNC-FEDECACAO, 2018).

The survey also showed that the average size of farms was 13.7 hectares, of which 2.9 hectares are utilized in cacao cultivation while the remaining area is dedicated to other uses[23]. In fact, 26% of farmers had cacao areas between 0.6 and 1 hectare and 49% between 1.1 and 3 hectares; only 24% overcome those dimensions. Besides, data revealed that 64% of farmers did not have credit; of those with credit, 79% used it for planting new areas, while the rest was used for maintenance (10%), infrastructure (6%), and renovation (5%) (FNC-FEDECACAO, 2018).

Regarding cultivation management, 83% of farmers execute pruning. Most of them (72%) implement this practice once a year, 25% make it twice, and 3% prune with a higher frequency. Fertilization practices are implemented by 57% of farmers which primarily focus on edaphic fertilization (93%). Regarding fertilization frequency, it is executed once a year in 56% of cases, 35% fertilize twice a year, and the remaining 9% make it three or four times. On the other hand, 84% of farmers declared they implement plant health controls, 21% apply this weekly, 49% biweekly, and 30% make it monthly. (FNC-FEDECACAO, 2018).

Finally, regarding fermentation and drying, 39% of farmers reported they did not have infrastructure for beans benefit. From those who declared having it, 95% use wooden boxes for fermentation. On the other side, 96% of farmers dry cacao beans by sun and 4% sell wet cacao. For drying, 38% utilize wooden trays, 31% plastic, 16% surfaces like *patios* and 10% use dryer

[23] According to Arboleda (2010) an important part of the cacao farms do not grow cocoa as their main activity. Using figures from the 1998 cacao census in Antioquia, cacao area share was approximately 10%, while in Nariño, it was 47.5% within farms. The most common complementary crops were pastures, coffee, banana, citrus, timber, cassava, maize, rice, and sugar cane.

marquees (FNC-FEDECACAO, 2018).

Since 1998, sectoral documents have pointed out the deficient level of technology adoption as the main restriction for increasing yields and benefits for cacao farmers in Colombia[24]. Among limitations scholars highlight the use of low-quality sowing material, scarce maintenance labors allowing pests and diseases emergence, and deficiency in post-harvest activities.

Regarding genetic material, besides the low use of quality seedlings (MADR & Consejo Nacional Cacaotero, 2012) farmers ignore what varieties suit their region (Castellanos et al., 2007) and which they currently have in their farms, especially when seedlings came from aid programs (Castellanos et al., 2011; Technoserve, 2015). Correctly-executed graft is also a critical condition for success in plants propagation; although farmers often receive training on it, application of grafting methods is significantly low (Acosta & Villarraga, 2006).

During the cultivation growth, practices and technologies less adopted are those aimed at water management (Arboleda & González, 2010; Castellanos et al., 2007), sowing density (Acosta & Villarraga, 2006; Barrientos & Gómez, 2017; García-Cáceres et al., 2014), and fertilization. In fact soil-testing-based nutrition plans are unusual (Abbott et al., 2018; Acosta & Villarraga, 2006; Arboleda & González, 2010). Likewise, maintenance pruning and disease controls are often poorly implemented because of their irregular frequency and wrong techniques (Abbott et al., 2018; ATKEARNEY, MADR, & Mincomercio-PTP, 2010; Correa et al., 2014a; Mantilla et al., 2000; Technoserve, 2015).

In harvest and post-harvest stages, limitations are related to the delayed harvesting that increases the vulnerability to diseases, and low technification of cacao fermentation and drying

[24] By 1998, 77% of farms only implemented basic practices of weed control, pruning, and harvesting; other 22% reached a medium level of technology by executing pest and disease control and shade pruning besides the basic practices, finally, 1% was classified as a high level of technification. They applied fertilization, irrigation, and re-planting (Espinal, Martínez, & Ortíz, 2005; SIC, 2012).

(Acosta & Villarraga, 2006; Barrientos & Gómez, 2017; Castellanos et al., 2007; Mantilla et al., 2000). Particularly, in the latter, the low availability of infrastructure and equipment (e.g., drying beds, fermentation boxes, small lab equipment, etc.) and the non-standardized application of methods affect the final quality of cacao beans (Abbott et al., 2018; Castellanos et al., 2007; Escobar, Santander, Useche, Contreras, & Rodríguez, 2020). Besides an additional problem is the mix of beans of different qualities during the post-harvest process (Abbott et al., 2018).

As mentioned above, only 16.5% of farmers declared having received technical assistance in 2014 (DANE, 2016). Although this service was enacted since the 1980s, as a public service that local and national governments should provide, neither institutional capability nor the official budget has been provided to accomplish such responsibility (Tami, Garrido, Uribe, Henao, & Rincón, 2013). In addition to the Municipal Units of Agricultural Technical Assistant (UMATA), other mixed, private, and NGOs accredited by the government could offer the service (Tami et al., 2013).

Thus, in the cacao sector increasingly varied stakeholders provide advisory, extension, and technology transfer services. Among them stand out Agrosavia, ADR, FEDECACAO, Red Cacaotera, Servicio Nacional de Aprendizaje (SENA), Programa de Transformación Productiva of Mincomercio, local governments through UMATAs and CPGAs, and grower´s cooperatives. In addition, others like Casa Luker, Compañía Nacional de Chocolates, CIAT-CGIAR, SwissContact, ProColombia, chambers of commerce, universities, UNODC, FAO and operators of aid funds from USAID, USDA, CAF, and SECO are also involved in these services through projects and programs (Abbott et al., 2018).

Despite multiple efforts, technical assistance services in Colombia are perceived as fractioned and with little effectiveness; besides, they are insufficient and low qualified in most of the regions (DNP, 2015a). Scarce funding, coordination, and political commitment have been the main constraints during the last few years (Corpoica, 2015a; OCDE, 2015; Tami et al., 2013). The cacao sector is not an exception, attempts to inventory and characterize their status have only captured a fraction of the intricate picture in this matter (Abbott et al., 2018). Thus, the most traceable figures are those reported by FEDECACAO as executor of levy resources of Fondo Nacional del Cacao[25].

By 2018, FEDECACAO had 25 technical units, eight experimental farms, 250 agricultural technicians, and lab infrastructures. Its technical assistance program incorporates advising for establishment, rehabilitation, renovation, and management of plantations (Table 7).

Table 7. Activities of Extension and Technology Transfer Executed by FEDECACAO.

		2016			2017			2018		
	Activity	**Number**	**Area**	**Farmers**	**Number**	**Area**	**Farmers**	**Number**	**Area**	**Farmers**
Advisory	Plantation		1,892	1,499		2,248	1,790		2,111	1,666
	Rehabilitation		2,346	1,796		1,367	1,089		1,691	1,195
	Renovation					1,179	992		1,194	1,006
	Maintenance & Management		12,218	5,196		11,624	5,107		11,489	5,368
	Total advisory activities		**16,456**	**8,491**		**16,418**	**8,978**		**16,485**	**9,235**
Training and Visit	Individual visits	25,974			23,780			25,161		
	Follow up visits	1,326			1,142			796		
	Farmers Characterizations			6,570			18,727			5,063
	Courses	15		488	19		613	14		424
	Method Demonstrations	744		9,774	708		8,796	995		11,256
	Method Demonstrations (Mobil Unit)	225								
	Field days	61		4,778	61		4,743	20		1,000
	Field days (regions without TU)	10								
	Technical Tours	18			16		421	3		53
	Trainings on farm							29		745
	Trainings for Technical Units	240			240			240		
	Training and Education Schools	24		628	35		555	22		729
	Demonstration Plots	5								
	Credit applications	1,437			1,604			1,302		

Source: Informes de Gestión 2016, 2017, 2018 Fondo Nacional del Cacao- FEDECACAO.

It also carries out education and extension activities based on training and visit models,

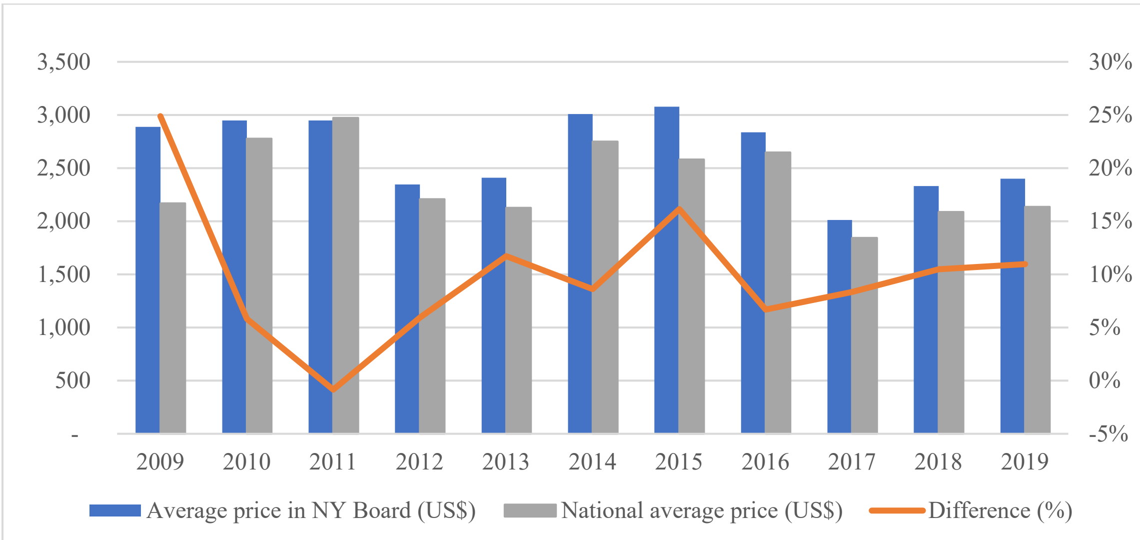
[25] Fondo Nacional del Cacao (FNC) is an account created to collect and administer resources from the cacao levy created by law 67 of 1983. The levy fixed for 3% over the sale value is discounted from farmers' payment for their harvests. Those resources are dedicated to financing programs and projects of research, technology transfer, and commercialization.

besides organizing events and, communication activities to disseminate information about products, services, and knowledge (FEDECACAO & FNC, 2016, 2018, 2019). The detail of activities carried out by FEDECACAO between 2016 and 2018 shows that advisory services and visits remained relatively stable, while training and complementary activities varied across the period. Regarding coverage, individual visits (25,161) reached about 48% of cacao farmers in 2018, although just 3% had a follow-up visit. On the other hand, advisory services served about 18% of farmers, and 58% of them focused on maintenance and management.

Regarding commercialization, cacao is primarily absorbed by the internal market despite increases in export transactions during the last years. National purchases are usually concentrated (between 75 to 77%) in a few processing companies (Compañía Nacional de Chocolates and Casa Luker) which historically offer a broad portfolio of products at local and international markets (Abbott et al., 2018; FNC-FEDECACAO, 2018; García-Cáceres et al., 2014). They purchase most of the cacao through authorized collector centers located in key municipalities; the rest of cacao is demanded by medium and small (about 50) processors offering chocolate products (Arboleda & González, 2010; Zamora & Ochoa, 2020). Although local prices respond to international references, they describe a permanent gap; between 2009-2019, it was 8% on average (Figure 11). Thus, the national market works as an oligopsony where buyers exercise dominant power over prices and quality requirements (Abbott et al., 2018; Arboleda & González, 2010; García-Cáceres et al., 2014).

Farmers cooperatives or associations and wholesalers are also relevant in this scheme of commercialization; they intermediate cacao collection while operating as credit agents or service providers for the farmers, who finally pay with coming harvests (Arboleda & González, 2010; García-Cáceres et al., 2014). Gathering cacao through cooperatives is also a strategy to increase bargains capacity with buyers (Rivera, 2012) since cooperatives act as quality inspectors checking the aroma, color, and moisture conditions established by Colombian Technical Standard NTC1252[26] (García-Cáceres et al., 2014).

Figure 11. National and International Cacao Prices (US$) 2009 -2019



Source: FEDECACAO, ICCO, Banrepública.

On the other hand, MADR (2020) data shows that cacao exports are a growing trend despite the ups and downs year by year (Table 8).

Table 8. Colombian Trade Balance of Cacao Beans. 2014 – 2019

	2014	**2015**	**2016**	**2017**	**2018**	**2019**
Exports (Ton)	8,018	13,744	10,550	11,926	7,056	9,116
% Change	4%	71%	-23%	13%	-41%	29%
FOB Value (US$)	24,352,661	41,740,158	35,516,633	27,949,319	16,779,747	22,783,988
Imports	6,688	5,891	4,643	488	670	402
% Change	189%	-12%	-21%	-89%	37%	-40%
CIF Value (US$)	19,373,343	16,723,926	13,185,699	877,320	1,484,426	1,093,294
Balance (Ton)	1,330	7,853	5,907	11,438	6,386	8,714
Balance (US$)	4,979,318	25,016,232	22,330,934	27,071,999	15,295,321	21,690,694

Source: MADR 2020, using Quintero Hermanos and DIAN.

[26] NTC 1252 classified cacao in three quality categories: premium, bulk and *pasilla*. Premium beans receive 3% overprice.

Between 2015 and 2019, the volume increased annually by 10% on average, with being 2015 the most outstanding year. Among destinations are Mexico, Malaysia, Belgium, the United States, Argentina, the Netherlands, and Indonesia. In contrast, imports have shown significantly lower levels. Between 2015 and 2019, they have fallen at an annual rate of 25%. Traditional providers have been Ecuador and Perú.

The traditional payment method for cacao producers in Colombia is cash (80%) regardless of the volume traded, followed by advance payments provided by cooperatives (Contreras, 2017). In contrast, foreign sales have payment systems for up to three months after export; thus, small farmers and associations cannot participate. In addition, cacao cultivation is isolated from market centers due to the deficit of roads and high transportation costs (Contreras, 2017; Escobar et al., 2020). These costs make an important difference in farmers' income since they increase according to the difficulty to mobilize the harvest (Abbott et al., 2018; Contreras, 2017; Rivera, 2012).

Finally, both national and international commercialization of cacao is increasingly influenced by other factors like certification schemes promoted among farmers to access markets with better prices, the classification of 95% of Colombian cacao as fine flavor by the ICCO, and the development of the *bean-to-bar* or *craft* trend as a luxury niche of the market.

Good Agricultural Practices (GAP), Rainforest Alliance, Fair Trade, organic production, Kosher, and UTZ stand out in terms of certifications. They entail that cacao beans are produced meeting certain criteria (origin, quality, fair prices, etc.) certified by a third party (Abbott et al., 2018; Contreras, 2017; Rivera, 2012). On the other side, a buyer is willing to pay a premium for them, estimated at 20 to 40% for organic/ecologic cacao and 15 to 30% for Rainforest certification (Rivera, 2012). Although some experts assert these certified attributes offer income improvements to farmers (Contreras, 2017; Rios et al., 2017), others consider this market niche as narrow, hence cacao ends up commercialized as bulk, losing the compensation to the costs incurred by farmers to obtain certification. Besides, such premium prices' stability over time is uncertain and is influenced by the whole market performance (Abbott et al., 2018; Arboleda & González, 2010). Abbott et al. (2018) found that overall Ecuadorian origin premium barely reached 10US$/ton, significantly far from the expected 100 to 300US$ /ton.

On the other hand, ICCO (2019) classified 95% of Colombian exports as *fine flavor* cacao[27] since 2016. According to this organization, the market distinguishes between fine flavor cacao produced mainly from *Criollo* or *Trinitario* varieties and bulk or ordinary cacao from *Forastero* (Contreras, 2017; ICCO, 2019).

Specialists estimate that cacao commercialized as a *fine flavor* represents 5 to 8% of the 4 million tons traded in the global market annually (Nieburg, 2016; Rios et al., 2017). By 2016, ICCO estimated a market of 12,000 tons of high-end fine cacao (0.3% of the total market), 210,000 tons of fine cacao (5.2%), 600,000 tons of certified bulk cacao (14.9%), and 3.2 million tons of bulk cacao (79.6%); the prices of those categories vary accordingly to quality (Johnson, 2018; Nieburg, 2016); Thus, prices for high-end fine cacao ranged between 5,000 to 10,000US$/ton, fine cacao between 3,700 to 5,000US$/ton, certified bulk cacao between 3,100 and 3,700US$/ton, and bulk cacao between 3,000 and 3,500 US$/ton (Nieburg, 2016). Like in certified cacao, experts consider the fine flavor market a niche promoted as an extremely optimistic alternative, while opportunities there are narrow (Abbott et al., 2018). At least 22 other countries have been included in the same category as Colombia (ICCO, 2019; Rios et al., 2017).

Finally, the *bean-to-bar* movement has been presented as an opportunity for farmers since they could receive the highest prices within the *fine flavor* segment (ACDI/VOCA Colombia,

[27] A combination of criteria is used to assess the quality of fine flavor cacao, among them are fruit, floral, herbal, wood, nut, and caramel notes, as well as rich and balanced chocolate bases.

2015; Stone, 2016). *Bean-to bar* is a luxury niche were processors pay detailed attention to all stages of chocolate production caring from the origin of cacao and post-harvest practices to packing and labeling, assuring to the consumer the quality and exclusiveness of chocolate bars (Rios et al., 2017; Stone, 2016). Besides, chocolatiers incorporate the formula of consumers' experience, culture, stories, and particular contexts of farmers who provide them (Stone, 2016).

The Colombian *bean-to-bar* segment has grown during the last years and has brands positioned (e.g., Origen cacao, Cacao Hunters, Santander, Juanchoconat) locally and abroad. For instance, Cacao Hunters, which has been one of the most active companies in Sierra Nevada is currently exporting bars to nine countries (ACDI/VOCA Colombia, 2015; Triviño, 2018).

Cacao in Alternative Development

In Colombia, the government and development agencies have invested many resources to expand and intensify smallholder cacao production in response to favorable markets and to the need of fostering income-generating options to discourage illicit crops (Abbott et al., 2018; Unidad de Consolidación Territorial, 2015; UNODC, 2018). According to UNODC (2018), cacao area supported by AD programs between 2003 and 2017 was 79,496 hectares, which corresponds to about 45% of national area reported in 2017 (MADR, 2020). Within such a period, area and production growth emerged partially from the USAID development projects (e.g., *ADAM - Áreas de Desarrollo Alternativo Municipal and MIDAS - Más Inversión para el Desarrollo Alternativo*) promoting planting between 2005 – 2011 (Abbott et al., 2018).

ADAM and MIDAS provided farmers free trees for planting, technical assistance, research and institutional support to identify feasible areas for cacao, cacao materials with higher yields, training on production methods and post-harvest practices. They also addressed activities around marketing and supporting farmers' organizations (Abbott et al., 2018; Meza Vargas, 2011).

Critiques to these projects pointed out its focus on planting while neglecting other critical aspects such as fertilization, grafting, and pruning. Likewise, the performance of farmers' organizations was mixed, while some were particularly well-run, most of them served only as channels to disperse funds to farmers (Abbott et al., 2018; Meza Vargas, 2011). On the other side, evaluations of ADAM and MIDAS accounted for difficulties in establishing cacao in areas without tradition, especially in post-conflict regions where training farmers was problematic since cacao had no local relevance. Aditionally, some regions were inappropriate due to lack of water (Abbott et al., 2018). The evaluation also calculated that over a third of the cacao trees planted had died by 2014, affecting production capacity. Besides, programs did not envision the permanence of technical assistance provision and labor availability once interventions finished. Therefore farmers abandoned pruning, weeding, controlling diseases, and post-harvest practices (Abbott et al., 2018).

Lately, besides planting new areas, AD programs also strengthened cacao organizations' financial and commercial capacity. Between 2012-2015 post-harvest protocols for export were implemented, while the government developed a label to identify products derived from illicit crop replacement (UNODC, 2016, 2018).

Finally, UNODC (2018) also reports increasing cacao promotion activities at national and international events like the *Salon du Chocolat* in Paris and *Chocoshow* in Colombia. In those, increasingly farmers linked to AD programs receive acknowledgments for cacao quality and have opportunities to exhibit artisanal chocolates and other products manufactured by them.

Discourses of cacao opportunities in the post-conflict era: Government, private sector and international aid.

More recently, the idea of cacao as an alternative to replacing illicit crops has been invigorated by messages derived from the Peace Agreement signing in late 2016. Both public and private actors and international aid agencies have declared their trust in cacao cultivation as a mechanism to promote rural development and replace coca in post-conflict regions.

In the IV International Seminar *Comercialización y Calidad del Cacao* two days after closing the peace negotiations, former President Santos declared:

> Cacao is a very special product in this historical moment of the country. Cacao has been fundamental in terms of crop substitution, and it will be even more so in the post-conflict period, for example, within the National Comprehensive Program for the Substitution of Illicit Crops that we are going to advance. If cacao is doing well, Colombia's peace is also doing well (Presidencia de la República, 2016b).

Interest in cacao as a strategy for peace is also embedded in multilateral organizations and international agencies agendas. In 2015, the MADR and the United States Undersecretary of State, signed the alliance *Cacao for Peace* to advance licit rural economic growth through cacao (FEDECACAO, 2015a; US Embassy in Colombia, 2016). In the same sense, the UNODC Program Officer in Colombia, expressed during the Seminar *Sabores y Saberes del Cacao, Aromas de Paz* in 2018: “Cacao is the hope for many producers in areas vulnerable to illicit crops. For that reason, it was included within national policies as a substitute crop and as a reconstructor of the social fabric” (Inizan, 2020).

On the other side, Eduard Baquero, president of FEDECACAO, has pointed out the cacao virtues for supporting rural development, on multiple occasions. His intervention in the *Chocoshow 2019* fair summarizes the messages he transmitted as a sectoral leader (Anadolu, 2020). According to the media coverage, Baquero highlights cacao's convenience for environmental sustainability, reforesting and soil-conserving species. Its requirement of other crops and plants functioning as a shade favors the diversity of flora, fauna, and microorganisms (Anadolu, 2020).

Baquero also mentions cacao commercialization advantage since local industry absorbs almost all production and remarks on the increasing number of chocolate new brands. Citing FEDECACAO data, he confirms that a decade ago there were only ten brands of table chocolate from traditional companies, and currently, there are at least 150 which are mostly community initiatives or products with special characteristics. Baquero also emphasizes that Colombian cacao has won important international awards opening market opportunities (Anadolu, 2020).

Finally, Baquero mentions that cacao is not perishable like many agricultural products which, once it is fermented and dried, allows it to tolerate transportation through roads in poor condition in many rural areas (Anadolu, 2020). Besides, cacao has brought peace to areas of violence, mainly because the cultivation of cacao occupies the labor of local communities (Anadolu, 2020).

Statement of the Problem

Although many programs have been deployed to support cacao farmers in accessing technologies (e.g. new cacao varieties, inputs, tools, fermentation equipment, etc), training, and extension services, increasingly questions are emerging about their low effectiveness in enhancing cacao production and farmers livelihoods.

Several stakeholders point out technology adoption gaps as part of the problem of cacao performance. However, there are few efforts to investigate which factors are influencing cacao farmers' decisions regarding the implementation of such technologies. To address this shortcoming, the current research focuses on the analysis around technological and socio-economic factors that influence technology adoption in cacao farms in Colombia. Thus, this research seeks to provide a more comprehensive reading of constraints and challenges that cacao farmers face to incorporate technology and practices adequately in their crops.

Due to the Colombian context, this questioning is especially relevant for regions where farmers face disproportionate poverty and conflict linked to the coca business, and cacao has been proposed as an alternative solution. Thus, in this research I have studied two post-conflict regions that comply with such conditions, Montes de María and Sierra Nevada de Santa Marta.

Specifically, the questions addressed in this work are:

1. What are the socioeconomic and technological factors that influence technology adoption in cacao farms of post-conflict regions in Colombia?

2. How have these factors influenced the technology adoption process in that context?

3. How might those factors constraining technology adoption in cacao farms be counteracted in such regions?

CHAPTER 3: THEORETICAL AND CONCEPTUAL FRAMEWORK

This research focuses on technological and socioeconomic factors influencing technology adoption in cacao farms. The next literature review describes the theoretical and conceptual framework through which the collected data on these specific cases are contrasted and revised.

The concept of innovation and agricultural innovation

For Schumpeter (1961), the production of goods in the economy implied the combination of material and immaterial forces. In the first group are included the classic factors of production: land, labor, and capital, while in the second are located "technical facts" represented by innovations, and the facts that arise from "socio-cultural forces" that also influence the level of economic development (CELAC-FAO, 2017; Montoya, 2004; Sabourin, 2017). Thus, Schumpeter described technological innovation as the fundamental force to move the productive system, foster its constant transformation, and mobilize economic development (Montoya, 2004). Within firms, such a process of "creative destruction" is driven by the force of the pursuit of profit in the market (Schumpeter, 1961) and supported by the application of a new combination of knowledge, resources, equipment, and other factors (Croitoru, 2012; Garrido, Martínez, Rendón, & Granados, 2016; Shah, Gao, & Mittal, 2015).

Later, neo-Schumpeterian authors used evolutionary assumptions to posit that individuals (or firms) do not have homogeneous behavior in innovation's decision processes (Esper, 2011; Howitt, 2009). On the contrary, in an environment with complete uncertainty, they present different behaviors and obtain different outputs of the technological change process (Esper, 2011). Thus, after Schumpeter's innovation definition and the flourishing of neo-Schumpeterian postures, the conceptual framework has been changing according to the transformation of the economy, the emergence of diverse inventions, and new business (Howitt, 2009; Possas, Salles-Filho, & Da Silveira, 1996; Ugochukwu & Phillips, 2018).

Particularly in agricultural production, due to the influence of a varied and complex set of farmers' characteristics and conditions, the innovation is configured, drawing different trajectories and scales (Bragdon & Smith, 2015; Possas et al., 1996). Therefore, it becomes necessary to elaborate on the concept of agricultural innovation for small-holder and family agriculture separately (Bragdon & Smith, 2015).

In smallholder agriculture, the scope of what is considered innovation is much broader than the development or use of new tools and techniques to improve the farm's productivity, profitability, or sustainability (Bragdon & Smith, 2015), farmers are motivated to innovate because of a variety of reasons that go far beyond opportunities to participate in markets (Bragdon & Smith, 2015; Escobal, 2017). In the same way, FAO (2018, p. 1) describes innovation for family farming as an action that can pursue more than one end :

> …the process whereby individuals or organizations bring new or existing products, processes or ways of organization into use for the first time in a specific context in order to increase effectiveness, competitiveness, resilience to shocks or environmental sustainability and thereby contribute to food security and nutrition, economic development or sustainable natural resource management.

FAO (2018) also points out that "innovation in agriculture cuts across all dimensions of the production cycle along the entire value chain," therefore can influence and generate changes in the management of crops, livestock or fishing, on the administration of inputs and resources, or the market access. Thus, agricultural innovation includes social, economic, institutional, organizational, and policy processes that impact the lives of family farmers (FAO, 2018).

Considering all the above, technological adoption could be considered a means or mechanism for innovation barely. While the innovation is a concept with a comprehensive and all-inclusive meaning with numerous applications, technological adoption is addressed to the process where a firm (or an individual) embarks on technical change for improving the production performance or attributes of its outputs. In summary, technological adoption focuses on the technological aspects of a product or service generation rather than technology innovation that is broader and more inclusive, covering the entire firm business model (Shah et al., 2015).

Classification of agricultural innovations (technologies)

The classification of innovations is useful for understanding the forces behind their generation and adoption. Sunding & Zilberman (2001) propose organizing categories related to their utility in the form of mechanical innovations (tractors and combines), biological innovations (seeds), chemical innovations (fertilizers and pesticides), agronomic innovations (management practices), biotechnological innovations, and informational innovations that rely mainly on ICTs.

From another perspective, innovations could be divided in embodied (capital goods or products such as machinery, fertilizers, and seeds) and disembodied (e.g., integrated pest management schemes), likewise could be distinguishable between process and product innovations (Sunding & Zilberman, 2001). Their impacts can also categorize them in groups focused on yield-increasing, cost-reducing, quality-enhancing, risk-reducing, environmental-protection, shelf-life enhancing, or labor-saving, among others (Sunding & Zilberman, 2001).

Besides, innovations could be split between divisible and non-divisible technologies (Feder, Just, & Zilberman, 1985; Feder & Umali, 1993; IFAD, 2016). Divisible technologies are those that can be separated into steps or dosed for their use, without losing their ability to achieve the function or objective for what has been created. Those that can be adopted by parts, but their application must be complete are non-divisible. When the technology is divisible, its degree of use may provide a quantitative measure of the adoption extent; while for non-divisible innovations, the adoption is necessarily dichotomous: adopt the technology entirely or not at all (Dimara & Skuras, 2003).

Finally, from the reflections of Nowak (1987) around the predictors of adoption of agricultural conservation technologies, we also could classify innovations as unprofitable or profitable. This classification reflects the capability of externalizing costs associated with its implementation (i.g. soil conservation practices, organic fertilization, or water harvesting). For Nowak (1987, p. 208) conservation technologies are less likely to be adopted because " The farmer is expected to bear the short-term costs for some ambiguous, long term, productivity goals".

Technology adoption and innovation process in agriculture

The innovation process literature is diverse. It is expressed in two major investigation lines: for one side research on technology generation and on the other, the study of adoption and use of innovation (Sunding & Zilberman, 2001).

Studies on technology generation are generally focused on how the creation stages happen and eventually encompass until its delivery to the users. Sunding & Zilberman (2001) describe it as research that attempts to depict the innovation production motivations and drivers from an offering perspective.

In general, the process of technology generation is depicted as a chain of actions where the initial step relies on the emergence of a concept that establishes the innovation (Universidad Nacional de Quilmes, 1996). The next step is the development, where the result moves from the laboratory to the field to be tested in uncontrolled conditions, then is scaled up, commercialized, and is integrated into the production process (Sunding & Zilberman, 2001). This linear model of generation and use of technology has been criticized by its top-down unidirectionality approach, in contrast to new proposals around systemic strategies like Agricultural Innovation Systems (Garrido et al., 2016; Universidad Nacional de Quilmes, 1996; World Bank, 2006) and others.

While the development of new technology occurs at a particular point in time and in a formal-relationships setting between actors, the awareness and use of technology take place over a more extended time period and involve a complex combination of factors, actors, and interactions (Ugochukwu & Phillips, 2018). With that premise, a growing number of studies that cut across many disciplines have identified several factors, including personal, cultural, social, and economic attributes, and the characteristics of the technology, which influence technology adoption (Ugochukwu & Phillips, 2018).

From the 1950s until now, numerous economists and sociologists have been suggesting different theoretical frameworks to depict the factors influencing not only individual decisions but also the spread of technology or innovation among individuals or firms (Feder et al., 1985; Sunding & Zilberman, 2001). The three major conceptual models proposed are 1) the *innovation-diffusion model* (Ferede, 2006; Rogers, 2003; Straub, 2009; Uaiene, Arndt, & Masters, 2009), 2) the *user perception model* (Davis, 1985; Pierpaoli, Carli, Pignatti, & Canavari, 2013; Straub, 2009; Sunding & Zilberman, 2001) and 3) the *economic and resources constraint model* (Cáceres, Silvetti, Soto, & Rebolledo, 1997; Calatayud, Pan-Montojo, & Pujol, 2002; CIMMYT, 1993; Feder et al., 1985; Ferede, 2006; Kebede, Gunjal, & Coffin, 1990; Sunding & Zilberman, 2001).

Innovation-diffusion approach

Studies of adoption and diffusion behaviors were undertaken initially by rural sociologists during the 1950s and 1960s. Rogers began conducting studies on the diffusion of hybrid corn in Iowa, comparing diffusion rates in different counties. From this and other empirical research, Rogers deployed his work around understanding technology dissemination in the agricultural sector, positioning it as Diffusion of Innovation (DoI) theory (Ferede, 2006; Straub, 2009; Sunding & Zilberman, 2001; Ugochukwu & Phillips, 2018).

Rogers (2003) defines the adoption decision process as an information-seeking and information-processing activity, where an individual is motivated to reduce uncertainty about the advantages and disadvantages of an innovation. For Rogers (2003), such a process occurs in five stages: 1) *Awareness* when the individual realizes the existence of innovation due to his/her personality traits, socioeconomic conditions, or contact with change agents that may influence this behavior. 2) *Persuasion* is when the individual obtains enough knowledge about the innovation's characteristics to make a personal judgment of it, either a favorable or unfavorable. 3) The *decision* is the moment when the individual chooses to adopt or reject an innovation. 4) *Implementation* is when the individual applies the technology. 5) In *Confirmation* the individual reflects on the implementation process and evaluates to continue or discontinue the innovation adoption.

Besides, Rogers (2003) pointed out a set of attributes of the innovation that feed users' perceptions and influence its potential adoption: first, the *relative advantage* is the individuals' perception that the innovation will be better or worse than others available to the same goal. *Compatibility* is the perception that a particular innovation fits properly into an individual's context-setting. *Complexity* refers to the perception of how difficult it is to comprehend and apply the innovation. *Trialability* refers to the individual's accessibility to innovation experimentation; try it out directly or vicarious could facilitate adoption. Finally, *observability* is how available one technology is for being observed in operation by individuals. This influences individuals' decisions from a social dimension (Pannell et al., 2006).

Regarding the diffusion of innovations, the phenomenon is depicted with the S-shaped curve (Straub, 2009; Sunding & Zilberman, 2001; Ugochukwu & Phillips, 2018). This trajectory describes the cumulative distribution of the adoption of innovation in time. The initial period shows a relatively low adoption rate but with a high speed of change. This tendency is followed by a take-off period in which the innovation penetrates to the “big piece” of the potential market with an increasing marginal growth rate until adoption reaches a period of saturation; At that moment, diffusion reaches its peak because marginal diffusion declines (Rogers, 2003). Other scholars conceived diffusion as a process of imitation wherein contacts with other individuals boost technology's spread (Sunding & Zilberman, 2001). The communication process is the basis for the diffusion of innovations; its channels are the means and mechanisms by which information about a particular innovation is passed from individual to individual (Straub, 2009).

Adoption behavior may be described by more than one variable This measure could be a discrete choice, namely whether the innovation is used or not, or by a continuous variable indicating the degree of utilization when it is a divisible innovation (Sunding & Zilberman, 2001).

The user perception model

The most relevant model used to evaluate people’s perceptions of innovations as a predictor of technology use is the Technology Acceptance Model (TAM) proposed by Davis (1985). This model finds support over the Theory of Reasoned Action (TRA) and the Theory of Planned Behavior (TPB) that allow an understanding that individual attitudes are constructed based on perceptions (Straub, 2009; Venkatesh & Davis, 2000).

According to Venkatesh and Davis (2000), numerous empirical studies have consistently applied TAM to explain about 40% of the variance in usage intentions and behavior. TAM theorizes that an individual’s intention to use a system is influenced by his/her perception of (1) *usefulness* and (2) *ease of use* (free of effort) of that system (Davis, 1985; Pierpaoli et al., 2013; Straub, 2009). Besides, TAM postulates that the effects of external variables (e.g., system characteristics, development process, training) on the intention to use are mediated by those perceptions (Straub, 2009; Venkatesh & Davis, 2000).

Although this approach is frequently applied to understand the adoption of information and communication technologies (ICT), it has also been extended toward the agricultural sector. Flett et al. (2004) consider TAM adequately explain agricultural technology adoption; in their work, they concluded that New Zealand farmers adopt technology to improve dairy production when they consider it ease of understanding and ease to use. Likewise, Pierpaoli et al. (2013), after reviewing several papers -where researchers used TAM - concluded that this could be a feasible method to do an ex-ante (predictive) evaluation about adoption attitude of technologies for Precision Agriculture (PA). In addition, contrary to Davis, they found that a deficiency in one of the constructs (usefulness and ease of use) is sufficient to negatively affect the potential users' attitude towards adoption.

Finally, extended versions of TAM have also been applied to evaluate technologies in agriculture. Haji, Valizadeh, Rezaei-Moghaddam, & Hayati (2020) analyzed acceptance of drip irrigation by sugar beet growers of Miandoab district, Iran. The results indicated that independent variables of an extended version of TAM could account for 59% of behavioral intention's variance toward its acceptance.

The economic constraint model (threshold models)

The economic constraint model argues that resource endowments' distribution determines the patterns of technology adoption by potential users (Ferede, 2006). Economic factors such as access to land, labor, and capital could significantly affect the decision to adopt new agricultural innovations (CIMMYT, 1993; Feder et al., 1985; Straub, 2009). For Feder et al. (1985) empirical studies suggest many relevant hypotheses linking the adoption of new technologies to critical economic and physical parameters, both at the micro and macro level, in static or dynamic analysis. Among them, scholars highlight farm size - that remains to be one of the factors on which research of technology adoption pay major attention-, credit, information, human capital, labor availability, land tenure, access to complementary inputs, extension service, and off-farm income (Feder et al., 1985; Feder & Umali, 1993; Nowak, 1987; Ugochukwu & Phillips, 2018).

In the same way, Kebede et al. (1990), based on their analyses of Ethiopian agriculture, indicate that adoption of agricultural technologies in developing countries is influenced by a wide range of economic and social factors as well as physical and technical aspects of farming. Therefore, it is vital to understand these factors to ensure the development of appropriate technologies and successful projects. Likewise, they propose that behavior regarding technology use is a decision-making process where inputs considered critical to farm survival are adopted first, followed by inputs that are believed to increase yield. For instance, the intention to use fertilizer depends on the decision made regarding how to prevent crop losses from insects and diseases if the latter is a priority problem. In making decisions about adopting a given technology, farmers are assumed to weigh the consequences of the adoption of innovation against its economic, social, and technical feasibility.

On the other hand, Cáceres et al. (1997) state that farmers are continuously immersed in technological change processes, mainly because they face permanent environmental, economic, social, and political changes occurring both within their agricultural production systems as in the environment that surrounds them. If farmers do not use technological innovations, they could hardly face the changes. For Cáceres et al. (1997) elements affecting this process are:

1) The scarcity of economic resources to adopt external technologies offered by projects.

2) No matter how big the promise of higher future income, farmers are more concerned about their immediate needs.

3) Farmers often receive a variable flow of technical information, in many cases, contradictory, so they get confused and cannot decide on technology adoption.

4) Small farmers take the least possible risk to their financial position. They are particularly reluctant to adopt those technologies that do not conform satisfactorily to their productive logic.

Considering all the above, we can say that agricultural innovation and technology adoption, in particular, are complex processes that justify the application of a comprehensive approach to interpreting them and find ways to make them more effective (CELAC-FAO, 2017; Straub, 2009; Sunding & Zilberman, 2001). Finding the critical factors associated with innovation increases the likelihood of improving conditions for rural communities dedicated to agriculture (CELAC-FAO, 2017; Feder et al., 1985; Ugochukwu & Phillips, 2018), therefore taking advantage of the different elements presented previously could be a strategy to advance in an integrative analysis.

After reviewing the mainstream approaches on this topic, three categories of characteristics that influence the adoption of innovation could be depicted by (Straub, 2009):

1) Individual's or Firm's traits: This category recognizes that individuals and firms have characteristics that shape their perceptions and reactions regarding everything surrounding them and, in this way, influence their decisions. Individuals' traits predispose them to seek out or reject any change. Therefore, they become factors that promote or inhibit the use of technology.

2) Innovations' characteristics: This category recognizes that innovations have features that make them attractive to users. For example, high water demanding technology will be perceived differently by a farmer whose availability and access to the water resource do not represent a

problem than a farmer who has access restrictions. It is convenient to take advantage of Rogers' (2003) and Davis' (1985) approaches regarding how technologies' characteristics get value for the innovation process through the individuals' perceptions. Thus, the attributes of relative advantage and compatibility proposed by Rogers could be related to the perception of usefulness; meanwhile, complexity, trialability, and observability could be associated with the perception of ease to use (TAM). Also, we should recognize that farmers occasionally adapt and develop their own versions of technologies to give them attributes that fit entirely with their needs (Cáceres et al., 1997).

3) Contextual characteristics: This category recognizes that the context or settings where individuals and firms are located are composed of elements whose characteristics could generate favorable or unfavorable conditions for the process of technological innovation. This involves not only economic and social factors but also environmental, political, and others.

In the same way, Alcon, Navarro, De-Miguel, & Balbo (2019) have proposed a revised classification, clustering factors under five categories, adding two to those described above: (1) farmer characteristics, (2) economic factors, (3) farm characteristics, (4) characteristics of the technology, and (5) environmental factors. For Alcon et al. (2019), farm characteristics are those (physical, technical, and locational features) that may affect the adoption of certain technologies. These include soil characteristics, such as slope, composition, or texture. Likewise, an agroclimatic setting has been claimed to influence the adoption. In some cases, crop type preferences have also influenced irrigation adoption. Finally, environmental factors are those not related particularly to the farmers, the farm, or the technology, but may still affect the adoption and diffusion of technologies. Among them are pollution regulations, natural phenomena occurrence, or water price policies. For instance, Alcon et al. (2019) assert that farms with several water sources have a lower perception of water scarcity and are less likely to adopt irrigation systems.

Particularly small-scale agriculture is characterized by its dependence on family labor and the low use of off-farm inputs, as well as by the application of various agricultural management practices, traditional knowledge, and flexible strategies, rather than specific technology package (Bragdon & Smith, 2015; Cáceres et al., 1997; Dominguez & Albaladejo, 1995; Olivier de Sardan, 1988). This selective behavior coincides with the analysis carried out by Chambers (1992), who points out that small farmers rarely adopt technological packages. Instead, they choose what they consider best suited to their socio-productive reality while extension services should take care of adverse effects such flexible implementation (Cáceres et al., 1997; CELAC-FAO, 2017).

Besides, the small farmer lives and works under a variety of socio-economic, political, and environmental conditions; differences in resource endowments and rights, labor relations, and ethnic affiliations contribute to inequality in terms of power relations, the means of subsistence within and between communities (Bragdon & Smith, 2015). Therefore, innovation among small farmers occurs by combining conventional and traditional knowledge and adopting decisions based on cultural preferences, assets, endowments, and local contexts.

Insights on extension, land tenure, market access, and association as drivers of technology adoption

This section deepens on four factors that represent important constraints in Colombian rural areas, and therefore they were proposed as strategies of Peace Agreement implementation to support agricultural communities: 1) land tenure, 2) access to agricultural extension, 3) fostering farmers organizations, and 4) market access (Gobierno Nacional de Colombia et al., 2016).

Research carried out by Schuck, Nganje, & Yantio (2002) in Cameroon evaluates how extension education influences adopting alternative agriculture methods (slash and burn, multiple crops, or mono-cropping). Results showed that although extension can have a significant impact in moving farmers away from slash and burn, and move into sustainable production methods, farmers with lower levels of land ownership are less likely to adopt them. For authors, this suggests land tenure issues limit the effectiveness of extension, reducing slash and burn practices; therefore, extension coupled to land tenure programs could be a solution (Schuck et al., 2002).

On the other side, Uaiene et al. (2009) inquired about determinants of agricultural technology adoption in Mozambique with a sample of 4,000 households; it showed that limited access to extension service and credit resulted in significant constraints technology adoption. Contrary, membership in an association influence positively because of improved information dissemination among participants.

Likewise, Boahene, Snijders, & Folmer (1999) found that credit and hired labor significantly positive impacted the adoption of hybrid cacao in Ghana. Also, education and information provided by extension agents were important in establishing whether a farmer would be an adopter. The study also found that resources obtained by farmers through their social networks (e.g., cooperative labor and information) are essential influencing adoption in a high manner for small-scale than for larger-scale farmers. On the other side, access to land, income, and skills did not significantly affect hybrid adoption (Boahene et al., 1999).

Maffioli, Ubfal, Vaquez-Bare, & Cerdan-Infantes (2013) inquired about the effectiveness of extension services to promote technical change to fruit producers (apples, peaches, and grapes) in Uruguay through a state program (PREDEG). Their findings point to a positive effect of extension over practices and technologies to increase plantation density. They also had positive effects but weaker on the adoption of improved varieties. Therefore, the positive impact on technology adoption partially supports the hypothesis of the effectiveness of extension services to motivate technical change. The authors also draw attention to the barriers that adoption studies face when crops correspond to long-term species, whereby it is not possible to know the full expected performance of the technology in the short term (Maffioli et al., 2013).

Abebaw & Haile (2013) investigates the impact of cooperatives promoted by the Ethiopian government on adopting technologies like fertilizers, improved seeds, and pesticides. Cooperatives exert a critical role delivering services to their members (provision of loans, inputs commercialization and marketing harvests). Results depicted cooperative members as primarily male-headed households with better access to agricultural extension services, off-farm work, and leadership experience. Conclusions indicated cooperative membership has a strong and positive influence on fertilizer adoption. The positive impact is more significant for illiterate households and farmers located farther away from a road. However, cooperative membership has a limited effect on adoption of improved seeds and pesticides (Abebaw & Haile, 2013).

Zeller, Diagne, & Mataya (1998) analyzed the adoption of hybrid maize and tobacco by small-holders in Malawi, focusing on aspects like access to commodity and financial markets, among other actions (e.g. credit and extension). The government undertook significant reforms to deregulate markets for tobacco commercialization.

First, the authors showed discernible patterns in factor endowment between families specialized in local or hybrid maize, and tobacco. Their principal findings regarding adoption stated that rapid adoption of tobacco responded to the market-policy reform and institutional changes oriented to open opportunities for smallholders' participation in commercialization. Also, households with small farm sizes and low risk-bearing ability were able to adopt capital-intensive crops, such as hybrid maize and tobacco, because of policies aimed at credit access, extension, input, and output markets (Zeller et al., 1998).

Finally, Zeller et al. (1998) found that households' transaction costs in accessing the nearest market for agricultural inputs and outputs have a negative influence on the share of the area planted with hybrid maize. Therefore, they conclude that access to agricultural markets and related improvements in rural infrastructure and marketing institutions are essential for the adoption of new technology in smallholder agriculture.

Kassie, Jaleta, Shiferaw, Mmbando, & Mekuria (2013) evaluated the diffusion and adoption of sustainable agricultural practices (SAPs) to tackle soil fertility depletion in Tanzanian smallholder farmers. Within SAPs were included conservation tillage, soil and water conservation, legume intercropping, crop rotation, chemical fertilizer, manure, and improved seeds. Their analysis reveals that, among other factors, effectiveness in the provision of extension services, land tenure status, social capital, plot location and size, and farmers organization into associations all influence farmer decision of investment in SAPs. Particularly the work underscores the significance of social capital in the form of membership in rural organizations and the number of salesmen that farmers know.

Besides, they suggest that security in land tenure fosters adoption because farmers associate that with enjoying benefits for over a long time (Kassie et al., 2013). The analysis also shows that the probability of a farmer adopting legume intercropping, manure for fertilization, and soil and water conservation, increases with participation in local collective action institutions. Similarly, the adoption of improved seeds is likely to increase with improved market integration by using the traders that farmers know in their vicinity. Finally, Kassie et al. (2013), indicates that opportunity cost of labor have significant implications on farmer decision to adopt.

The literature reviewed allows us to posit some ideas about socioeconomic factors of interest. Concerning extension services, as was suggested by Feder et al. (1985), they positively influence the adoption of technologies. However, it is necessary to consider its practical link with credit and financing services; Thus, the extension seems to be a complementary factor that could hardly promote the adoption of any particular technology by itself. Among the factors that seem to couple better with extension are credit, land tenure and association or cooperative membership.

On the other hand, there is evidence of the challenge of technological adoption analysis in agriculture when applied to long-term productive activities (cacao, fruit trees, palms, rubber, and others) whose adoption evidence is only possible to see partially in a study that does not involve analysis over time.

From the literature, market access can also be a relevant variable for technology adoption analysis in agriculture. Consequently, investigating the distance producers travel and the costs they incur to access both the markets where their crops' marketing occurs and where inputs are acquired is appropriate within these analyses. Like agricultural extension, market access variables are related to others like credit, and transport infrastructure; The latter facilitates interaction between farmers either with the extension agent or with the buyer of the harvest or the seller of inputs.

In the same way, belonging to some organization, either a cooperative or an association, can also positively affect adoption of technology. However, this relationship has as a prerequisite that the organization effectively executes tasks that promote and facilitate farmers' access to the services and products that support innovation process. It is also relevant that the organization serves as a space for interaction or a way to share the costs of technology adoption, investing as a group of members so that the replication of an innovative decision can be motivated.

Finally, land ownership has a positive link with technological adoption, especially when the farmer's investment will only bear fruit in the medium or long term. Titling provides certain degree of security that the investments made by a producer will be profitable for him or herself.

Other socio-economic factors that appear recurrently in the literature as a driver of technology adoption are labor availability when the technology is labor-intensive, level of household off-farm income, and farm size.

Particularly regarding cacao farmers, recent studies in Ghana indicated several socioeconomic factors as positive or negative drivers for technology adoption. Baffoe-Asare, Danquah & Annor-Frempong (2013) examined the adoption of practices to pest and disease control introduced by the government using an index of adoption. In this case, results yielded that the experience, training, age of household head, household size, and social capital influences positively on adoption. In contrast, the age of the farm had a negative effect.

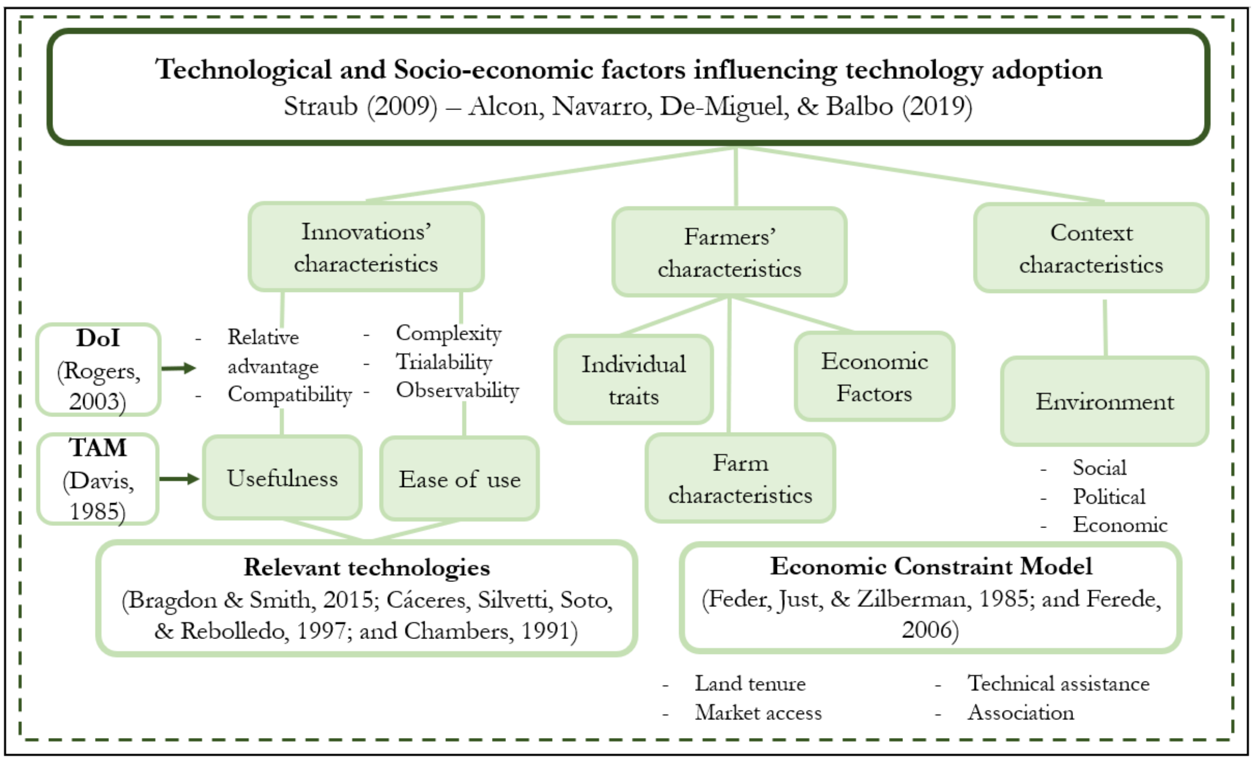
On the other side, Aneani, Anchirinah, Owusu-Ansah & Asamoah (2012) used a multinomial logistic regression approach to identify factors influencing the level of adoption of practices related to controlling of capsids and black pod disease, weed control, planting hybrid cacao varieties, and fertilizer application. Results showed that credit, number of cacao farms owned by the farmer, gender, age of the cacao farm, migration status of the farmer, cacao farm size, and cacao yield significantly affected adoption levels amongst these farmers.

In conclusion, technological innovation cannot be studied without contextualizing it within more holistic socio-economic and technological processes (Bragdon & Smith, 2015; Straub, 2009; Torres & Rodriguez, 2015). It means that studies on this topic need a global explanatory/exploratory framework that recognizes that the multiplicity of factors (farmer characteristics, economic factors, farm characteristics, characteristics of the technology, and environmental factors) build conditions that farmers assess in their technology adoption processes (Alcon et al., 2019; Straub, 2009; Torres & Rodriguez, 2015). Such factors limit or enhance the match between technologies and technical-productive scenarios that individuals or firms face.

Therefore, using complementary approaches (Nowak, 1987; Straub, 2009), understanding the technology adoption decision-process moves toward more accurate scenarios that are more likely to be adequate inputs for policy-making.

To summarize the theoretical and conceptual framework of this research, Figure 12 show the integration of concepts and approaches mentioned above in a more comprehensive framework aimed at understanding socioeconomic and technological factors influencing technology adoption in cacao farms in Colombia.

Figure 12. Theoretical and Conceptual Framework



Using Straub (2009) and Alcon et al. (2019) this research studies technology adoption considering innovations' characteristics, farmers' characteristics (individuals' traits, farm qualities and economic factors) and context characteristics. Within the first group, the attributes of technologies and practices were evaluated according to the concepts contributed by Rogers (2003) and Davis (1985) in the theory of Diffusion of Innovation (DoI) and Technology Acceptance

Model (TAM), looking for technologies farmers perceived usefulness and ease of use. Such technologies and practices were declared as relevant technologies, considering they fit properly to farmers context (Bragdon & Smith, 2015; Cáceres et al., 1997).

On the other side Feder et al. (1985) provide the Economic Constraint Model as proper approach to understand how context and individual characteristics (socioeconomic factors) influence technology adoption. The economic constraint model argues that the distribution of resource endowments determines the patterns of technology adoption by potential users. Factors such access to land, labor and capital could significantly affect the decision to adopt agricultural innovations (Feder et al., 1985; Feder & Umali, 1993).

CHAPTER 4: RESEARCH DESIGN

Research Questions

The research questions that guided this study are:

1. What are the socioeconomic and technological factors that influence technology adoption in cacao farms of post-conflict regions in Colombia?
2. How have these factors influenced the technology adoption process in that context?
3. How might those factors constraining technology adoption in cacao farms be counteracted in such regions?

Methodological Rationality

Considering the theoretical framework and problem background, this research project was set up as a multi-case study with an exploratory sequential mixed-methods design.

According to Yin (2014), a case study could be defined as "an empirical inquiry that investigates a contemporary phenomenon (the 'case') in-depth and within its real-world context, especially when the boundaries between phenomena and context may not be clearly evident." (p. 16). Therefore, a case study inquiry must rely on multiple sources of evidence covered in a triangulation fashion through a logical design with data collection techniques and analysis methods which are pre-described (Yin, 2014). Besides, Yin (2014) explains that case study research is pertinent to answering questions that ask how or why, and where the researcher has little control of events that are happening at that moment. Finally, he also highlights the substantial benefits from the analytical standpoint of having two or more cases in the same research; multiple-case study design offers the possibility of direct replication or, conversely, contrasting situations.

On the other hand, mixed methods provide a complex approach advantageous to having a more comprehensive understanding of research questions since it involves collecting quantitative and qualitative data, integrating the two forms of data in the analysis, and the possibility of using diverse designs to respond to assumptions and theoretical frameworks (Creswell, 2014; Meissner, Creswell, Klassen, Plano, & Smith, 2011; Singleton & Straits, 2010). It also allows the researcher to advance a contextualized approach to phenomena, as well as the ability to triangulate and complement information (Singleton & Straits, 2010). "Social science has embraced triangulation as a research practice that uses more than one method in a study to double-check research results" (Firebaugh, 2008, p. 64).

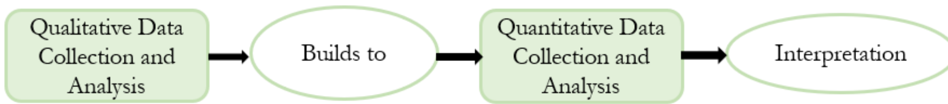
According to Creswell (2014), qualitative methods contribute to reporting multiple perspectives, identifying the factors involved in a situation, and sketching a holistic understanding. Likewise, they allow directly capturing the meaning that participants have about the problem; therefore, research design involving qualitative methods usually requires changes in some phases of the process. Among relevant characteristics for case studies, these methods use purposeful sampling, collection of open-ended data, analysis of text or pictures, and personal interpretation of findings (Creswell, 2014).

Conversely, quantitative methods provide data to numerically describe variables associated with social background, behaviors, trends, attitudes, or opinions and examine the relationships between them in a population by studying a sample of it (Creswell, 2014; Singleton & Straits, 2010).

Considering all the above, Creswell & Plano Clark (2011) indicate that mixed-methods could have different configurations. According to research objectives, questions, and theoretical frameworks, the way qualitative and quantitative methods are designed and developed varies. In the group of alternatives described by them, the *exploratory sequential mixed methods* design arose as the most proper to address this research study's cases (Creswell & Plano Clark, 2011).

An *exploratory sequential mixed methods* design begins by exploring qualitative data and analysis and then uses the findings in a second quantitative phase which builds on the results of the initial findings (Figure 13) (Creswell, 2014). The strategy generally looks to develop better measurements with specific samples of populations and to evaluate if preliminary findings can be generalized to a large sample (Creswell, 2014; Singleton & Straits, 2010).

Figure 13. Exploratory Sequential Mixed Methods Design



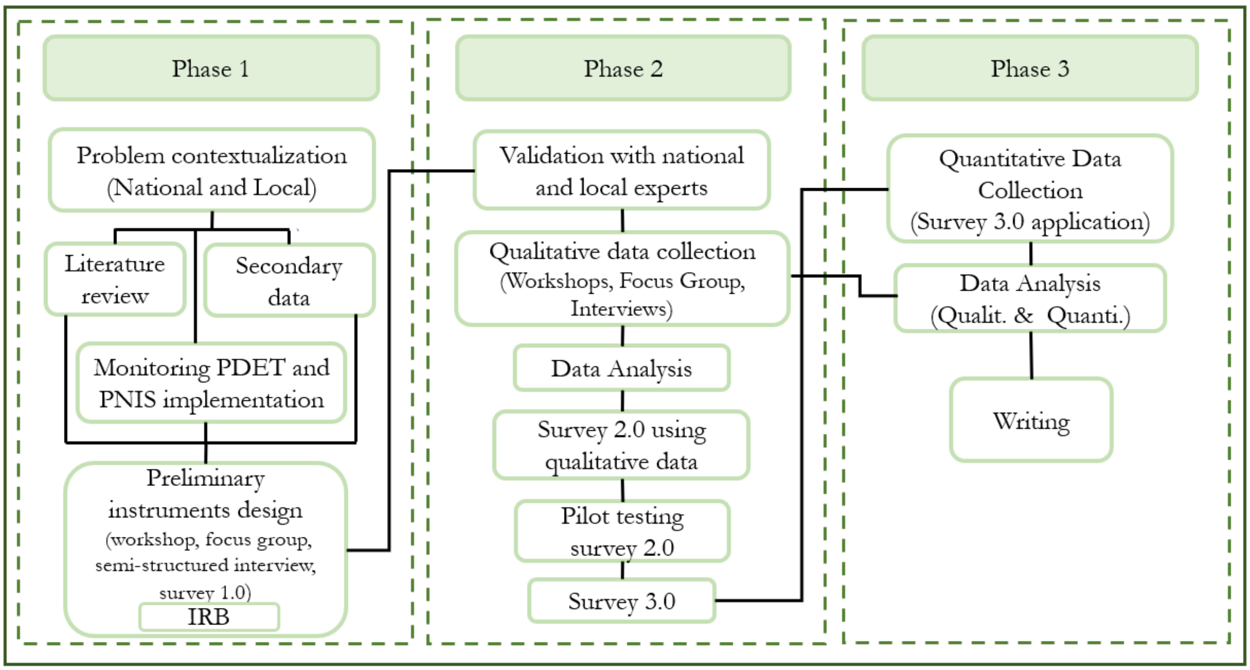
Source: Creswell (2014)

According to Creswell (2014), researchers using this approach employ a three-phase procedure with the first phase as exploratory, the second as instruments development, and the third as applying the instrument to a population sample. In this case, first data analysis enables the researcher to develop instruments with good measurement capacities and comprehensiveness by identifying new variables or types of scales and establish categories of information that will be explored in the quantitative phase (Creswell & Plano Clark, 2011; Singleton & Straits, 2010).

This study addressed the understanding of technological and socioeconomic factors influencing technology adoption in Sierra Nevada de Santa Marta and Montes de María by collecting and analyzing qualitative and quantitative data. The exploratory sequential mixed-methods-design focused its qualitative component on identifying, based on participants' perceptions and experiences, both the pool of technologies and the socioeconomic factors that would later be examined in the quantitative component. This design supported by different methods (workshops, interviews, focus groups, surveys, and secondary data) was organized in three phases (Figure 14). Before proceeding to fieldwork, I completed the Internal Review Board

(IRB) protocol for human subject research, and I received approval from the Office for Research Protections of the Pennsylvania State University (STUDY00009974).

Figure 14. Research Implementation Phases



During the first phase, the research focused on the problem contextualization, both regarding the technological perspective at the sectoral level and from socioeconomic conditions at the regional level. Using secondary data and literature review, I drew up the historical socioeconomic context of the Sierra Nevada de Santa Marta and Montes de María region and a general understanding of the technology adoption problem pointed out previously by stakeholders. In addition, I examined Alternative Development (AD) programs and the evolution of cacao as part of illicit crop substitution strategies, including PDET and PNIS implementation during the first two years after the Peace Agreement signing. The result of this preliminary context review is addressed in Chapters 1 and 2 of this work.

To understand which of those socio-economic and technological factors influence technology adoption in cacao farms of post-conflict regions in Colombia and how they do that, I had to address first the question of what technologies would be analyzed in this work. Usually, technology adoption studies start with a pre-determined technology; however, in this case, technologies and practices of interest were identified during research through a participatory process with farmers, technical assistants, and cacao researchers.

Regarding the socioeconomic factors to evaluate, I arrived at this topic with research interests in factors like land tenure, market access (in terms of commercialization and logistic conditions), access to extension services, and association, due to my previous experience in Agrosavia. However, during the research, they became just a part of a broader pool of factors. Therefore, the analyzed socioeconomic factors were finally those identified in my dialogue with technical assistants, farmers, local experts, policymakers, practitioners, and researchers.

As mentioned above, this research was framed in a mixed-methods approach. The theoretical and conceptual framework selected guided the design of data collection instruments and their implementation. As a result, in the first phase, I designed the methodological guides for the workshops, focus groups, semi-structured interviews, and the first version of the survey to be applied in the fieldwork. In the second phase, I validated these instruments with local experts and colleagues of Agrosavia; then, such improved versions were applied to the samples identified; finally, some qualitative data was processed to be input into the final version of the survey.

In the third phase, the final survey was implemented in the Sierra Nevada de Santa Marta region to collect the quantitative data. Unfortunately, in Montes de María it was suspended due to logistic issues. However, during fieldwork, I obtained a database collected by the Public-Private Cocoa Alliance for Montes de María (APP of Cacao MdM) [28] between 2016 and 2018.

[28] APP of Cacao MdM is integrated by USAID, governments of Bolívar and Sucre, FEDECACAO, Corpoica, Compañía Nacional de Chocolates, Cámara de Comercio de Cartagena, ICA, SENA, CGyP Consultores, Prodesarrollo, Asociación de productores de cacao (APROCASUR), Asociación de productores de cacao de Montes de María (ASPROCAMM), Asociación de productores agroforestales de Santo Domingo de Meza y veredas circundantes (ASOPAGRO SDM) and Asociación de productores agropecuarios de María La Baja, Bolívar (ASPROAGROMAR).

Finally, I advanced the analysis of the second part of qualitative data collected by the interviews and focus groups. Likewise, I undertook the quantitative analysis through descriptive analysis and statistical regression models to examine the relationship between adoption measurement of different technologies and socioeconomic factors identified.

Instrument development

Workshops.

The workshops (one in each region) were selected as the participatory method to co-construct with local farmers and technical assistants the *regional catalogs of technologies* from the base of the individual technical itineraries as identified by Calatayud et al. (2002).

According to Calatayud et al. (2002), the itinerary of technical change or technical itineraries are understood as “the set of particular techniques that the different production units have been choosing over time, in a dynamic context characterized by the permanent transformation of the constellations of available techniques” (p.16). Besides, through their reviewing of multiple works of innovation and technical change in agriculture, they found two common characteristics of the technical change process: first, technological options for each agricultural activity do not behave as a static package but as technological itineraries constructed throughout the years of experimentation and decisions of farmers; and second, technological elections in agricultural activities respond to economic circumstances, institutional framework, technological interrelations, and environmental conditions (Calatayud et al., 2002). This selective and transformative behavior coincides with the analysis carried out by Cáceres et al. (1997) and Chambers (1992); the latter points out that farmers are good at making choices and can experiment with different options, not taking things at face value. Hence, farmers need a “basket of choices” or a “set of principles” rather than fixed technological recommendations or even domains of recommendation (Chambers, 1992; Scoones, Thompson, & Chambers, 2008).

Two workshops were implemented to go beyond the prescribed technological package's perspective for cacao cultivation. These were designed to undertake five activities to identify the technologies to evaluate in this research: 1) Individual description of technical itinerary; in this activity, each participant described his/her own set of technologies and practices applied for cacao cultivation; 2) Group consensus of technical itinerary at municipal level. Participants belonging to each municipality reached a consensus around the common technologies and practices utilized locally. They also discussed the divergences; 3) Group consensus about technical itinerary at the regional level (catalog construction). All participants reached a consensus around technologies and practices utilized in each region; 4) Participatory selection of relevant technologies. From the catalog were participatory selected those technologies considered relevant to the target proposed, in this case, increasing yields and/or improve cacao quality; 5) Ranking of *relevant technologies and practices* according to the attributes or criteria proposed by Diffusion of Innovation theory (DoI) (Rogers, 2003) and the Technology Adoption Model (TAM) (Davis, 1985).

Thus, with the last activity, I separated from the analysis those technologies that, by users' judgment, have insufficient attributes to be considered proper for adoption. Namely, technologies that, even with an auspicious context to adopt them, their relative advantage, complexity, compatibility, trialability, observability, ease to use, and/or usefulness were found to be insufficient and were thus discarded. The technologies were ranked using a matrix to evaluate all those attributes. Technologies at the top were the technologies included in the analysis.

Interviews.

I also designed a semi-structured interview to be administered to the staff of national and international agencies, farmers, technical assistants, processers, associations, and researchers, involved in the cacao sector at the national and regional level (Montes de María and Sierra Nevada). Interviews aimed to inquire about their perspective on socio-economic factors that influence technology adoption in cacao farms. Primarily to confirm or discard the factors *a priori* I proposed as explanatory variables (land tenure, participation in farmers organizations, technical assistance, and market access). Thus, I created a pool of questions to inquire about such topics.

Questions included focused on: 1) goals, actions, and regions covered by the organization including cacao fostering, illicit crops replacement, or rural development interventions in post-conflict regions; 2) socio-economic conditions of communities served; 3) strategies executed for technological improvement in cacao cultivation; 4) factors (e.g., demographic, socioeconomic, technologic) they consider favorable for or constraining productive activities in rural areas; 5) factors influencing technological adoption in cacao farms. 6) opinion about my *a priori* factors; 7) organizational/individual experience tackling or fostering such factors; 8) ideas, suggestions, and recommendations to counteract factors that discourage technology adoption.

Focus Groups.

I also elaborated a methodological guide for focus groups to be applied with participants linked to the regional cacao value chain in each case. Through the guide, I proposed a set of topics of discussion, although emergent topics also were incorporated during the discussion:

- ✓ Main agricultural activities in the region and their historical development. In particular, how cacao production arose in this region. Is it a traditional or introduced crop?
- ✓ Regional or local characteristics that favor agricultural development in general. What are

those favoring cacao production?

- ✓ Regional or local characteristics constraining agricultural development in general, and which are those that limit cacao cultivation?
- ✓ How would they describe the following elements concerning cacao in the region? Which of these elements can be considered factors that influence technology adoption in cacao farms locally? If we had to define the most relevant factors, what would they be?
    - Access to technical assistance or extension service
    - Land tenure
    - Association (farmers' participation in associations, cooperatives or other schemes aimed at collective purchase of inputs, cultivation management, or marketing)
    - Market access (commercialization and logistic conditions)
    - Access to credit
    - Marketing (producer price, sale agreements, local markets, purchase schemes)
    - Labor availability (familiar or waged)
    - Training and technology transfer
    - Farm and cacao plantation sizes
    - Level and sources of family income
    - Demographic characteristics of those who make the decisions on cacao farm.

Survey (Pilot and Final)

As mentioned above, in the first phase, I designed first version questionnaire using the theoretical framework proposed, as well as the preliminary literature review about the problem. During the second phase, it was re-elaborated in real-time, incorporating new ideas I got from observations and conversations during workshops, focus groups, and interviews, as well as using suggestions from local experts. Such incorporations produced a more robust second questionnaire; it was utilized during the pilot testing at the end part of the fieldwork. This experience provided additional information and practical ideas for tuning the final instrument. Ultimately, the questionnaire had a final round of re-design, incorporating results of qualitative data analysis. This was the version implemented in the three municipalities selected (Santa Marta, Ciénaga, and Dibulla) in Sierra Nevada de Santa Marta during the third phase of this research.

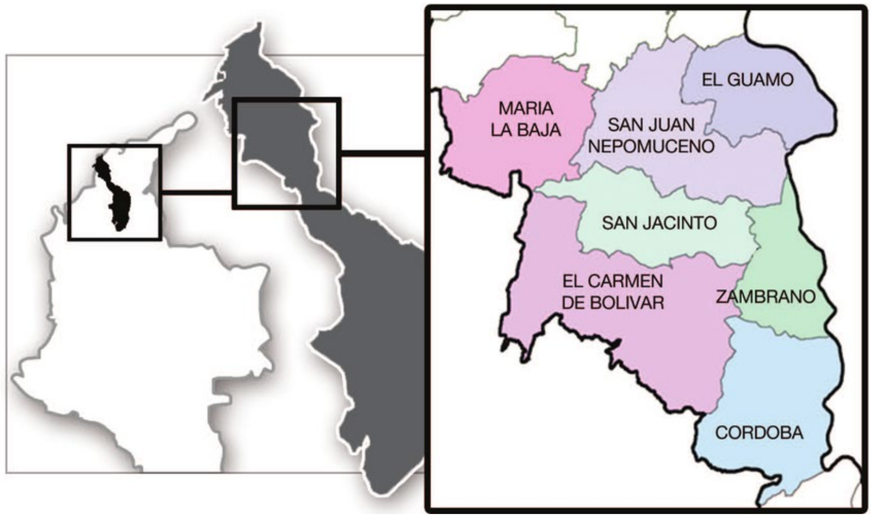
The final questionnaire ended up organized in eight sections: 1) demographic and socioeconomic data of farmer, such as age, gender, education, household composition, and income; 2) farm information, such as location, area, land tenure, water sources, etc.; 3) cultivation characteristics, like plantation age, type of sowing material, complementary crops, yields, etc.; 4) technologies use for detailing tools, practices and protocols, timing, labor, etc.; 5) participation in organizations; 6) access to technical assistance or extension services; 7) market access including data of marketing channels, purchase conditions, transport logistic and costs, etc.; finally, 8) other aspects like credit access, labor availability, and general comments. (See Appendix A).

Setting

This research was held in six municipalities belonging to two regions in the north of Colombia. To select them I considered three elements: 1) municipalities included in post-conflict regions by Decree 893/2017 with cacao areas, 2) municipalities with experience on AD interventions including cacao projects, and 3) municipalities of interest for the Cacao for Peace project. These municipalities did not necessarily correspond to the major areas at a national level.

Thus, on one hand, I selected San Juan Nepomuceno, San Jacinto and El Carmen de Bolivar, pertaining to Montes de Maria region in the department of Bolívar (Figure 15). They are located about an hour and a half or two hours from Cartagena, the capital city of the department.

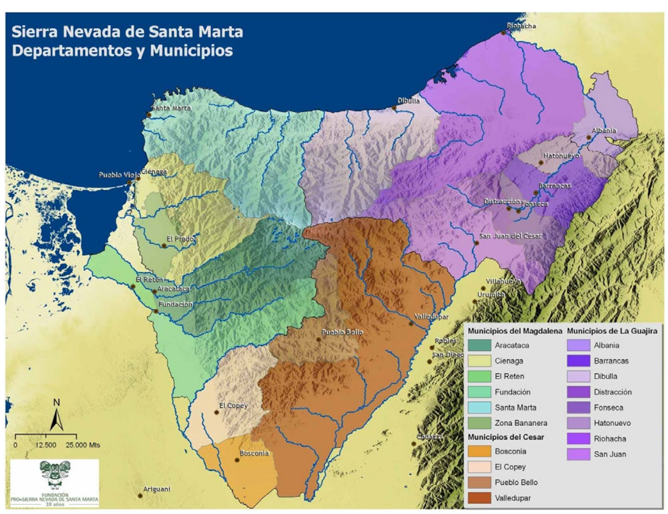
Figure 15. Municipalities of Montes de María Region (Bolívar)



Source: Gobernación de Bolívar through Maza et al., (2012)

Santa Marta, Ciénaga, and Dibulla belong to Sierra Nevada de Santa Marta region (Figure 16), in the departments of Magdalena and La Guajira. Santa Marta is also the capital city of Magdalena. Ciénaga is located about 45 minutes to the southwest of Santa Marta on the Caribbean Highway while Dibulla is about two and a half hours to the east.

Figure 16. Municipalities of Sierra Nevada of Santa Marta



Source: Fundación Prosierra (2018)

Relevant information on both regions is provided in Table 9, in addition to the extensive characterization elaborated on Chapter Two.

Table 9. Characteristics of Montes de María and Sierra Nevada de Santa Marta Región

<table>
<tr><th>Region</th><th>Municipality</th><th>Population
(2018)</th><th>N° of villages
2018</th><th>Cacao Area (Ha)
2017</th><th>Cacao Organizations*
2018</th></tr>
<tr><td rowspan="4">Montes de María</td><td>El Carmen de Bolívar</td><td>Total: 70,131
Rural: 28%
Fem: 44% Male: 56%</td><td>145</td><td>279</td><td rowspan="4">ASPROCAMM
CORINTEGRAL
ASOPRAMM19V
ASOAGRO
RUTA DEL SOL
ASPROAGROMAR
ASOPAGRO - SDM
ASOPROCOAS
ASICHAD
ASOFRUSAN
APROCASUR-MM</td></tr>
<tr><td>San Jacinto</td><td>Total: 23,913
Rural: 13%
Fem: 41% Male: 59%</td><td>39</td><td>441</td></tr>
<tr><td>San Juan Nepomuceno</td><td>Total: 36,874
Rural: 26%
Fem: 47% Male: 53%</td><td>40</td><td>88</td></tr>
<tr><td colspan="3">Total</td><td>807</td></tr>
<tr><td rowspan="4">Sierra Nevada de Santa Marta</td><td>Santa Marta</td><td>Total: 499,192
Rural: 9%
Fem: 49% Male: 51%</td><td>3
1 reserve
4 districts</td><td>1,360</td><td rowspan="4">GUARDABOSQUES DE LA SIERRA
APROCOSNET
ASOARHUACOS
COOAGRONEVADA
ASOCAESMIC
APOMD
CAMPROACTIVO</td></tr>
<tr><td>Ciénaga</td><td>Total: 70,131
Rural: 28%
Fem: 47% Male: 53%</td><td>17
1 reserve</td><td>430</td></tr>
<tr><td>Dibulla</td><td>Total: 37,740
Rural: 87%
Fem: 49% Male :51%</td><td>7
1 reserve</td><td>363</td></tr>
<tr><td colspan="3">Total</td><td>2,153</td></tr>
</table>

Source: DANE (2016, 2019b), MADR (2019), *Fieldwork.

Sampling

This research utilized two sampling procedures: 1) a non-probability sampling to select participants of workshops, focus groups, interviews and pilot studies (Palinkas et al., 2015; Singleton & Straits, 2010), and 2) cluster sampling for the survey (Singleton & Straits, 2010).

According to Palinkas (2015), sampling must be consistent with the aims and assumptions inherent in the use of either method. Qualitative methods are, for the most part, intended to achieve a depth of understanding, while quantitative methods are intended to achieve a breadth of understanding. Each methodology, in turn, has different standards for arranging the number of participants required to achieve its aims (Creswell & Plano Clark, 2011; Palinkas et al., 2015).

For the non-probability sampling, I selected a stratified purposeful sampling for workshops and interviews and convenience sampling for focus groups and the pilot survey. In general, for Palinkas (2015):

> ...purposeful sampling is widely used in qualitative research for the identification and selection of information-rich cases for the most effective use of limited resources. This involves identifying and selecting individuals or groups of individuals that are especially knowledgeable about or experienced with a phenomenon of interest" (p. 2).

Particularly, stratified purposeful sampling helped to capture variations rather than to identify a common core, although it may emerge in the analysis (Creswell & Plano Clark, 2011; Palinkas et al., 2015). In this case, the stratification of non-farmers (technical assistants and researchers) and farmers in a ratio of 1:5, as well as municipal representativeness in the workshops, contributed to identifying the differences and commonalities regarding technological catalogs and ultimately the selection of relevant technologies. Additionally, the stratified selection of participants in the interviews sought a representation of different stakeholders both at the national and local level for the identification of socioeconomic factors to be evaluated (Table 10).

The convenience sampling for reaching the participants of the focus groups and pilot survey was based on local actors linked to the regional value chain that were available and willing to participate, communicating their experiences and opinions in an articulate, expressive, and reflective manner. In both cases, these groups included participants directly involved with agronomic and administrative activities at the farm level (Table 10). Finally, the number of cacao growers to be surveyed was defined under the cluster sampling scheme, with a 95% confidence level and a 5% error.

Table 10. Non-probability Sampling for Qualitative Data

Technique	Region	Sampling technique	N° of Participants	Goal
Workshop	Sierra Nevada	Stratified purposeful	**25** (20 farmers + 5 non-farmers)	Relevant Technologies identification
	Montes de María		**17** (14 farmers + 3 non-farmers)	
Focus Group	Sierra Nevada	Convenience	**6**	Socioeconomic factors identification
	Montes de María		**12**	
Interviews	National level	Stratified purposeful	**13**	
	Local level (Sierra Nevada and Montes de María)		**5**	
Pilot Survey	Sierra Nevada	Convenience	**9**	Validation questionnaire survey
	Montes de María		**7**	
		TOTAL	**94**	

The clusters were the municipalities of each region (Singleton & Straits, 2010). The sample allocation was established in proportion to the number of farmers estimated in each municipality. In the municipalities of Sierra Nevada de Santa Marta, the number of farmers was estimated using data of hectares by 2017 (MADR, 2019) and the average cultivation size in Magdalena and La Guajira calculated by FNC-FEDECACAO (2018). In the final sampling, there were 213 farmers surveyed in the region cluster (Table 11).

In Montes de Maria, the cacao farmers population accounted in the research corresponds to the reported by Prodesarrollo (2017) in the diagnosis of cacao cultivation in the region published by the Land & Rural Development Program (USAID). The sample of the database provided by APP of Cacao MdM was enough to comply with the cluster proportions needed for the analysis.

Finally, the selection of participants was established after consultations with local experts, extension agents, and researchers; this strategy was successful for contacting cacao farmers.

However, access to the indigenous territory in Santa Marta was highly restricted to due to their difficult contact protocols; therefore, the sampling lacks representation of this population[29].

Table 11. Cluster Sampling for Final Survey

Region	Municipality	Estimated number of farms/farmers	%	Surveyed Farmers	%
Sierra Nevada[30]	Dibulla	182	18%	36	17%
	Ciénaga	198	20%	45	21%
	Santa Marta	627	62%	132	62%
	Total	**1,007**	**100%**	**213**	**100%**
Montes de María*	El Carmen de Bolívar	208	51%	203	51%
	San Jacinto	144	35%	144	36%
	San Juan Nepomuceno	54	13%	54	13%
	Total	**406**	**100%**	**401**	**100%**

Source: (FNC-FEDECACAO, 2018; MADR, 2019) *APP of Cacao MdM

Data Collection and Measures

The fieldwork of this research took place in two stages according to the *exploratory sequential mixed methods* design. The first stage was executed during the summer of 2018 when I collected qualitative data in Bogotá, Santa Marta, Ciénaga, Dibulla, San Jacinto, San Juan Nepomuceno, and El Carmen. During this period, I was supported by Agrosavia researchers from regional branches; as well as the staff of UNODC, FEDECACAO and Peace Corps working for the Cacao for Peace project at those municipalities.

Thus, throughout eight weeks, two workshops were accomplished, along with two focus groups, 18 interviews, and the visitation of 16 farms to implement the pilot survey with farmers. The pilot also allowed me to make a first-hand observation of territory, households, and crop conditions. I conducted all activities face-to-face in Spanish.

[29] Participation of indigenous people in cacao production was estimated in 30% by local experts during focus group.
[30] According to FNC-FEDECACAO (2018) the average cacao plantation size in Magdalena and La Guajira was 2.17 and 1.99 hectares respectively. To estimate the number of farms/farmers, it was divided the total municipal planted area by such averages.

Technological Catalogs and Relevant Technologies

Data from workshops were captured in several forms developed as worksheets, which were easy to fill out and guided by straightforward instructions. Technical assistants helped participants who did not read or write. In all cases, forms were identified by numbers instead of names in order to maintain confidentiality. All the forms were recovered at the end of the activity, processed, and analyzed. Memos and notes also were taken by me, capturing comments and discussions among participants.

In that way, workshops concentrated on constructing regional technological catalogs and selecting the relevant technologies to be incorporated later in the final survey. Table 12 and Table 13 summarize regional catalogs for Montes de María and the Sierra Nevada de Santa Marta, corresponding to the pool of technologies and practices that local farmers have chosen over time, rather than representing a recommended manual for cacao cultivation in such regions.

Table 12. Technological Catalog for Cacao Cultivation in Montes de Maria.

STAGE	ACTIVITY	TECHNOLOGY / PRACTICE
Site selection and preparation	Soil preparation	Soil analysis
		Minimum tillage
		Fertilization plan (organic and chemical)
		Amendments
	Plot layout cacao (density)	“3 bolillos” (triangle 3x3x3) in plots with slope ≥10%
		Square 3x3 in plots with slope < 10%
	Plot layout plantain (density)	Triangle 3x3x3 or square 3x3
Shade establishment	Transitory shade sowing	Plantain
	Permanent shade sowing	Avocado and timber (e.g., matarraton, campano, mahogany, guacamayo, cedar, Colombian oak, nogal cafetero)
Associated crops	Complementary crops	Sapote, mango, borojó, lemon, orange, maize, cassava, and yam
Cacao establishment	Seedlings	CCN51, ICS95, ICS60, TSH565, EET8
	Grafting	Approach grafting method
	Grafted seedlings	Rootstock or pattern: IMC67
		bud: ICS95, ICS60, CCN51, ICS39, EET8, TSH565
		Grafting budding method

	Transplanting	Manual digging holes (post hole digger) and transplanting
	Fertilization	Fertilization plan (organic and chemical)
	Pruning	Pruning for shape
Integrated crop management	Pruning	Maintenance or regular pruning
	Monilia or frosty pod rot disease (*moniliophthora roreri*).	Cultural control (diseased pods removal) by weekly
	Witches' broom disease (*moniliophthora perniciosa*)	Cultural control (diseased pods and tissues removal) during pruning
	Black pod disease (*phytophthora sp.)*	Chemical control
		Cultural control (diseased pods removal)
	Squirrels' control (squirrels)	Diversified fruit trees - traps
	Cacao weeds control	Manual control (twice a year)
	Plantain - black sigatoka (*mycosphaerella fijiensis morelet*)	Cultural control (pruning and cut off affected leaves)
	Avocado - phytophtora (*phytophthora cinnamomi*)	cultural control
Water management	Rain collection	Rain harvesting: reservoirs, ponds and jagüeyes
	Irrigation	Gravity irrigation
Harvest	Harvesting	Manual harvesting (using cutting tools)
Post-harvest	Fermenting*	Manual fermenting method (time, tools, and equipment)
	Drying **	Sun drying methods
Marketing		Commercialization through association

* Varied containers (e.g., Wooden boxes, plastic containers, fiber sacks, and buckets)

**Varied infrastructure and containers (e.g., solar dryers, wooden boxes, beds, pallets, or patios)

Table 13. Technological Catalog for Cacao Cultivation in Sierra Nevada de Santa Marta.

STAGE	ACTIVITY	TECHNOLOGY / PRACTICE
Site selection and preparation	Soil preparation	Non tillage
		"Socolado": Manual selective clearing of plot
	Plot layout cacao	"*3 bolillos*" (triangle 3x3x3) in plots with slope ≥10%
		Contour lines planting in plots with slope ≥10%
		Square 3x3 in plots with slope < 10%
	Plot layout plantain	Triangle 3x3x3 or square 3x3
	Fertilization	Organic fertilization
Shade establishment	Transitory shade sowing	Plantain and banana
	Permanent shade sowing	Avocado, timber (e.g., matarraton, campano, mahogany, guacamayo, cedar, wild cashew, Colombian oak, nogal cafetero, and guácimo), and fruit trees (guamo, soursop and guava)
Associated crops	Complementary crops	Sapote, mango, maize, and cassava

Cacao establishment	Seedlings	CCN51, ICS95, ICS60, ICS39, Hybrids, *Criollos*.
	Grafted seedlings	Rootstock or pattern: IMC67
		bud: ICS95, ICS60, CCN51, ICS39, SCC61, Hybrids, Criollos
		Grafting budding method *"injerto de parche"*
	Transplanting	Manual digging holes (post hole digger) and transplanting
	Fertilization	Fertilization plan (organic and chemical)
	Pruning	Pruning for shape
Integrated crop management	Pruning	Maintenance or regular pruning
	Monilia or frosty pod rot disease (*moniliophthora roreri*).	Cultural control (diseased pods removal) by weekly
		Chemical control
	Black pod disease (*phytophthora sp.)*	Chemical control
		Cultural control (diseased pods, branches, roots and tissues removal)
	CERATOSYSTIS (*Ceratocystis fimbriata*)	Cultural control (plant eradication)
	MONALONION *(Monalonion spp.)*	Cultural control
	Plantain - black sigatoka (*mycosphaerella fijiensis morelet*)	Cultural control
		Biological control
		Chemical control
	Avocado - phytophtora (*phytophthora cinnamomi*)	Cultural control (cutting or "*soqueo",* eradication, pruning)
	Sapote - fruit fly (*tephritidae ceratitis capitata)*	Cultural control
	Weed's control	Manual cultural control (twice a year)
		Chemical control
	Fertilization	Fertilization plan (organic and chemical)
	Plant's rehabilitation	Side-grafting technique *"injerto malayo"*
Water management	Irrigation	Gravity irrigation
		Drip irrigation
		Sprinkler irrigation
	Drainage	Drainage (Plain lands)
Harvest	Harvesting *	Manual harvesting (using varied cutting tools)
Post-harvest (1)	Fermenting **	Manual fermenting methods (time, tools, containers, and equipment)
	Drying ***	Sun drying methods
Administrative		Farm Book / Accounting Book

* Varied tools (e.g., Machete, "lamina", scissors)

**Varied containers (e.g., Wooden boxes, plastic containers, fiber sacks, and buckets)

***Varied infrastructure and containers (e.g., solar dryers, trays, beds, pallets, plastic, or patios)

After the catalogs' configuration, participants selected five technologies they considered relevant to increasing productivity and improving cacao quality; then, participants assessed each one using the matrix to evaluate technology attributes (Table 14). Finally, participants ranked them according to the points obtained.

Table 14. Matrix for Evaluation of Technology Attributes

Response Category / Technology: X	Totally Disagree (A)	Disagree (B)	Indifferent (C)	Agree (D)	Totally Agree (E)
Using technology -X- has **relative advantages** (cost-benefit ratio) compared to other technologies that seek the same objective in the cultivation of cacao.	1	2	3	4	5
The technology -X- is **compatible** (fits appropriately) with the needs and conditions of the farms and cacao growers in the region.	1	2	3	4	5
The application of technology -X- is **complex** (difficult to use) for producers who grow cacao in the region.	5	4	3	2	1
It is easy for cacao growers in the region to **try or test** the technology -X- on their farms.	1	2	3	4	5
It is easy **to observe** how other cacao growers in the region use the -X-technology.	1	2	3	4	5
The application of technology -X- is **useful** to achieve the expected results (higher yields, better quality, lower costs, other) in the cultivation of cacao in this region.	1	2	3	4	5
It is **easy** for cacao growers in the region (**accessing**) to the -X- technology	1	2	3	4	5

Through the instrument, participants valued all the attributes of each technology from 1 to 5 according to the scale and criteria provided. By the end, participants calculated the total points they conceded to each technology, the maximum score obtained could be 35 and the minimum 7. Then I calculated the average of their responses to rank the list of relevant technologies in the region. During the workshop, we got in-depth about concepts of relative advantage, compatibility, complexity, trialability, observability, usefulness, and ease of use, in the terms proposed by Rogers (2003) and Davis (1985) described in Table 14.

As a result, Table 15 reports the ranking obtained in each workshop. For Sierra Nevada de Santa Marta (Santa Marta, Ciénaga, and Dibulla), participants agreed that relevant technologies to increase cacao productivity are disease control (specifically Moniliasis cultural control), maintenance pruning, and irrigation, while technologies and practices to improve quality are fermentation practices despite the growing trend towards selling wet cacao *(baba)*. Sowing material and density are considered relevant for both goals.

Table 15. Ranking of Relevant Technologies Selected by Participants (Workshops).

Technology	Sierra Nevada		Montes de María	
	Score	Ranking	Score	Ranking
Moniliasis control (cultural method)	33	1	27	3
Maintenance pruning	31	2	32	1
Post-Harvest practices (Fermentation)	29	3		
Genetic material and density	26	4	27	4
Post-Harvest practices (Fermentation + Drying)			31	2
Irrigation	24	5		
Pest control (Squirrels)			23	5

Conversely, in Montes de María (San Jacinto, San Juan Nepomuceno, and El Carmen de Bolívar), participants considered that relevant technologies to improve productivity are maintenance pruning, diseases control focused on cultural control of Moniliasis, control of squirrels, and adequate sowing material and density. They also agreed on post-harvest practices for quality improvement, especially because cacao in the region is sold entirely dry.

To analyze adoption of the relevant technologies, it was necessary to have a detailed understanding regarding the protocols, instructions, or technical guidelines for the implementation of such technologies. For that, I identified technical manuals and booklets frequently utilized by technical assistance services and cacao projects in Colombia. Some of them refer particularly to the Sierra Nevada de Santa Marta and Montes de María regions. Such information was validated with local experts and researchers. Table 16 summarizes the description of relevant technologies.

Table 16. Description of Recommended Implementation of Relevant Technologies

<u>Moniliasis or Frosty Pod Rot control</u> Moniliasis or Frosty Pod Rot is the most widespread disease in the entire Colombian territory (Jaimes & Aranzazu, 2013). The causative fungus attacks only the fruit (pods) at any stage of its development; the intensity of the attack varies according to climatic conditions (Pinto & Saurith, 2006). Excess shade or competition for nutrition and water also contribute to generating a proper environment to the disease (FEDECACAO, 2015b). In the fieldwork carried out in 2018, the farmers interviewed reported Moniliasis as the main source of plant health problems, with significant decreases in obtaining viable fruits and, therefore, lower productivity. Visited producers also reported they do not carry out control actions in the intensity required, and some do not even recognize the disease in the early stages of development. According to Correa, Castro, & Coy (2014b), various practices and technologies allow for controlling the disease. Among them are cultural practices, chemical management, biological management, and the use of resistant genotypes. All have the objective of eliminating the inoculum of the pathogen; however, so far, they have only been able to reduce the damage (Correa et al., 2014b). The information collected in workshops indicates that the cultural method is the most used in both regions, although, in Sierra Nevada, chemical control was also identified. The latter was rejected in the evaluation of relevant technologies because it is perceived as more expensive. Sixty years after the first recommendations of Instituto Colombiano Agropecuario (ICA) for cultural control of moniliasis, there is a relative technical consensus on the practices that should be implemented (Correa et al., 2014b). They are: 1) Maintain trees less than 3.5 m tall; 2) Carry out maintenance pruning at the beginning of dry periods; 3) Cut the fruits with initial symptoms like oily bumps and spots weekly in months of harvest season (April to October) and every other week for the rest of the year. 4) Diseased fruits must remain on the ground to be covered with weeds or leaf litter (Correa et al., 2014b). Besides, Jaimes & Aranzazu (2013) indicated that disinfection of tools is necessary to avoid spreading diseases

during pruning. FEDECACAO (2015b) recommends monitoring the disease with a weekly frequency in the rainy season and every 15 days in the dry season.
<u>Maintenance pruning</u> In cacao plants obtained by asexual propagation (by grafting), two types of pruning are practiced (Pinto & Saurith, 2006). On the one hand, the *training pruning* that is used to give balance and order to the plant while it begins its productive stage (approx. 36 months); on the other, *maintenance pruning* to preserve the proper shape of the tree (FEDECACAO, 2015b; Pinto & Saurith, 2006). Pruning in clones is less demanding and light than in grafted materials (Pinto & Saurith, 2006). In the fieldwork of this research, participants recognized the relevance of this practice to improve their productivity. According to Jaimes & Aranzázu (2013), the benefits of maintenance pruning are: 1) stimulating growth of vegetative shoots; 2) remove senescent, damaged or diseased parts; 3) manipulate the canopy microclimate; 4) improve flowering; and 5) increase pod production. Thus, this pruning focuses on eliminating dry, diseased, deformed, intercrossed branches, cutting the lateral branches and those too dominant. Also this helps to maintain the main stem and eliminate what is known as "*chupones*" (branches growing from the base of the stem) (FEDECACAO, 2015b; Pinto & Saurith, 2006; Valenzuela, Fernández, Restrepo, & Aristizábal, 2012). Besides all cuts in the tree must be executed using proper tools (e.g., hand and aerial pruning shears, machete and pruning saw) and covered with healing paste or a fungicide to prevent pests and diseases from attacking (FEDECACAO, 2015b). In Colombia, two prunings are carried out each year, both immediately after harvest to prepare the trees for the next production (Jaimes & Aranzazu, 2013).
<u>Post-Harvest practices (Fermentation and Drying)</u> The post-harvest process of cacao includes various activities: the classification, cutting, and shelling of pods, besides fermentation, drying, cleaning, and packing of beans. Pods' collection must be done, ensuring they have the proper maturation degree. Green or overripe pods must be removed since the flavor of their beans can spoil the quality of an entire batch (FEDECACAO, 2015b; Pinto & Saurith, 2006). For small-or-medium-size farmers, it is recommended to collect every two or three weeks to avoid overripe fruits (FEDECACAO, 2015b). The cut must be done with pruning shears; pulling pods may destroy the flower cushion and causes injuries to the trunk of the tree (FEDECACAO, 2015b).

Although some producers store pods, this practice is not recommended when storage conditions are poor (Contreras, 2017). In contrast, manuals recommend stacking the pods by variety in the plot they were harvested and separating diseased or damaged ones from the pile (FEDECACAO, 2015b). Then, open pods (with mallet, machete, or machine), separate the beans from the vein or placenta, and place them in a clean container (FEDECACAO, 2015b; Pinto & Saurith, 2006; Valenzuela et al., 2012). Beans should begin their fermentation within six hours after shelling (Cáritas & CRS, 2017). The vein and the husks must be disposed of in the plot for decomposition and use as organic fertilizer (FEDECACAO, 2015b).

Several systems are used for fermentation. Among them, the pile system, platforms, fermenting beds, rectangular boxes, natural fibers baskets, and rotating drum stand out. Their size, design, and construction material influence cacao quality differently (Contreras, 2017).

Most used systems in Colombia are *rectangular fermenter box* and *boxes of two or three levels* (ladder type), made of wood according to the available species in the area (e.g., cedar, oak, teak, melina, ceiba, caracolí, walnut, guindo.)[31] (Contreras, 2017). Although there is a knowledge gap regarding sizes, shapes, characteristics, and installations for fermentation receptacles, general recommendations are (FEDECACAO, 2015b; Pinto & Saurith, 2006):

- Wooden boxes must have holes, well-distributed, in the bottom and sides to facilitate the exit of liquids draining from the mucilage.
- Boxes should be placed between 10 and 15 centimeters above the ground.
- Boxes' installation sites must be covered and protected from cold drafts and light that alter the stability of the temperature required for fermentation.
- Number of boxes varies according to the size of the farm's harvest.
- The most common dimensions and capacity are:

Dimensions (LxWxD)	Volume (cacao wet / dry)	Source
0.9m x 0.3m x 0.6m	(120 Kg wet)	(Pinto & Saurith, 2006)
0.75m x 0.75m x 0.75 m	(210 Kg wet)	(Pérez & Contreras, 2017)
1.0m x 0.4m x 0.6m	(378 Kg wet - 141 Kg dry)	(FEDECACAO, 2015b)
1.5m x 0.8m x 0.8m	(648 Kg wet – 246 Kg dry)	
2.0m x 0.8 x 0.6m	(756 Kg wet - 288 Kg dry)	

[31] In each case is necessary to verify that the woods do not alter cacao's taste or smell (Pérez & Contreras, 2017).

After shelling, next step is to place the wet cacao mass in the fermenters, where the microorganisms (yeasts and bacteria) and temperatures generated act on the mucilage favoring biochemical and physical changes essential to form the aroma, flavor, and the beans' structure (Contreras, 2017; FEDECACAO, 2015b; Pérez & Contreras, 2017; Pinto & Saurith, 2006).

Although establishing fermentation protocols is difficult due to the multiple variables determining the activity of the microorganisms involved in the process, experts provide a series of general recommendations (Pérez & Contreras, 2017).

According to Fedecacao (2015b), beans must not be turned during the first 36 to 48 hours, a time that covers the anaerobic fermentation phase; in this, microorganisms operate under conditions of low oxygen, high sugar content and medium or high temperature (Contreras, 2017; FEDECACAO, 2015b; Pérez & Contreras, 2017). Then it is necessary to give the first turn to cacao mass; this procedure is repeated every 24 hours to allow the release of CO2 and its replacement with air that guarantees the oxidation process (FEDECACAO, 2015b; Pérez & Contreras, 2017). The main indicator of the fermentation process is temperature; hence it must be monitored every 12 hours (Pérez & Contreras, 2017). The total fermentation period takes from 5 to 6 days (120 to 144 hours) after depositing the cacao mass in the containers. Therefore, beans harvested on different days should never be mixed (FEDECACAO, 2015b).

Depending on the management given during fermentation, the cacao goes to the drying phase with about 65% humidity (Pérez & Contreras, 2017). To store the cacao without food safety problems (e.g., fungi), farmers need to reduce humidity to 7% (FEDECACAO, 2015b; Valenzuela et al., 2012). At this stage, practices aim to dry beans while completing biochemical transformations that began in the fermentation. In this process, organoleptic characteristics of cacao are consolidated (Contreras, 2017; FEDECACAO, 2015b; Pérez & Contreras, 2017).

The most favorable drying conditions are obtained when using systems that take advantage of the sun as an energy source since its results are cheap and adequate (Contreras, 2017; Pinto & Saurith, 2006). The other alternative is implementing artificial drying using mechanical dryers that produce hot air from gas or other energy sources (Contreras, 2017).

For solar drying, structures such as *paseras* (wooden drawers), *elbas*, greenhouses or drying canopies, raised wooden beds, among others, are used (FEDECACAO, 2015b; Pérez &

Contreras, 2017; Pinto & Saurith, 2006). According to Contreras (2017), the most widely-used systems in Colombia are *elbas* and drying canopies. *Elbas* are systems composed by a platform made of wood that moves through rails under a fixed roof; In some cases, it is part of farmer's house (commonly called *casa-elba*). There are also ones with mobile roofs of zinc, which slide over the platform. Contreras (2017) also reports platforms on various levels which move one above the other. Finally, drying canopies are greenhouse-type structures supporting a plastic cover that protects the wooden platforms generating a controlled temperature environment (Contreras, 2017). Although there are contrary opinions, manuals often discourage drying on surfaces like patios, mats, or paved areas due to the high probability of contamination (FEDECACAO, 2015b). In *elba* systems, cacao beans are spread on the platform, forming a 3cm layer that can thicken (up to 10 cm) as the climatic conditions become drier. In the first 24 hours, it is recommended to expose beans directly to the sun only during the first and last hours of the day for a maximum of six hours in total; also, turn them frequently (every 30 to 35 minutes) with a wooden rake to distribute heat evenly (Contreras, 2017; Pérez & Contreras, 2017). On the second day, cacao should be turned every 60 minutes maintaining the layer height. Frequency should go up to one turn every 45 minutes if the sun is extreme. From the third day onwards, the turning should be reduced in frequency by doing it every 2 hours, even if the sun is strong. From the fourth day, monitoring humidity every 12 hours is recommended (Pérez & Contreras, 2017). Manual testing is possible to assess moisture percentage (squeezing beans and recognizing if they crack like gravel) (FEDECACAO, 2015b). The total time spent drying cacao in *elba* systems is a function of the climatic conditions (Contreras, 2017). After drying, cleaning is carried out to classify beans according to size and conditions, and eliminate residues (FEDECACAO, 2015b; Pérez & Contreras, 2017; Pinto & Saurith, 2006). It is also recommended to carry out the *cutting test* to review internal characteristics; by splitting them longitudinally farmers can observe the characteristics judged by industry in accordance with Colombian Technical Standard NTC1252 (FEDECACAO, 2015b).
<u>Genetic material and density</u> Since 2000, Colombia started propagating cacao through grafted plants of clones to overcome the aging of hybrid plantations and the irregular result of the seeds. (FEDECACAO, 2015b). By 2010, the Consejo Nacional del Cacao, made up of the cacao-chocolate value chain

actors, issued Agreement 003, indicating cacao materials recommended to be established in the country (FEDECACAO, 2015b). According to such guidelines, materials for rootstock are: P7, PA 46, PA150, PA 121, EET 400, EET 96, CAU 39, CAU 43 and IMC 67; while the materials to obtain buds to grafting are: ICS 1, ICS 6, ICS 39, ICS 40, ICS 60, ICS 95, TSH 565, TSH 812, EET 8, EET 96, UF 650, SCC 61, FLE 2, FLE 3, FSA 11, FSA 12, FSA 13, FEAR 5, FTA 2, CAU 39, CAU 43, FSV 41 and FEC 2 (Consejo Nacional Cacaotero, 2010).

According to Pinto & Saurith (2006), the genetic materials recommended for the Sierra Nevada by the 2000s were TSH565, CCN51, ICS95, TSH812, EET8, IMC67, F303, TSH792. However, during the pilot survey in 2018 fieldwork, we could establish that genetic materials sowed in the visited farms are IMC67, ICS39, ICS60, ICS95, CCN51, SCC61, and others recognized as criollos. In the case of Montes de María, materials were IMC 67 ICS95, ICS60, CCN51, ICS39, EET8, TSH565

Experts indicate that besides the agroecological conditions of the farm and market trends, election of material for sowing should consider its performance indicators (Jaimes & Aranzazu, 2013; Rojas, 2016). Among them are productivity, tolerance to diseases, and compatibility for pollination (Aranzazu, Martinez, Rincón-Guarin, & Palencia, 2009; FEDECACAO, 2015b; Perea, Martinez, Aranzazu, & Cadena, 2013)

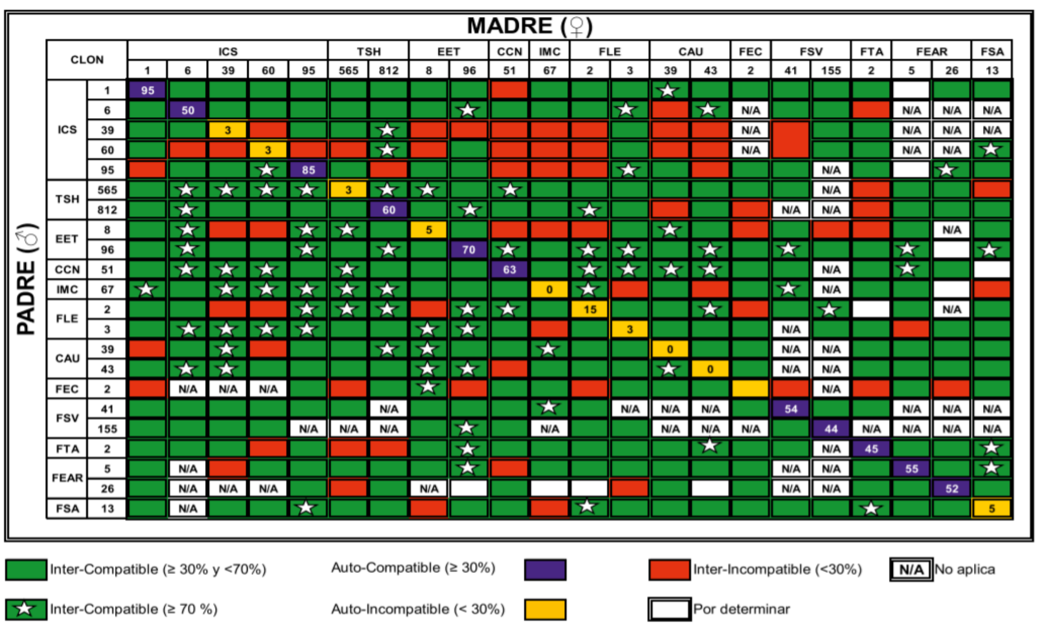
Regarding productivity, experts highlight indicators like yield, pods index, bean index, and number of pods per tree by year (Perea et al., 2013). Yields are measured in Kg / Ha / Year; pods index refers to the number of pods needed to obtain one Kg of dry cacao; the bean index is the average weight of a dry bean in grams, from a sample of 100 beans. The following table summarizes productivity and sanitary indicators for most of the cacao materials in both regions:

Material	Yield (Kg/Ha/Year)	Pods/Tree/year	Pods Index	Bean Index	Reaction to Moniliasis
ICS39	1,589	24	14	2	S
ICS60	1,076	22	15	2.1	S
ICS95	905	22	21	1.3	R
CCN51	1,441	26	15	1.6	MR
IMC67	975	21	20	1.2	MR
SCC61	1,090	21	14	2	S
EET8	1,235	22	14	2.2	S
TSH565	1,212	27	18	1.4	S

S: Susceptible, MR: Moderately resistant, and R: Resistant.
Source: Perea et al. (2013).

Each cultivar can be classified according to the ability to self-pollinate or to give or receive pollen from other material: self-compatible (AC), which means it is capable of pollinating itself; self-incompatible (AI) when it is not capable of pollinating itself; inter-compatible (IC) when it admits pollination by another individual; or inter-incompatible (II) when it does not admit being pollinated by another individual. The knowledge of such capacities is relevant at the moment of defining the quantity and distribution of planting materials in each plot. To guide these decisions, FEDECACAO (2015b) published the Matrix of compatibility of the most-used clones in Colombia.

Finally, the sowing density depends on conditions of each locality. FEDECACAO's (2020) recommendation is from 1,250 to 1,350 trees per hectare in non-shaded plantations or 1,000 to 1,100 trees per hectare when using a shaded layout (e.g., square, triangle, contour lines, etc.) with uniform distances and asymmetric arrangements. However, high-density demands more attention since it increases disease susceptibility (Jaimes & Aranzazu, 2013).



Source: FEDECACAO (2015b)

<u>Irrigation</u>

As mentioned above, cacao is a tropical species that requires 1,500 to 2,500 mm of well-distributed water per year (FEDECACAO, 2015b). Water deficit affects the growth of branches and leaves, which impacts photosynthesis and production (García, 2014). Although crops' ability to tolerate a deficit depends on multiple variables, water management is a crucial element (FEDECACAO, 2015b).

The Sierra Nevada massif functions as a regional climate regulator that generates various climate types. The rains there are described with a unimodal regime (CTC & PNNC, 2020). In general, the northern slope is the wettest with rainfall that can exceed 2,500 mm, the southeast and eastern slopes register rainfall around 1,500 mm in more than 100 days a year, while the western one presents intermediate levels. On the other hand, in the Montes de María, the climate is also influenced by the Alisios winds, which present conditions of dry to a humid climatic regime of two thermal floors and bimodal rainfall regime. Annual precipitation is, on average, 1,500 mm (Aguilera-Díaz, 2013).

According to Climate-Data.org (2020) register[32] summarized in the next table, it is possible to say that although the annual rainfall volume is in the range required, in most of the municipalities (except for Santa Marta and Dibulla), the distribution is not adapted to crop needs, therefore application of water management technologies may be necessary.

		J	**F**	**M**	**A**	**M**	**J**	**J**	**A**	**S**	**O**	**N**	**D**	**Annual**
Ciénaga	Precipitation (mm)	23	20	43	148	310	224	185	228	317	406	300	96	**2,300**
	Rainy days (d)	3	3	5	13	18	16	14	16	18	20	17	10	**153**
Santa Marta	Precipitation (mm)	4	3	6	38	132	99	80	88	148	213	141	25	**977**
	Rainy days (d)	1	1	1	3	8	9	7	9	12	15	10	3	**79**
Dibulla	Precipitation (mm)	44	51	113	223	422	239	175	273	367	442	293	121	**2,763**
	Rainy days (d)	7	7	12	17	21	17	17	20	20	21	18	12	**189**
San Jacinto	Precipitation (mm)	20	26	60	147	222	161	121	161	198	236	162	67	**1,581**
	Rainy days (d)	4	5	9	17	20	16	15	18	19	20	17	10	**170**
San Juan Nepomuceno	Precipitation (mm)	22	27	64	158	251	180	145	190	223	264	183	80	**1,787**
	Rainy days (d)	4	5	9	17	20	17	17	19	20	20	17	10	**175**
El Carmen	Precipitation (mm)	20	26	60	147	222	161	121	161	198	236	162	67	**1,581**
	Rainy days (d)	4	5	9	17	20	16	15	18	19	20	17	10	**170**

Source: Climate-Data.org (2020)

The table shows that from December to March rainfall are scarce in both regions. On the other hand, May and June and September to November show a higher volume and rainy days throughout the month. This coincides with Castellanos et al., (2011) and Prodesarrollo (2017) who indicate that, in Montes de Maria, conditions are favorable for cacao cultivation as long as contingency measures are available for drought periods in November to April. In the

[32] Data is generated using Copernicus Climate Change Service information for 1999- 2019 period.

case of the Sierra Nevada, concerns about the availability of water has grown due to the greater frequency of extreme climatic phenomena in the last decade (CTC & PNNC, 2020). Technologies used to manage water locally are sprinkler and micro-sprinkler irrigation, surface flooding, drip irrigation, or micro hose (FEDECACAO, 2015b). In general, they have a water reservoir (*Jagüey* type), pumping tool or motor, conduction pipe, distribution pipe or hoses, and sprinklers (Castellanos et al., 2011). In sprinkler irrigation systems, water is provided by simulating rain by sprinklers connected to tubes that conduct the water driven by motor pumps or by gravity. The water, in this case, can be applied to the surface of the leaves or sub foliate (FEDECACAO, 2015b). Irrigation by gravity is done on plants' root areas using the flow of water on the surface advancing towards the lower part aided by the force of gravity. In this case, the water is controlled with channels along rows of trees or by flooding the surface in the slope's direction. The water flow must have adequate volume and speed not to cause erosion or exposing the root system of trees. The dikes should be made in the opposite direction of the slope; however, it is not recommended for steep slopes. The efficiency of this method is estimated at 60% (FEDECACAO, 2015b). Irrigation through hoses and micro-hoses arises from drip irrigation to make it cheap and maintain its efficiency. The method consists of mobilizing the water by gravity through a hose network from three to two inches in diameter. From there, the distribution feeds various sections of the crop through half inch hoses. This system is fed by a water tank located in the upper part of the plot (FEDECACAO, 2015b). Regarding irrigation frequency, the weather, the soil, and the intensity of drought must be considered. However, the application in a dry month should be three or four times a month by flood irrigation. In the case of drip irrigation or micro hose, it should be every three days (FEDECACAO, 2015b).
<u>Pest control (Squirrels)</u> Cacao is exposed to attack by squirrels that mainly affect the pods by biting them searching for food. Although the damage to cacao is not considered widespread, experts agree that it results from inadequate crop management. Among the recommended practices to control squirrels' attacks are reducing trees' height by pruning, reduction of shade plants, timely harvest of pods, elimination of overripe fruits, expansion of feeding options for these species

(planting fruit trees), or using chili pepper at the edge of the plantation. (FEDECACAO, 2015b; MADR & FNC-FEDECACAO, 2013; Weaver, Perez, & Ismael, 2009).

At this point is necessary to remember that technologies can be organized in categories according to their utility (Sunding & Zilberman, 2001); In that sense, moniliasis control (cultural method), maintenance pruning, fermentation, drying, and squirrel control respond to the group of agronomic innovations in the form of management practices, although they can be mediated by infrastructure, equipment, tools, and inputs utilization. Therefore, those technologies are divisible, which means they can be separated into steps for their implementation (Feder et al., 1985; Feder & Umali, 1993).

On the other side, irrigation and genetic material are mechanical and biological innovations that are non-divisible, even though sowing density could be evaluated by the extent of following the technical recommendation. When the technology is divisible, its degree of use may provide a quantitative measure of the adoption extent; while for non-divisible innovations, adoption is necessarily dichotomous since individuals have only a discrete choice: either adopt the technology entirely or not at all (Dimara & Skuras, 2003).

In that sense, to evaluate the adoption of relevant technologies in the cacao farms of the regions selected, I proposed to calculate the Degree of Technological Adoption (DTA) based on an adjusted version of the methodology applied by Baffoe-Asare et al. (2013).

Considering all the above, the Degree of Technological Adoption (DTA) of technologies composed by two or more practices is the relationship between the practices applied and the total set of practices that make up each technology[33]. Thus, DTA is expressed with a ratio Y^*, where $0 \leqslant Y^* \leqslant 1$, and operates as the independent variable of statistical regression analysis. In the

[33]Although some practices may have a higher level of importance in the technology implementation and weighting this relevance can help to calculate a more accurate DTA, in this study, it is assumed that all practices have the same level of importance.

case of technologies that are not subject to division, $Y^* = 0$ is assigned to a technology that is not implemented and 1 when it is.

As an example, Table 17 shows the set of practices involved in the moniliasis control according to the technical manuals validated with researchers. In this case, if a farmer accomplishes three of five practices that made up the protocol, he or she has a Yt (DTA) of 0.6.

Table 17. Practices of Frosty Pod (Moniliasis) Control

	Practice	
1	Identification of fruits with disease symptoms or signs (humps or deformations or black spots with spores)	☑
2	Cutting of fruits before sporulation (release of contaminating spores)	☑
3	Disposal of diseased fruits on the ground	☑
4	Covering of diseased fruits with leaf litter	☒
5	Increase monitoring frequency of the disease in rainy and humid seasons	☒

According to the previous considerations, Table 18 summarizes the treatment given to the top three relevant technologies (moniliasis control, maintenance pruning, postharvest) evaluated in this research through the DTA calculation.

Table 18. Outcome Variables Description (DTA)

Technology	**Type**	**Number of practices** [a]	**Variable (DTA)**
Moniliasis control (cultural method)	Divisible	5	Y^*, where $0 \leq Y^* \leq 1$
Maintenance pruning	Divisible	7	Y^*, where $0 \leq Y^* \leq 1$
Post-Harvest (Fermentation)[b]	Divisible	7+5	Y^*, where $0 \leq Y^* \leq 1$
Post-Harvest (Fermentation + Drying)	Divisible	7+5+6	Y^*, where $0 \leq Y^* \leq 1$

a. Number of practices according to the description of technology in the final questionnaire (Appendix A).
b. In post-harvest practices were included 7 practices corresponding to pre-fermentation preparation of cacao mass.

Socioeconomic factors

To examinate the socioeconomic factors influencing technology adoption, I executed 18 interviews both in Bogotá and the two regions of analysis. They covered people involved in cacao cultivation, processing, commercialization, as well as people operating projects in post-conflict regions related to cacao or illicit crops substitution. Such interviews provided diverse standpoints around constraints and factors favoring cacao cultivation, primarily in post-conflict regions. Likewise, those conversations also addressed recommendations and suggestions to manage those factors considered restricting. Thus, interviews resulted in several memos, bibliographical material provided by interviewees, and approximately 13 hours of recording (only 12 interviewees accepted being recorded) that ended up being about 150 pages of transcriptions.

Finally, by focus groups, actors linked to the regional cacao value chain shared opinions, experiences, and perspectives about the socio-economic and technological factors influencing technology adoption in cacao farms. This space was useful for hearing the local perspective and identifying factors overlooked in the interviews. In this activity, data were captured by notes elaborated with colleagues of Agrosavia, and 7.5 hours of audio recording. After the transcription and organization of focus groups and interviews data, the coding process was divided according to research questions in two different moments.

The first moment (during the second phase of the research) focused on the analysis of socioeconomic conditions that might influence technology adoption. The second moment (third phase of research) was oriented to analyzing the recommendations provided by participants to tackle factors that discourage technology adoption.

Besides, the analysis strategy in this part of the research was configurated as two-cycle coding analysis that drew upon elemental and exploratory methods, particularly on descriptive and hypothesis codes (Miles, Huberman, & Saldaña, 2014; Saldaña, 2016); thus, the analysis built upon codes and categories that respond to the groups of contextual and farmers' characteristics included in the theoretical framework, while recognizing potential emergent topics as factors that might explain the technology adoption process in cacao farms.

According to Miles et al. (2014), the first cycle of coding initially focused on summarizing segments of qualitative data into codes expressed as words, phrases or segments, then pattern coding, as a second cycle method, grouped those summaries into categories, themes, or constructs offering explanatory or inferential codes that identify an emergent theme, configuration, or explanation.

Particularly, the descriptive code summarizes data in basic topics providing an inventory of them for indexing and categorizing subsequently, while hypothesis coding refers to applying a researcher-generated predetermined list of codes or topics into qualitative data specifically to assess an hypothesis (Miles et al., 2014; Saldaña, 2016). This method's results are appropriate for the search for rules, causes, and explanations in the data. Likewise, to confirm or disconfirm any assertions, propositions, or theories developed (Miles et al., 2014).

In that sense, a list of codes was constructed using the theoretical framework and preliminary ideas about potential socioeconomic factors influencing technology adoption in cacao farms in Colombia. Such a list of hypothesis codes included: 1) Technical assistance and agricultural extension; 2) Market access, prices, niches, and commercialization conditions; 3) Association, farmers' association, organization, and cooperative; 4) Land tenure, land title, and property rights; 5) Labor availability, labor costs, hired labor, and family labor; 6) Investment capacity, credit access, funding, subsidies, and donations; 7) Farm size, cacao area, assets; 8) Age; 9) Education; and 10) Experience in cacao cultivation.

The coding process ended up with a new list of factors including some of the previous ones and additional new ones which emerged from descriptive coding. Figures 17 and 18 show the results of this analysis. Both figures describe a summary of data segments coded, first cycle codes, pattern codes or topics and the category associated with the theoretical framework.

Figure 17. Results of Two-cycle Coding of Potential Socioeconomic Factors Influencing Technology Adoption (Context Characteristics).

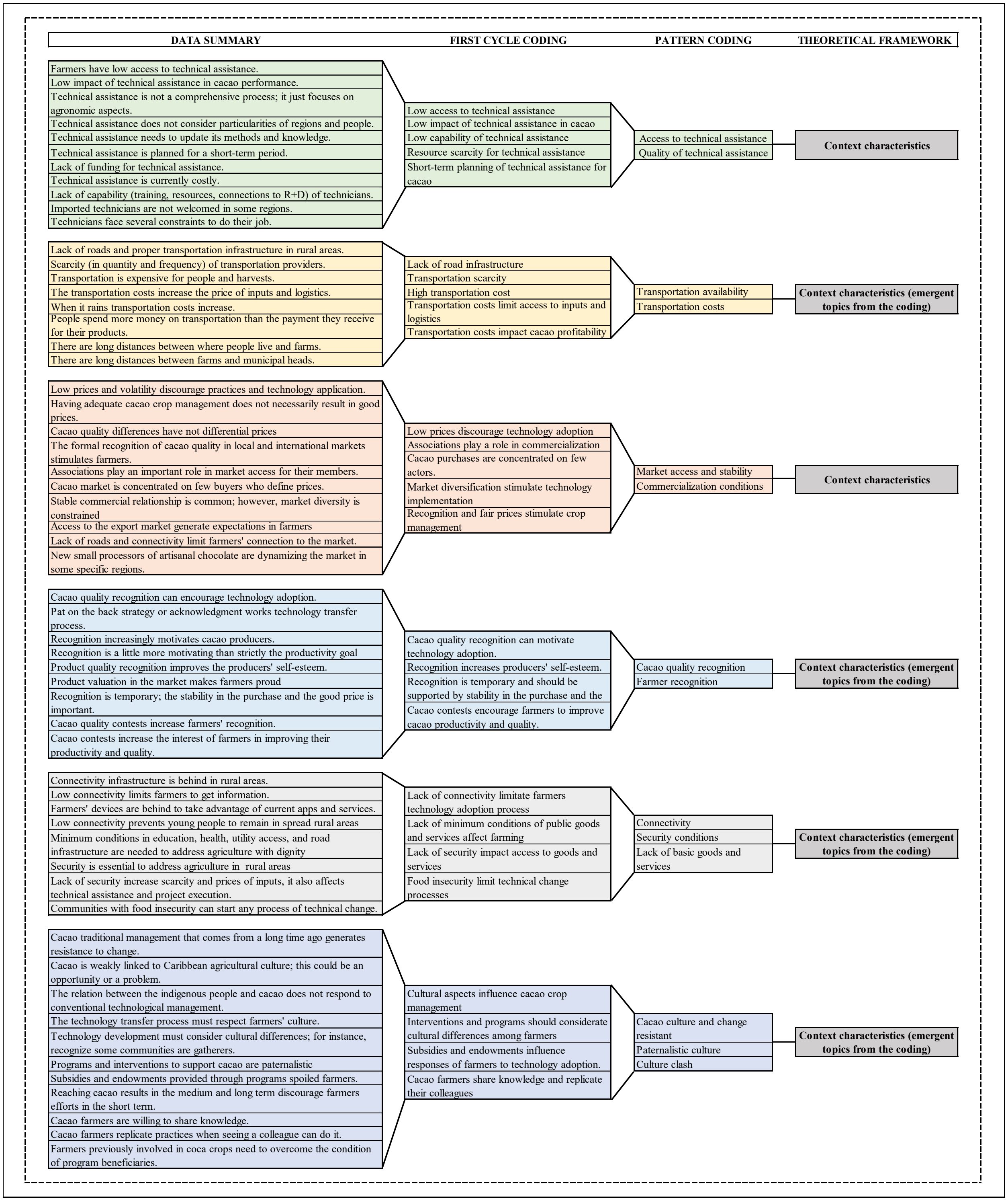
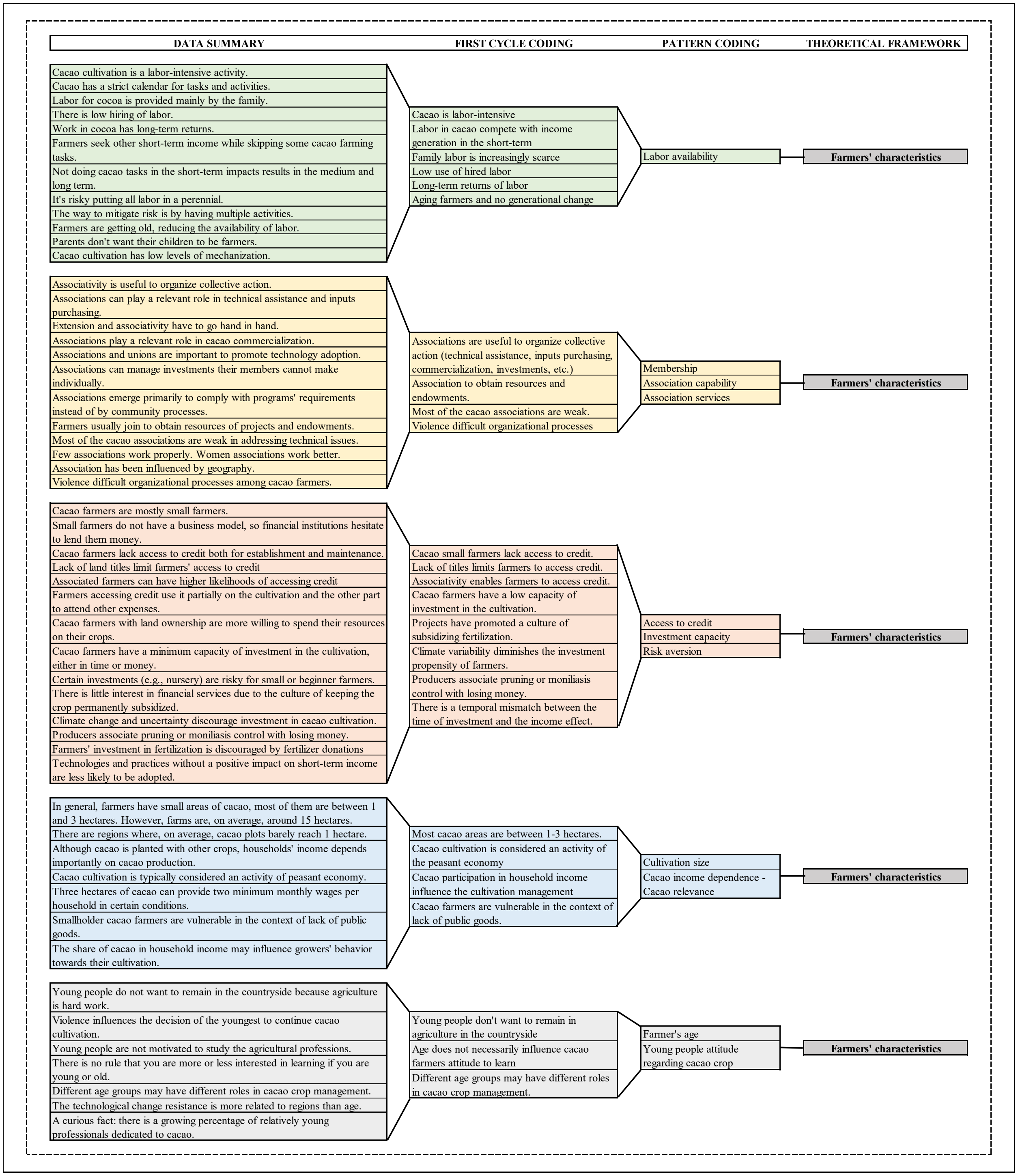


Figure 18. Results of Two-cycle Coding of Potential Socioeconomic Factors Influencing Technology Adoption (Farmer´s Characteristics).



From this group of topics or potential explanatory variables of technology adoption was selected a set to be included in the final survey (Appendix A), Table 19 shows the principal socioeconomic factors were operationalized.

Table 19. Incorporation of Socioeconomic Factors into Final Survey Design

	Factors	Question	Responses
Farmers' characteristics	Labor availability	4.1a This labor (pruning) is carried out with:	[] Own [] Hired [] other
		4.2a This labor (monilia control) is carried out with:	[] Own [] Hired [] other
		4.6a This labor (postharvest) is carried out with:	[] Own [] Hired [] other
		8.3 Is there a shortage of labor on the farm?	[] Yes [] No
	Membership	5.1 Do you belong to some type of farmer organization?	[] Yes [] No
	Association capability	5.6 How would you rate the performance of the organization?	[] Poor [] Fair [] Good [] Excellent
		5.3 Type of Organization	[] Association [] Cooperative [] Foundation [] Other
	Access to credit	8.1 Have you used credit to finance cacao cultivation during the last 5 years?	[] Yes [] No
	Cultivation size	3.1a Total area for cacao plantation	______Hectares (Ha)
	Cacao income dependence	1.12a What are the most important items in the generation of family income?	1.________, 2. ________ 3. ____________
	Farmer's age	1.5 Birth date	DD / MM / AAAA
Context characteristics	Access to technical assistance	6.1 Have you received any technical assistance service for CACAO in the last 5 years?	[] Yes [] No
		6.2 How often did you receive or receive the service?	[] Weekly [] Biweekly [] Monthly [] Quarterly [] Semiannual [] Annual [] Other
	Quality of technical assist.	6.8 How would you rate the service you have received?	[] Bad [] Fair [] Good [] Excellent
	Transportation availability	7.7 Type of transportation for harvest between the farm and the site of commercialization	[] Animal [] Moto [] Car [] Bus [] Truck [] Other
		7.8 How do you consider the transport supply?	[] Sufficient [] Fair [] Insufficient
	Transportation costs	7.6 Cost of transportation between the farm and the site of commercialization ($ / Sack)	($ / Sack)
	Market access and stability	7.1 Marketing mechanism for cacao	[] Directly [] Association or Cooperative [] Middlemen [] Other Which?
	Commercialization conditions	7.3 Information about the last cacao sale	Date Price / Kg Volume (Kgs) [] Wet [] Dry
		7.2 Do you have commercialization agreements? (You can mark more than	[] Volume [] Price [] Quality [] None
	Connectivity	1.11 Do you have cellphone?	[] Yes [] No
		1.11a Type of Device	[] Conventional [] Smartphone
	Lack of basic goods and services	2.14 Access to utilities in the farm	[] Aqueduct [] Sewerage [] Landline [] Electricity [] Internet [] Natural Gas [] Propane

Qualitative data was collected during the second stage of research between July and September of 2019. The survey was applied to 36 farmers in Dibulla, 45 in Ciénaga, and 132 in Santa Marta. In total, 213 farmers answered the questionnaire. For that, I was supported by a field technician that assisted me in conducting the survey in the Sierra Nevada de Santa Marta region during this period. I trained him in the management of the instrument and guided its implementation.

Finally, I elaborated on the descriptive statistical analysis in combination with qualitative data obtained through interviews and focus groups. Besides, I applied simple linear regression models to establish statistical correlation between the Degree of Technological Adoption (DTA) of relevant technologies and the explanatory variables identified in qualitative analysis.

Alternatives to Counteract the Socioeconomic Factors Influencing Technology Adoption in Cacao Farms

The second moment of coding (within the third phase of research) was concentrated on analyzing the experiences, suggestions, and recommendations to tackle factors that discourage technology adoption, as well as in recapping those mechanisms proposed by participants to support factors contributing to it. The analysis of these factors was elaborated through the same logic of two-cycle coding but using the evaluation coding proposed among the affective methods (Miles et al., 2014; Saldaña, 2016). According to Miles et al. (2014), this method applies primarily nonquantitative codes onto qualitative data that assign judgments about the merit, worth, or significance of actions, programs, or policies. Descriptive coding was also used to complement this analysis.

CHAPTER 5: DATA ANALYSIS AND RESULTS[34]

This chapter presents the detailed analysis of qualitative and quantitative data collected with fieldwork and complementing what was presented in Chapter 4.

First, I present the main findings regarding relevant technologies in Sierra Nevada and Montes de Maria, using descriptive analysis of data collected by the survey and the database of APP of Cacao MdM 2018. This analysis is combined with details about technology utilization in cacao farms of such regions from the qualitative data. Then I boost with quantitative analysis the results obtained through the coding process about socioeconomic factors. Thus, I respond to the first question *What are the socio-economic and technological factors that influence technology adoption in cacao farms of post-conflict regions in Colombia?*

I also describe a pool of bivariate linear regression models results aimed at answering the second research question, *How have these factors influenced the technology adoption process in that context?* Although data provided by APP of Cacao MdM were useful to contrast with qualitative information there was insufficient data to include in the regressions, so this analysis is only executed for the Sierra Nevada region. Finally, I briefly resume the qualitative analysis to recap the different proposals and suggestions shared by participants around the third question, *How might those factors constraining technology adoption in cacao farms be counteracted in such regions?*

What are the Socio-economic and Technological Factors that Influence Technology Adoption in Cacao Farms of Post-conflict Regions in Colombia?

[34] All quotes from focus groups and interviews included in this section have been translated from Spanish by me.

To increase the understanding of cacao farmers' decisions around technology adoption in post-conflict regions, I conducted a survey of 213 farmers located in Santa Marta, Ciénaga, and Dibulla, between July and September of 2019. The questionnaire included several questions regarding the use of relevant technologies selected by participants and other technologies of interest (e.g., grafting, soil analysis, GPA protocol, etc.), as well the socioeconomic factors identified as influencers of technology adoption, and demographic data.

I also used data from the APP of Cacao MdM survey executed in 2018 to addressing the quantitative analysis of San Jacinto, San Juan Nepomuceno, and El Carmen de Bolivar in Montes de Maria. This database was structured in a less detailed way than the one I developed in the Sierra Nevada region. Therefore, the calculation of DTAs was possible only for a few technologies.

Sierra Nevada de Santa Marta

Descriptive data analysis shows that 74% of farmers who responded to the survey were men and 26% were women; the average age they had at the moment of data collection was 49.4 years, with an age range between 20 and 83 years old. Analysis indicates that 88% declared being residents of cacao farms, while 12% were living in a different place, either in the municipality or on another farm. Regarding schooling level, 23% of farmers did not have any formal education, 41% had some elementary school, 11% completed elementary school, 12% had some high school, and 11% completed high school. Barely 3% reported a technical degree. Farmers also declared having, on average, 8.6 years of experience planting cacao.

On the other hand, responses about housing conditions showed the low quality of life of these communities: 64% of farmers did not have access to any of the basic utilities (clean water, electricity, sewage, landline, gas, or internet), only 33% had electricity, and barely 3% had clean water service and electricity. Interestingly, about 10% reported having alternative energy in their farms, like solar panels or Pelton turbines for hydraulic energy generation. In addition, only 11% indicated they considered their housing conditions were good or excellent; about 48% thought they were fair, and 32% considered their conditions to be bad or poor; 9% did not answer.

Farms sizes ranged between 0.5 and 166 hectares, with an average of 14.5 hectares since they are concentrated in small properties. Besides, 89% of farmers declared their land tenure condition was certified possession. Regarding cacao area, data shows extensions between 0.5 and 15 hectares, although the average barely reaches 1.97 hectares. This figure is close to the regional average size of cacao cultivation (1.9 hectares in Magdalena and 2.17 in La Guajira) revealed by the FNC-FEDECACAO (2018) survey. On the other hand, sowing density was calculated for each farmer, using both the area and number of cacao trees reported. Thus, the average sowing density in the region was 697 trees per hectare, although 49% of farmers reported having fewer.

Complementary species cultivated included avocado, plantain, fruit trees, maize, cassava, and citrus. More than 56% reported growing one or more of those crops, with 90% reporting avocado and 84% plantain. Yam, beans, and coffee were also declared (between 16% and 26%). Despite this, 71% of farmers indicated cacao as their primary income source. Another 9% said cacao was their second or third; finally, 20% had cacao as a less relevant household income source.

Montes de María

On the other side, the database of Montes de María had few demographic variables available. From 401 farmers included in the sample for San Jacinto, San Juan Nepomuceno, and El Carmen, barely 14% of farmers were women while 86% were men. Although the data do not report farm sizes, the size of areas occupied in agriculture and stubbles was calculated on average as 11.97 hectares per farm. However, they ranged between 1 and 120 hectares.

Regarding the size of cacao cultivations, the area was, on average, at 1.36 hectares which corresponds to 52% of the size reported by FNC-FEDECACAO (2018) for Bolivar. However, cacao cultivation sizes in the three municipalities had a range from 0.13 to 6.75 hectares. For land tenure, 61% of farmers reported they had legal property, 20% were occupants, 13% were land holders, and 5% were possessors[35].

Among complementary crops, plantain, timber, avocado, and yam stand out; They were reported by 98%, 74%, 21%, and 18% of farmers respectively. Finally, cacao was ranked as the principal crop for 85% of farms, and the secondary crop for 6% of farms. In 9% of them, it was less relevant.

Relevant Technologies According to Farmers' Perspective in Sierra Nevada de Santa Marta and Montes de María

As Chapter Four explains, during workshops implemented in 2018, the relevant technologies for cacao production were selected, both in Montes de María and Sierra Nevada. At that point, farmers and technical assistants agreed on technologies recognized as feasible alternatives for local farmers. Participants built *regional catalogs of technologies*. Later, they selected and ranked the relevant technologies for cacao management in each region, using criteria of usefulness and ease of use of technologies, among others.

Table 15 reports the ranking obtained in each workshop. In the case of the Sierra Nevada de Santa Marta participants agreed that relevant technologies to increase cacao productivity were disease control (specifically cultural control of Moniliasis), maintenance pruning, and irrigation, while technologies to improve quality were fermentation practices despite the growing trend

[35] According to the Agricultural and Rural Planning Unit (UPRA), 1) a possessor is one who uses a private property without being legally registered as owner. 2) An occupant is one who lives and operates a vacant property without it having been granted. 3) A holder is the person who, through a verbal or written contract (lease, sharecropping, loan of use, etc.), obtains permission to use a property individually or collectively. (see http://abc.finkeros.com/la-upra-aclara-terminos-sobre-tenencia-de-tierra/).

towards selling wet cacao (“*baba”*). Sowing material and density were considered relevant for both goals.

On the other hand, Montes de María participants considered relevant technologies that improve productivity to be maintenance pruning, disease control (also focused on Moniliasis), control of squirrel damages, and adequate sowing material and density. Finally, they also agreed on post-harvest practices relevant for quality since regionally cacao is entirely sold dry.

Moniliasis Cultural Control.

As mentioned before, to evaluate the degree of adoption of cultural control for Moniliasis, the recommended protocol was divided into five practices (Table 18 based on section 4.2b in Appendix A). Data reported by farmers indicates a Degree of Technology Adoption (DTA) of 0.67 on average for the Sierra Nevada region, which means that farmers usually execute between three and four of the five practices made up of monilia cultural control protocol. By municipality, farmers from Dibulla showed an average DTA of 0.81, followed by those located in Ciénaga with 0.78, and finally, farmers from Santa Marta had 0.59.

The practice less adopted was the increase of monitoring frequency in rainy seasons. This practice was controlled by questions around how often farmers execute moniliasis controls in dry and rainy seasons. Only 21% of farmers declared increasing the frequency of monitoring infected fruits, either passing from one control every other week to weekly as it is recommended, or from monthly to biweekly controls. Local experts confirmed that the last monitoring alternative was also correct for areas with high temperatures in the dry season and less humidity in the wet season.

Data shows that about 21% of farmers in the Sierra Nevada region do not execute moniliasis controls. Of those implementing the control, 90% use family work to accomplish the practices, 8% use hired work, and 2% used both.

APP of Cacao MdM reported answers regarding disease control implementation in general for the Montes de Maria region; barely 12% of surveyed farmers declared executing phytosanitary management in cacao cultivations, despite 49% having declared they had received training for that. At the local level, farmers implementing such practices barely reached the 17% of surveyed farmers in El Carmen, 13% in San Juan Nepomuceno, and 4% in San Jacinto.

Qualitative data collected through interviews and focus groups indicated that moniliasis control represents the most challenging plant health issue in both regions; even experts calculated that approximately half of the pods in Montes de Maria and Sierra Nevada are spoiled by the disease every season. However, they recognized there are many variables to consider for increasing the implementation of practices, mainly when farmers depend on cacao production for family income. A researcher implementing a project in technology adoption with cacao farmers said:

> ... we noticed that some producers associate pruning for monilia control with losing money. For them, each fruit they discard is a tough decision because that is where the investment goes, so cutting the monilia generates great frustration ... so [they] prune, but not how they should do that ... they analyze and say that throwing away one fruit is losing at least 1,000 pesos, ... there is a rationality behind that.

On the other side local advisors and technical assistants highlighted the importance of having the capability of identifying the disease in its initial stage, one of them working for multilateral organization in the Sierra Nevada indicated:

> Having the scissors is not enough for them... every program provided scissors but many of them (the programs) did not invest sufficiently in technical assistance, and the farmer does not have someone who can guide them… having the tools does not mean that they can

> recognize the bad pod in a timely manner... besides, they do not clean the tools, so they spread the problem.

Participants also noted the relationship between the moniliasis and status of the genetic material, which also influences the plant performance beyond the moniliasis control implementation, an expert working for 20 years fostering cacao cultivation from NGOs and government programs stated:

> … we cannot ignore, the advanced age of the plants and the type of materials we planted 15 years ago... these make them more susceptible to monilia compared to new materials, so what we have to do is renew plantations with more resistant materials ... there are farms where although frequent control is done, the susceptibility is very high.

Maintenance Pruning.

Maintenance pruning was selected as relevant technology in both regions. This technology was split into seven practices to evaluate its execution (see section 4.1b in Appendix A). Thus, the DTA for the Sierra Nevada region was, on average, 0.71. At the municipal level, DTA varies as in the moniliasis control. For farmers in Dibulla, I calculated an average of 0.81, 0.82 for Ciénaga, and 0.65 for Santa Marta.

According to the survey, about 2% of farmers implement maintenance pruning more than twice a year; 52% do this following the technical recommendation, namely, twice a year; 29% only prune cacao trees once a year; and 16% of farmers do not even execute a maintenance prune annually. This is consistent with the fact that only 80% of farmers declared that they perceive pruning practices as an important or very important activity for cacao management.

Regarding the source of labor dedicated to the maintenance pruning, 92% indicated they use family labor, 6% contract waged labor, and 2% use both sources. The survey also asked them about their capabilities to prune: 75% of surveyed farmers declared having received training about pruning, while 25% have never received any. Consistently, 77% considered themselves able to execute the pruning practices, while 23% do not consider themselves able to do it.

An important part of pruning is the application of sealer which prevents the contamination of exposed parts of plants after cuts. According to experts, pests and diseases take advantage of uncovered parts during pruning to attack the tree; therefore, treatments to protect them are critical (FEDECACAO, 2015b). The question associated with this practice included queries about inputs used for its implementation. Only 8% of farmers declared making these protection applications. Finally, the most common tools used by farmers to prune are those of manual operation like a pruning saw (47%), hand or aerial pruning shears (44%), and a machete (11%).

Regarding Montes de Maria farmers, although 56% of them reported they had received training in pruning techniques, only 27% responded that they execute maintenance pruning on cacao trees. At the municipal level, those implementing it were 38% of farmers surveyed in El Carmen, 20% in San Jacinto, and 13% in San Juan Nepomuceno. The scarce implementation of these practices occurs despite seemingly sufficient tools and equipment available for pruning. According to the survey, 94% of farmers had a machete, 70% had hand shears, and 24% had aerial shears.

During focus groups, interviews, and workshops, participants highlighted the critical implications over the productivity of executing maintenance pruning. Among the relevant characteristics of this labor, experts mentioned the opportunity and proficiency to prune. One of the participants in focus group working for a local NGO in Montes de María said:

> Sometimes they do not do a task in the cacao, and you ask them, why you have not done it, what's going on? Then they respond, mmm ... it's that I've been with the cows, or I had

> to pick another crop or whatever... so the pruning month is passed. So, they no longer prune, or they do it out of time. The calendar in cacao crop is demanding, and this affects productivity.

Another participant working as a technical assistant in the Montes de María region said:

> Cacao pruning is time-consuming labor and requires ability if you want to do it quickly...so if they have to attend to many household needs... they prioritize working on what is going to give them the money they need in the short-term. So, they prune like in patches... when they have the time they go and prune while they survive with the money, they got... then, they stop the labor again and resume it when they can.

Visits to farms during the pilot survey were also revealing regarding the relevance of pruning; one of the interviewed farmers in the Sierra Nevada de Santa Marta said:

> I know that I have to prune the cacao because the plant needs its space to grow, to birth pods, and to reduce the monilia... so, I try to do it before rains begin, but I never can finish all the plot, I do not have enough time.

Post-harvest (Fermentation and Drying).

As it was detailed before, cacao post-harvest practices are those labors associated with bean fermentation and drying. This research also included the pre-fermentation practices (activities to prepare cacao mass for fermentation), considering their implications in the quality of cacao beans.

Thus, for pre-fermentation and fermentation processes assessment, I examined seven[36] and five[37] practices, respectively. Likewise, I evaluated the drying process on six practices (see sections 4.6b, 4.6c, and 4.6d in Appendix A). In that sense, the survey asks for the execution of 12 different practices for pre-fermentation and fermentation processes. According to data collected in

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[36] For the final analysis calculations, I discarded data of the first practice (a) described in the section 4.6b of the survey.
[37] For the final analysis calculations, I discarded data of the first practice (g) described in the section 4.6c of the survey.

the Sierra Nevada region, the DTA for this set of practices was, on average, 0.85. At the municipal level, it was 0.82 in Santa Marta, 0.86 in Ciénaga, and 0.91 in Dibulla (For separate calculations for each process see Table 25).

Three practices stood out as the less-commonly executed. Number one was the temperature measurement of beans in the fermentation container. That was followed by classification of fruits by variety, and the third was cleaning and disinfection of harvest receptacles. Not classifying and separating pods before putting the beans into cacao mass is likely to generate quality problems. According to experts, this industry requires certain materials (e.g., CCN51 or criollos) be processed independently. Likewise, harvest containers with a scarce cleaning process could also affect cacao masses.

Regarding the drying process, the DTA calculation was, on average, 0.85 for the region. Locally, the farmers of Santa Marta obtained, on average, a DTA of 0.83; those from Ciénaga obtained 0.88, while in Dibulla, DTA obtained 0.90. In this case, the practice least executed was cleaning impurities, or moldy, broken, and void beans before packing to commercialize.

The survey also provides detailed information about the conditions in which the farmers executed such practices. The regularity of pod harvesting triggers the frequency of fermentation and drying practices. According to farmer responses, fruit harvesting is completed primarily biweekly (59% of farmers) and monthly (25%) in low production seasons; however, about 16% of farmers collected pods weekly. The frequency increases partially in high production seasons; from those doing harvesting monthly in low production season, 59% increase the frequency to biweekly or weekly, while 41% do not change their frequency despite the negative implications of overripe pods for plant health and cacao quality. On the other side, from those doing harvesting biweekly, 85% maintain such regularity, and 11% increase the frequency toward weekly harvesting.

Regarding tools and infrastructure, the survey produced relevant results. For farmers surveyed in Sierra Nevada de Santa Marta, about 87% declared utilizing machete or hand shears; other tools are scarcely used. Thus, 22% did not report any type of tool for harvesting which may raise concerns about the caring of flower cushions and their conditions to continue producing.

Likewise, fermentation infrastructure has limitations. Although 71% of farmers declared using wooden fermenter boxes, only 1% were installed with the elevation recommended (between 10 to 15cm). Having wooden boxes without elevation complicates the drainage of liquids from cacao mass, affecting the fermentation process. More importantly, 25% of farmers indicated they use plastic buckets and boxes which, according to experts, are not adequate to address this part of post-harvest management.

Those using wooden fermenter boxes used different sizes and volume capacities; however, the most common dimensions were 1.5m x 1.0m x 1.0m (500 Kg of wet cacao) and 1.0m x 1.0 m x 1.0m (300 Kg of wet cacao). Materials used to manufacture such boxes in Sierra Nevada were primarily wild cashew or caracolí (*Anacardium excelsum*) wood, followed by campano (*Samanea saman*) and nogal cafetero (*Cordia alliodor*a) wood. Finally, although there are contrasting opinions about the utilization of coverages over cacao mass during fermentation (Cacao Móvil, 2020; FEDECACAO, 2015b; Pérez & Contreras, 2017), some technical manuals suggest the use of plantain leaves and other materials. About 85% of farmers indicated they use covers; of them, 58% prefer fiber or "*fique"* sacks, and 41% use plantain leaves.

The drying process is executed using different infrastructures. About 24% of farmers reported using two or more types that they combine according to their needs, weather conditions, and labor availability. The most common is the wooden drawers ("*paseras*") or wood trays; 53% of farmers use them. Another group of farmers (47%) said they dry the beans on patios or canvas mattings ("*lonas*") over the floor, and 7% reported having a solar dryer ("*marquesinas solares*"). The traditional *Casa Elbas* or *Elbas* are really scarce in the region. Finally, 97% of farmers using tools for cacao turning during drying utilize wooden rakes.

Concerns about low infrastructure capacity in Sierra Nevada was expressed by participants of the workshops and focus group. Interviewees declared the need to foster collective mechanisms for fermentation and drying. Three of the comments captured during data collection were as follow:

A women farmer from Santa Marta reflected about that:

> Recently people buying wet cacao is increasing, but we do not know if it is just a trend ("*una moda*") or this will be the new way to sell our cacao... I am very worried because people are excited about it... I don't want us to forget how cacao dries, what happens if tomorrow they do not come to buy the wet cacao?

Another woman farmer from Ciénaga during the workshop indicated:

> ...sometimes one has problems drying cacao... I don't want to put it on the pavement, on the road near to my house...but I do not have any other option, my '*pasera*' is damaged and I have not fixed it because I have no money."

Finally, one member of the board from a farmers' association in the Sierra Nevada stated:

> "We have the hope to have a community center for fermentation, processing, and storage in this side of the Sierra, but we don't have the capital to invest in the adequations this place needs...people can't spend money on that...we have received government aid, but nothing is sufficient..."

Regarding post-harvest practices in Montes de María, APP of Cacao MdM survey reported that only 44% of farmers in the study area utilize hand shears to harvest cacao pods from the trees;

likewise, 52% of them reported being trained in cacao harvesting practices. This situation may raise concerns about the implications of pulling fruits from the plants as it was mentioned above. Likewise, data indicate that only 25% of farmers reported using a proper tool to open the pods, which are primarily “machetes” and “laminas.”

About fermentation processes, farmers in Montes de María indicated they spend, on average, 4.5 days implementing the practices. However, only 23% of farmers reported processing cacao mass in wooden fermenter boxes (elevation conditions are not specified). The other 29% use various plastic containers, and 7% use synthetic or natural fiber sacks. The remaining 41% do not report any type of receptacle or infrastructure to address the cacao fermentation.

Finally, regarding cacao drying, 20% of farmers reported they use wooden drawers (“*paseras”*) or wood trays, 18% use a solar drying canopy *(“marquesinas solares”*), and 3% reported having *Casa Elbas*. 11% of farmers reported other types of mechanisms like patios, mattings, plastics, pallets, tables, and zinc roof tiles, which are less-recommended practices. The remining 47% of farmers did not report infrastructure. About their capabilities to address post-harvest processes, 50% of farmers declared they received training in fermentation practices, 49% in drying techniques, and 18% in cacao beans selection by quality.

Genetic Material and Density.

According to data reported by interviewed farmers in Santa Marta, Ciénaga, and Dibulla, the genetic material utilized in the region is diverse. However, this diversity is not necessarily reflected within their farms. There were 12 materials reported for rootstock, including CAU-37 and IMC 67, which are planted in 48% of farms, followed by P-7, PA-46, and CAU- 43 in 18%, 18%, and 12% respectively. On the other hand, budwoods correspond to 24 genetic materials;

however, 11 are the most common (Table 20). The latter group respond to the Consejo Nacional del Cacao in 2010 and other technical manuals recommendations (Pinto & Saurith, 2006).

Table 20. Genetic Material Planted in Sierra Nevada de Santa Marta.

Rootstock				Budwoods*					
Material	**Farmers**	**%**		**Material**	**Farmers**	**%**	**Material**	**Farmers**	**%**
CAU-43	54	25%		CCN-51	53	25%	ICS-40	9	4%
IMC-67	50	23%		ICS-1	37	17%	ICS-60	8	4%
P-7	38	18%		TSH-812	19	9%	ICS-95	6	3%
P-46	38	18%		ICS-6	15	7%	PA-46	6	3%
CAU-37	26	12%		SCC-61	12	6%	TSH-792	6	3%
Other	7	3%		ICS-39	12	6%	Other	30	14%

* This data corresponds to the principal material farmers have in their plots; 23% of them reported having a second material.

To understand farmers' rationality in selecting genetic materials, I used six criteria to identify the most proper material (agroclimatic conditions required, pods index, beans index, diseases tolerance, compatibility, and market requirements). Each of these criteria was marked as relevant by between 94% and 96% of farmers. However, when the questionnaire asked about their knowledge on the compatibility matrix of genetic materials released by FEDECACAO (Perea et al., 2013), 98% of farmers declared they did not use or know about such a planning tool.

The survey also inquired about the source of budwoods used by farmers and their capability to implement grafting practices. Their responses indicate that 29% of farmers use budwoods from their own plots while 67% do not, and 4% did not respond. Besides, 77% of them claimed they lacked the ability to graft cacao plants, although 66% declared they had received training previously.

Finally, the DTA of sowing density reaches, on average, 0.63. This ratio indicates that farmers of the Sierra Nevada region have, on average, a number of cacao trees per hectare that correspond to 63% of sowing density recommended (1,100 trees per hectare) by FEDECACAO

(2015b; Zamora & Ochoa, 2020)[38]. Individually, calculations of DTA for sowing density showed that farmers in Santa Marta reported fewer cacao trees per hectare (0.58) than Ciénaga (0.65) and Dibulla (0.79).

On the other side, farmers from El Carmen, San Jacinto, and San Juan Nepomuceno in Montes de María reported at least 14 genetic materials, although about 34% of farms did not have grafted trees.

Materials that stand out are ICS-95, ICS-60, CCN-51, TSH-565, ETT-8, and ICS-39. Table 21 shows the most-planted material is ICS-95 which was reported in 61% of sampled farms, followed by ICS-60 and CCN-51. Likewise, it is relevant to highlight that 98% of farms have IMC-67; in farms with grafted trees, IMC-67 works as a rootstock for cultivation. According to the APP of Cacao MdM survey, farms in Montes de Maria depict several genetic material combinations; 24 farms reported having at least five materials, and 163 farms four reported materials combined within their agricultural arrangement.

Table 21. Genetic Material Planted in Montes de Maria.

Budwoods*					
Material	**Farmers**	**%**	**Material**	**Farmers**	**%**
ICS-95	246	61%	ICS-39	58	14%
ICS-60	168	42%	Other	98	24%
CCN-51	175	44%	IMC-67 (Pattern)	396	98%
TSH-565	138	34%	No grafted	136	34%
ETT-8	80	20%			

Source: APP of Cacao MdM

Regarding capabilities, 51% of farmers declared they had received training in grafting techniques; however, only 13% of them applied such practice in their cultivations. Thus, the DTA

[38] The FEDECACAO (2015b) recommended sowing density for cacao is between 1,100 and 1,500 trees per hectare. On the other hand, Zamora & Ochoa (2020) recommend density between 1,000 and 1,100. The selected parameter was 1,100 trees per hectare.

of grafting utilization, calculated with the numbers of grafted trees and the total per farm, reach 0.39. This means that about four of 10 cacao trees in the region are grafted plants.

Finally, the DTA of sowing density in Montes de Maria reach, on average, 0.84 when comparing the number of cacao trees per hectare on each farm with the recommendation of 1,100 trees per hectare provided by FEDECACAO (2015b; Zamora & Ochoa, 2020). Individually, calculations of DTA for sowing density showed that farmers in San Juan Nepomuceno reported fewer cacao trees per hectare (0.76) than San Jacinto (0.87) and El Carmen (0.84).

Qualitative data also provided insights about genetic materials and sowing density. In focus groups and interviews some of the comments about it referred to type and quality of materials currently planted in Colombia. A member of Consejo Nacional del Cacao stated:

> Colombia has a very wide and good set of genetic materials, but sometimes there are fads and myths ... So, there are many possibilities, and it happens that one producer wants the market x, another wants a different one, and so on… but sometimes you have to make country decisions ... so the country has tried to align decisions about materials toward establishing those considered fine flavor types.

From a standpoint of a researcher implementing a project with cacao communities in Montes de Maria:

> ... we found that some producers had patches in their plots with plants that had never produced a pod. The issue is that some materials introduced at the beginning of the 2000s were not sufficiently known, it was not clear the compatibility between them or their performance in the different regions... anyway they were promoted.

Likewise, a representative of a farmers' organization talked about their experience in other post-conflict region:

…we followed planting models from manuals... we started implementing the 3X3 model... and after a few years when the humidity began to affect us, we realized we had a high density for that weather, so, we had to eliminate some trees and the following sowings we did with another distribution; this has been a trial and error ... we do not believe that an arrangement can be standardized, even in the same region.

Irrigation.

As I mentioned above, although the volume of rainfall seems enough to supply cacao water requirements in most of the municipalities analyzed (except in Santa Marta), it is very poorly distributed throughout the year. Therefore, the survey in the Sierra Nevada asked about the utilization of irrigation systems. About 41% of interviewed farmers reported they use irrigation for cacao cultivation; 74% had springs or natural seeps as water source, 25% use streams, and 1% practice rainfall harvesting. Regarding type of irrigation, 100% of farmers reported sprinkler irrigation as the main technology used for water distribution to cacao trees. Locally data show that in Ciénaga 84% of farmers used irrigation, in Santa Marta only 34%, and barely 9% in Dibulla.

Finally, in Montes de María, APP of Cacao MdM registered that 61% of farmers reported having water availability for cacao cultivation; one in three farmers reported having an underground well.

Other Technologies.

Besides the technologies selected by participants, some interviewees also suggested the relevance of knowing farmers' use of soil testing for fertilization decisions, and the implementation of Good Agricultural Practices (GPA) that appear as a strategy of differentiation in cacao market.

Regarding soil testing, data collected for Sierra Nevada showed that about 51% of farmers had done soil testing; however, 89% of them declared it was within a project implementation, so this is not a regular practice among cacao farmers, and only 33% indicated they knew the results. Finally, barely 8% of farmers reported that they implement the GPA norms and are certified.

Socioeconomic Factors that Influence Technology Adoption in Cacao Farms in Post-conflict Regions: Local and National Actors' Standpoint

This section presents results from qualitative and quantitative analysis of socioeconomic factors identified as drivers or constrainers of technology adoption in cacao farms in Colombia.

According to the coding process of interviews and focus groups explained in Chapter 4, the second phase of this investigation ended up with a list of socioeconomic factors that potentially influence technology adoption in both regions (Figures 17 and 18). Although many of these factors emerge from the conversations about the problem of technology adoption in cacao in general, many of the participants emphasized that they could have higher incidence in post-conflict regions.

Those topics were operationalized through a set of variables (Table 19) described in the survey applied to 213 farmers in three municipalities of Sierra Nevada de Santa Marta during 2019. Thus, although the quantitative results focused on the Sierra Nevada region, the qualitative analysis addresses data collected in both regions. Table 22 summarizes the main results from the survey; however, more details are provided in the description of each factor throughout this section.

Labor Availability

According to many participants from interviews and focus groups, cacao is recognized as an activity primarily of peasants or small holders' economies, which implies that it is cultivated in contexts where household income relies significantly on agricultural activities and family on-farm labor. Besides, cacao is highly labor-dependent for technology implementation due its low mechanization. Thus, family labor availability is critical for executing cacao cultivation properly.

Furthermore, cacao farmers usually have a diversified strategy of income generation, not just because cacao is 95% cultivated in multi-cropping arrangements but also because farmers need to face the non-production period of cacao (36 to 48 months) responding to their needs of food security and basic goods. The multiplicity activities on their farms create a labor burden hard to conciliate, especially to balance farming tasks and results in the short, medium, and long-term. During one of the interviews, an expert working for a government agency in agriculture sector with projects in the studied regions indicated:

> Cacao is a typical crop of the peasant economy, where the labor is familiar because hired labor is minimal…the family nucleus does the practices such as harvesting, processing, and disease control...

Likewise, two of the focus groups participants (one agronomist and educator, the other one was working by a local NGO) relating to local projects with cacao farmers said:

> Their daily needs make them allocate their labor to what gives them money in the short term. They see the income from cacao in the very long term, so when it's time to begin the production, those trees are full of monilia, and without prune, so the harvest is poor. It is a vicious circle…
>
> They always have some other activity as an income alternative because that is a way of mitigating risks; they distribute their labor in various activities that give them income. Cacao is very demanding of time; it does not have enough labor to attend it as they should.

Moreover, participants also mention the effects of cacao prices on the farmers' labor allocation decisions among different tasks within the farm. In that regard the expert working for a sectoral agency pointed out:

> When the price of cacao goes down significantly, they begin to earn less money than their theoretical labor wage, so they distract their labor in other activities that generate livelihoods ... they focus on chickens or cows, other short-term plants.

In the same sense, the 20-years expert fostering cacao across the country stated:

> When prices are low, nobody prunes cacao, nobody fertilizes, nobody does anything. Because their strategy is to try to locate their work capacity in other sources or in another way to ensure subsistence.

Finally, participants mentioned the implications of the difficult scenario for cacao farmers, regarding the labor allocation-income generation relationship on future generations' decisions around their occupation in cacao cultivation. A professional related to an agency for rural development have observed this during her experience of the last 10 years, she explained:

> There is also a labor issue, there is a decrease in the family size, therefore a reduction in the availability of labor. Besides, they do not want their children to be farmers, they have to be a driver or whatever but not a farmer because agriculture is difficult.

According to the survey results, labor use in cacao production in the Sierra Nevada is primarily from family members. Both in maintenance pruning and monilia control, and post-harvest activities, more than 90% is provided by the family, only between 6 and 9% is hired labor. Therefore, barely 4% of farmers considered they lack labor in their farms.

Association.

Regarding farmers' association, participants highlighted two elements relevant for the analysis. First, associations in the cacao sector have been seen as a mechanism to access projects and subsidies without necessarily reaching better performance in their crops, despite most of them addressing technological interventions. This relates, according to the participants, to the weak capability of associations to harness properly such initiatives. The expert working for a sectoral agency indicated based on his experience leading several programs including cacao cultivation:

> In Montes de María, the organizations are mentalized to take advantage of any project, their concern is the short term, and they lose their goals. For example, they do not consolidate revolving funds to pay for technical assistance or finance fertilization... associations often are looking for a new project that will give them the fertilizer or give them something.

Likewise, some participants highlighted organizations that have demonstrated the benefits of association to manage technological change and practices implementation. A coordinator of official programs for agricultural value chains improvement commented:

> I think associations and sectoral unions are important to promote the [technology] adoption… for instance, the coffee federation provide extension, good or bad, but they have been doing it for many years and have found a formula…results have been palpable in tree renovation and pest control. To achieve something like that in cacao, we need strong associations from a technical point of view.

Similarly, the 20-years expert fostering cacao across the country stated about associations:

> Some associations are mature today to address, for example, monilia control or better fermentation and drying. We need to strengthen associations, so their affiliates believe in them and participate in these processes; *Red Cacaotera* is an interesting case to understand how associations can work.

From the survey results (Table 22), I found that about 48% of surveyed farmers indicated they belonged to an organization.

Table 22. Survey Results of Principal Variables in Sierra Nevada de Santa Marta Region.

	Factors	Question	Results Sierra Nevada de Santa Marta							
Farmers' characteristics	Labor availability	4.1a This labor (pruning) is carried out with:	Own Labor	92%	Hired	6%	Both	2%		
		4.2a This labor (monilia control) is carried out with:	Own Labor	90%	Hired	8%	Both	2%		
		4.1a This labor (postharvest) is carried out with:	Own Labor	91%	Hired	9%	Both	0%		
		8.3 There is a shortage of labor on the farm?	Yes	4%	No	96%				
	Membership	5.1 Do you belong to some type of farmer organization?	Yes	48%	No	44%	NR	8%		
	Association capability	5.3 Type of Organization*	Association	9	Cooperative	0	Foundation	1	Other	0
		5.6 How would you rate the performance of the organization?	Excellent	9%	Good	73%	Fair	19%	Poor	0%
	Access to credit	8.1 Have you used credit to finance the cacao crop during the last 5 years?	Yes	5%	No	95%				
	Cultivation size	3.1a Total area of cacao plantation (Hectares)	Average	1.93	Min	0.5	Max	15	Stand Dev	1.97
	Cacao income dependence	1.12a What are the most important items in the generation of family income? (Ranking 3)	Cacao in Rank 1	71%	Cacao in Rank 2	7%	Cacao in Rank 3	2%	None	20%
	Farmer's age	1.5 Birth date (result ins years)	Max	83	Min	20	Average	49.4	Stand Dev	15
Context characteristics	Access to technical assistance	6.1 Have you received any technical assistance or agricultural extension service for cacao in the last 5 years?	Yes	54%	No	46%				
		6.2 How often did you receive the service?	Weekly	0.0%	Biweekly	1%	Monthly	11%	Quarterly	54%
			Semiannual	4%	Annual	30%	Other	2%		
	Quality of technical assistance	6.8 How would you rate the service you have received?	Excellent	24%	Good	59%	Fair	16%	Poor	1%
	Transportation availability	7.7 Type of transportation available to move the harvest between the farm and the site of commercialization	Animal	16%	Moto	86%	Car	5%	Bus	0%
			Truck	0%	Other	0%				
		7.8 How do you consider the transport supply?	Sufficient	31%	Fair	58%	Insufficient	11%		
	Transportation costs	7.6 Cost of transportation between the farm and the site of commercialization ($ / Sack)	Max	30,000	Min	2,000	Average	11,546	Stand Dev	5,769
	Market access and stability	7.1 Marketing mechanism for cacao	Directly	14%	Association or Cooperative	36%	Middleman	46%	Other	1%
			NR	3%						
	Commercialization conditions	7.3 Information about the last cacao sale (price $/Kg)	Max (Wet)	3,000	Min (Wet)	2,000	Average (Wet)	2,255	Stand Dev	367
			Max (Dry)	7,500	Min (Dry)	4,000	Average (Dry)	5,075	Stand Dev	633
		7.2 Do you have commercialization agreements? **	None	99%	Quality	1%				
		7.4a Are you looking for another cacao buyer?	Yes	23%	No	72%	NR	4%		
	Connectivity	1.11 Do you have cellphone?	Yes	82%	No	18%				
		1.11a Type of device	Conventional	59%	Smartphone	41%				
	Lack of basic goods and services	2.14 Access to utilities in the farm.	Aqueduct	3%	Sewerage	0%	Landline	0%	Electricity	36%
			Internet	0%	Natural Gas	0%	Propane	0%		

* This indicator count the number of organizations reported instead of times reported

** Options for responses are described in Table 21.

Among them, farmers mentioned GUARDABOSQUES DE LA SIERRA, APOMD, CAMPROACTIVO, FUNCA, AGROAZUL, ASOCAESMIC, ASOARHUACOS, SAN RAFAEL, APROCOSNET, and SIERRA MAR).

Farmers were also asked about their perception around organizations' performance. Most of them (73%) rated these organizations as having good performance, 19% rated them as fair, and 9% as excellent. Regarding second level organizations, 35% of surveyed farmers indicated they were related in some regard to FEDECACAO and 18% to Red Cacaotera.

Access to Credit – Funding – Investment Capacity.

Although the survey included few questions about credit for cacao cultivation, in the focus groups and interviews, the conversation turned around a set of considerations regarding cultivation funding beyond credit access and its implications on technology adoption.

From the survey, results show that scarcely 5% of farmers in the Sierra Nevada region had had any credit for cacao funding during the last 5 years. Therefore, most farmers lacked such an alternative to allocate monetary resources to technology implementation. In the same sense, participants of focus groups and interviews not only recognized limited credit access but also the limited capacity that farmers have to manage them properly. One of the ideas about this was pointed out by the coordinator of official programs for agricultural value chains improvement:

> ...the small farmer scarcely obtains a credit for maintenance and less for the establishment; besides, giving credit to them does not guarantee greater productivity. Those who access it invest little in their cultivation and use the rest to survive, to pay debts here and there, to buy things or for a new motorcycle, ... if they calculated to applied 1.5 tons of fertilizer per hectare; finally, they use half of it.

In addition, participants also mentioned that cacao farmers perceive as risky the resources invested in the crop, especially when they lack formal ownership of land; they do not have experience in certain practices/processes or have concerns about climate change. Some of the comments around this were mentioned by the 20-years expert:

> Beginners or small producers cannot set up a nursery; it requires an investment that is risky if you do not have experience. Cacao is technologically more capricious than other crops, so the selection of the pattern, the appropriate bag, the soil, the polisombra (shade cloth) is complex… it's too risky when a project is starting.

And the researcher implementing a project in technology adoption with cacao farmers in two different moments of the interview said:

1) In the program, we saw that the farmers who were owners were more willing to spend all the subsidy and their own resources on the crop ... it seemed logical to us, right? Finally, this will be inherited by their children.
2) The drought was extreme for a couple of months in the area at the end of the project, there was already the death of plants, and … there was also a loss of production. So, logically this also made them think, will it be good to spend on that? I can't predict the climate, and the climate is changing; I'm going to invest my money here? But what if my plants die? ... It was good to work in the middle of El Niño and understand that.

Second, the relationship between land ownership and credit access was highly recognized for many of the participants, for a technical assistant in Sierra Nevada de Santa Marta:

> The problem of titling and lack of succession processes in peasant households limits their access to credit; many times, the land is inherited from their parents, but there is no paper that guarantees it ... banks require a formal title.

Finally, some of them exposed their concerns about the influence of certain interventions on the investing propensity of farmers in cacao crops. Particularly, the proliferation of projects and programs in Montes de Maria municipalities based on input donations seems to discourage their own resources investment in some practices. The expert working for a sectoral agency commented:

> ... they (farmers) have little interest in going to bank services. You will find it in Montes de María... if you ask them, do you invest? Do you apply for credit for cultivation? They are going to say no or for what?... There is a perverse incentive that encourages them to jump from project to project. From an NGO to cooperation, from cooperation to the government, from the government to the Ministry, and from the Ministry to any agency... this is the way to keep the crop permanently subsidized.

In the opinion of the researcher implementing a project in technology adoption:

> …farmers invest the least in fertilization; we introduced it with irrigation, in a '*fertirriego*' trial we made, and it worked well, ... but they do not want to buy it; ... it is normal cause projects use to give them some packages [of fertilizer], this is when they apply it, otherwise, they do not.

However, other executors talked about how focusing on developing proposals that are affordable and adequate for farmers' budgets provides more relevant results for them. The 20-years expert talked about his most recent experience in a post-conflict region different from those included in this study but focused in enhancing cacao cultivation:

> We understood that the process of engaging them with technology implied prioritizing some practices that were high impact and low cost; it is useless to propose a list of activities that they cannot pay for. We focus on the control of monilia. It is a cost in time but

ultimately it is not money that comes out of pocket, it is a little more manageable for the producer.

Cultivation Size and Cacao Income Dependence.

Comments regarding cultivation size and income dependence coincide with the idea that most cacao farmers are small and embedded in the peasant economy. Many of interviewees and participants involved in programs execution at a national and local level indicated figures that provided a sense of the type of cacao areas in Colombia, and how it influences in technology adoption decisions. The 20-years expert indicated about records in his organization:

> ... we have more or less 3,000 records of farmers of our projects, 95% have less than five hectares, 80 to 85% have less than three and a half hectares, they are small. For that reason, we made a segmentation of beneficiaries because clearly, there are different responses depending on the region, the production system, and size…”.

Similarly, the member of the Consejo Nacional del Cacao indicated:

> ... in average terms, the cacao farm is 15 hectares ..., but of those only three have cacao. The other twelve hectares are in stubble, pastures, and some other permanent crops ..., a short group exceed that and reach five or seven [hectares], the one with 10 [hectares] is considered large. And there are others, such as in Cauca, where the average size is less than one hectare ... Most of these families have incomes that depend 70% on cacao... That is, it is a typical crop of the peasant economy.”

In the opinion of the technical assistant in Montes de Maria:

> ...all depends on the weight of cacao in their income; if it is very important then they are interested in caring for the crop: they occupy a good part of the workforce in it. If not, then they focus on the things that provide them the livelihood.

The professional related to an agency for rural development considered that:

> …cacao farmers contribute to rural development in a very important way, but their areas are very small, and that makes them very vulnerable to the lack of public goods, education, roads, health centers; that diminishes their capacities to improve their crops...

Finally, the 20-years expert also indicated that:

> The area and number of cacao trees per farm is very important data to evaluate any potential intervention... when you talk about cultivation practices or labors, what a farmer with 2 hectares and 1500 trees thinks and is able to do, is completely different compared to another with 700 trees in the same 2 hectares...

Data from the survey, for the region, were presented at the beginning of this Chapter; however, at the municipal level, they show that the average farm size and cacao area in Santa Marta are 11.32 and 1.66 hectares, respectively. In Dibulla, those figures are larger with 17.5 and 3.25 hectares, and finally, Ciénaga had the higher average in farm size (21.4 hectares) but the lower average in cacao area (1.64 hectares). On the other side, 71% of surveyed farmers indicated cacao as their first source of income, 7% ranked cacao as the second source, and 2% as of third. The remaining 20% have cacao cultivation as a minor item on their farm.

Gender.

Although during the focus groups and interviews, participants raised few comments about gender as a factor influencing technology adoption, during workshops and pilot survey, I could notice the higher participation or recognition of men as cacao farmers even though the women (some of them present at the moment of the visit) are involved in many of the farming tasks. As mentioned above, from surveyed farmers in the Sierra Nevada de Santa Marta, 74% of farmers who responded to the survey were men, and 26% were women. In the Montes de María region, participation was even more scarce, reaching barely 14%. These results are close to the data of the FNC-FEDECACAO survey for Bolivar and higher than those for Magdalena and La Guajira.

Regarding gender, as a factor influencing technology adoption, I could recover from the interviews two relevant comments. First, came from the volunteer working with cacao farmers' associations in Montes de María told me:

> ...women, mainly the spouses of farmers, are more interested in administrative tasks for the association and cacao transformation processes. Although they are an important part of the workforce for open pods and collect the cacao mass during harvest season, they are not usually executing the pruning or the fertilization... they prefer to work on adding value to the cacao, mainly motivated by commercial fairs promoted by the SENA.
>
> Most of my work has been finding training opportunities for them, both in chocolate bars elaboration and the development of other by-products... they are excited by producing cacao wine. However, we need to find funding to acquire some equipment and pods.

On the other hand, during a visit to applying the pilot survey, I could talk with a couple of farmers; when we were talking about the relevance of cacao within the income, she told me that she used to roast part of the cacao in their stove to elaborate nibs with her grain manual mill... part of this was for self-consumption.

Farmer Age and Generational Replacement

Participants mentioned two considerations regarding the implications of age of farmers and generational replacement in technology adoption. The first is related to the perception that the older the farmer, the less likely they are to change agricultural practices or integrate new technologies in cacao cultivation. Some of them referred to traditional cacao regions (e.g., Santander) as regions where any technological improvement will be tough. An expert working for multilateral agency focus on agricultural development told me:

> ... age is a key aspect, mainly because of the roles that different groups may have in the management of a production system ... if you are going to do some innovation in cacao, I would not recommend going to Santander. There are four generations doing everything in the same way, so proposing something different is complicated.

Besides, there are expectations for both children and parents, that the former will be occupied in activities different than agriculture; In this sense, experts raised concerns about no generational replacement in cacao farms. Consequently, incremental aging of a regional workforce can affect the execution of practices demanding physical efforts like pruning or disease control. The participant working for a local NGO in Montes de María mentioned:

> ... the older population it no longer has the same strength, …rural work is very hard and young people do not want to stay in the countryside, not only because of how hard it is but because of all the violent conditions that we lived in the past, so they don't feel secure there.

In addition, the expert working for a sectoral agency think that:

> Few people want to dedicate to agricultural activities, and those doing it today are people of a certain age, who want to continue with what they have done, and it has worked for them. We do not see that great changes will happen there.

Finally, the member the of Consejo Nacional del Cacao summarized:

> In general terms, in cacao farmers, there is nothing different from the typical Colombian rural man with a low level of education and little access to training. Perhaps there is a curious fact: in recent years, we have found a growing percentage of relatively young professionals dedicated to cacao... “

Although Montes de Maria and Sierra Nevada are no traditional cacao regions and cacao farmers' age is, on average, below (49 years) the national average (53 years), what is considered as a discouraging aspect for young people is the lack of basic goods and services besides secure conditions in rural areas.

Technical Assistance.

Technical assistance was one of the most-mentioned factors in each activity of data collection. In general, participants considered that technical assistance contributes to several goals associated with technology adoption. Among them are increasing farmers' capability for crop management through training and access to new technologies and practices, connecting farmers to other services like credit or public programs, and stimulating knowledge exchange between farmers. Farmers participating indicated that in some cases, having an extensionist makes them feel more accompanied and confident in their agricultural labor.

Despite the ideas above, participants indicated various elements of technical assistance that can discourage technology adoption. Those can be grouped into two topics: lack of farmers' access to technical assistance services, on the other side, the low quality of such services.

Regarding access, participants indicated that usually there is irregular or no provision of technical assistance services; besides, when it is available, its duration is extremely short, contrasting to the cacao crop cycle. Most of them mentioned that services operate between 6 and 18 months, negatively impacting results in the medium or long-term. Often processes are interrupted by administrative, funding or planning issues. In that regard, the professional related to an agency for rural development commented:

> In cacao, projects usually finance one year of serve or a maximum of 1.5 years, and the cultivation requires a minimum of three [years] to begin to bear fruit. That year or year and

a half without advisory are key; they (farmers) relax the effort [cause] they have no one to consult on emerging problems. Project planners believe that because it is intangible, it is not important, it is not like giving a tool or seedling.

Finally, about the quality of technical assistance, participants indicated they have some problems that affect their effectiveness over technology adoption. Among them, technology transfer capacity, lack funding and extensionists training and precarious labor conditions for technicians stand out. The member of the Consejo Nacional del Cacao expressed his concern about the way knowledge and technology development is delivered:

> We see a problem with the criteria unity of those who produce knowledge and technology and therefore of those who transfer it. When the knowledge is still in development, it is risky to put the farmer to try things that are not clear. The technical assistant is not prepared to react to the emerging questions. Look what is happening with the Cadmium issue... There is one genetic material that is more prone to [absorb] cadmium and another does not. Who says that? Where is the information? That is still under investigation. Others say that fertilizers also influence ... We do not know the truth about that, so there is a lot of uncertainty. That puts the credibility of the technical assistants at risk.

In the opinion of the researcher implementing a project in technology adoption:

> “Public technical assistance is generally very precarious; the UMATAs’ technicians do not even have the resources for gasoline for the motorcycle. They are very politicized, and sometimes the secure conditions limit them a lot. With difficulty, they obtain a project for the peasants.”

The expert working for a sectoral agency mentioned his experience regarding technical assistance quality:

> … salaries and conditions of the technicians in the field are terrible. They received, at that time, about 800,000 to a million pesos, and they had to pay for transportation. They are also very poorly prepared; when we hired the project technicians, there were few good agronomists, but they knew very little about cacao. We had to train them at La Granja Luker and Fundación Manuel Mejía.

Data from the survey in Sierra Nevada shows that 54% of surveyed farmers indicated they had received technical assistance in the last five years. Those having service, reported that attention was primarily quarterly (54%) or annual (30%). In fewer cases (11%), farmers reported monthly attention. Among the providers, farmers indicated they were FEDECACAO, Red Cacaotera, local associations, processors, ICA, ASOHOFRUCOL, SAG, and NGOs, among others.

According to farmers' responses, technical assistance scheme in the regions responds to training and visit model. Regarding confidence in technical recommendations provided by those services and their implementation, 75% of farmers said they always trust and implement, 14% almost always, 8% sometimes, and 4% almost never. Finally, 59% of them rated the service as good, 24% as excellent, 16% as fair, and 1% as poor.

Transportation.

Participants of all data collection activities expressed their concerns about constraints that transportation imposes on cacao sector. Particularly regarding the influences of transportation issues (low quality and high costs) over technology adoption, participants highlighted the effect on inputs access and prices, the logistical cost for technical assistance, limitations on cacao commercialization, and hiring labor. In that regard, the professional of the agency for rural development commented:

> “Transportation is a constant headache in rural areas; sometimes, it is so expensive that people prefer to lose production rather than harvesting it. If they go to the town and sell it, sometimes they cannot even cover the transportation expenses. They cannot recover all they have invested, so who can be motivated in that way?”

The researcher implementing a project in technology adoption referred his own experience regarding logistical constraints faced by rural population:

> ... In this project we had farmers located six hours away from the town. It was a few kilometers, but the road was destroyed, so we exceeded the transportation budget very quickly. In addition, it was necessary to find a middle point for meetings ... otherwise the peasants could not participate.”

During the pilot survey of the local farmers, one of my gatekeepers working as a volunteer related to a cacao farmers’ association told me:

> When it rains in Montes, the transportation cost rises from 20,000 to 30,000 pesos. The more mud there is on the road, the more expensive it gets. The same happens with the transportation cost per sack, whatever you sell. In addition, the cacao harvest season coincides with the rainy season.

Likewise, a technical assistant in the Sierra Nevada explained while showed me road issues:

> In the rainy season, the cars hardly come up to this part of the Sierra; it is dangerous [cause] the road is in poor conditions. That is why people have their 'burrito' (donkey) or a mule; some even have a horse to carry the harvest at least as far as the road below to reach the main [road]; otherwise, it is impossible...”

Finally, participants highlighted that especially in the post-conflict scattered rural areas, transportation could be a serious constraint to performance cacao cultivation properly. Thus, a professional working in the Peace-Agreement programs implementation indicated:

> There are serious problems of lack public goods in post-conflict municipalities... road infrastructure is a drama, farmers are left with the harvest stuck because the roads are terrible... besides, this situation increases prices of inputs, not to mention the difficulties for services provision, the technical assistance, etc.

Also, the 20-years expert indicated that:

> … verifying the suitability of the soil to grow cacao is not enough; you have to see that the logistic is at least manageable ... I understand that programs have to plant something to help them eliminating the illicit [crops], but that should not create a new problem for them. There are areas where there is no way to take the cacao out...In Vichada, the Air Force ended up subsidizing cacao transportation. This logistic is unsustainable.

From the survey, data shows that about 86% of farmers in Sierra Nevada primarily use motorcycles as transportation for cacao, 16% use animal transportation, while barely 5% use cars. Other types of transportation are not available in the region unless it is by special contracts. Besides, only 39% reported ownership of transportation, primarily horses, mules, and motorcycles.

Regarding the types of roads available in the area, 5% of surveyed farmers had mainly paved roads, 69% unpaved roads, and 26% had footpaths *("trochas").* The conditions of those roads were rated by farmers as poor 11%, bad 35%, acceptable 27%, good 26% and excellent 1%.

Finally, costs of cacao transportation ranged between 2,000 and 30,000 pesos per sack; it depends on distance, transportation type, and road conditions. Weather is a relevant factor in the transportation tariff (e.g., in rainy days costs may be different than in dry days). The next table shows average costs by distances.

Table 23. Transportation Costs in Sierra Nevada de Santa Marta

	Distance from farm to sales site	Time (min)	Costs ($/sack)
		Average	
Sierra Nevada de Santa Marta (Santa Marta, Ciénaga and Dibulla)	0.5-5 Km	32	6,702
	> 5-10 Km	47	9,000
	>10-20 Km	89	14,571
	> 20 Km	122	16,897

Considering all the above, only 31% of surveyed farmers considered transport supply as sufficient, 58% saw it as fair, and 11% claimed it was insufficient.

Market access and commercialization conditions.

Participants provided several ideas around how cacao commercialization conditions can influence technology adoption throughout focus groups and interviews. Most of these ideas are related to how levels and volatility in prices discourage farmers from producing higher volumes of cacao or cacao with better quality. In that way, low prices discourage proper application of practices and technologies for fermentation, drying, pest control, selection, etc. Some of the comments in that regard were from the member of the Consejo Nacional del Cacao and the leader of farmers association in the Sierra Nevada, respectively:

> 1)... they (the farmers) think about whether it (the price) compensates for the time and money put into the crop. When the payment is not good for them, it becomes a vicious cycle; they relax the tree care, then production and quality down, this affects their income, so, again they continue neglecting the cultivation, and so on. By the end, they become cacao collectors instead of farmers".

> 2)... the Colombian market is very perverse because you bring the sales site a well-dried, clean, and unmixed product, and they pay you 5,000 pesos per kilo, or you bring cacao full of stones, garbage, and they pay you the same 5,000 pesos, so what does the cacao farmer think? ... this arbitrary market management is a big discouragement".

Technical assistant from the Sierra Nevada also mentioned:

> "In the case of monilia, it is hard work that they have to do, all the time there watching and pruning, so they are waiting for a good reward, a good price."

On the other hand, participants highlighted the expectation of exporting cacao and receive better prices, as factors that encourage farmers to improve cacao management, in addition to the emergence of the niche markets for special cacao, and social recognition of farmers and their products. However, they also pointed that the effect of such factors is short-lived unless those are materialized through better prices or purchase guarantees. Regarding market strategies, the 20-years expert indicated:

> "... for me, there are two ways of technological change, one stimulated by the market and the other by productivity. In Colombia, the one that is working best is the market. If one tells them (farmers) ... we are going to export, then they ask what they should do?... they know that if they do not implement the postharvest correctly, they do not achieve the quality to export".

In that sense, the professional working in the Peace-Agreement programs implementation indicated, while referring an experience in Caquetá:

> Farmers begin to say ... since my cacao is exported, I cannot neglect my farm because foreigners come to visit it. So, you can find those farmers leaders who buyers visit in each

area; then the neighbor begins to be envious because they (foreigners) do not visit him. So, they think, ... I'm going to make my farm beautiful; I'm going to start copying my neighbor, I'm going to go to the workshops, etc.”.

Likewise, participants indicated that interlinkages between market access, technical assistance, and associativity could foster technology adoption. The expert working for a sectoral agency stated:

> Associations play a very important role in connecting with the market, for example, Aprocasur, which is the strongest in the entire southern area of Bolívar is buying 60 or 70% of the cacao from Santa Rosa, Simití, San Pablo… so they are very solid in the marketing at the national level. They have been working with Luker for a long time, breaking that is difficult. In addition, Luker is giving them a premium for not having cadmium.

At this point, the 20-years expert indicated:

> New small processors who are buying mainly wet cacao to do artisanal chocolate are becoming relevant to some regions' cacao market; I would highlight Sierra Nevada, Tumaco, Arauca, and Tolima, among others...The purchasing of wet cacao increases its quality because those who buy it dry it in better conditions.

Finally, quantitative data indicates that 14% of farmers reported they commercialize directly to processors, 36% commercialize through their associations, and about 46% depended on an individual middleman with tradition in the region. The survey also asked them if they were seeking different buyers; 23% of farmers declared they were doing it. Besides, cacao farmers in the Sierra Nevada region reported no having commercialization agreements with buyers, except 1% of them that declared having quality agreements.

Regarding cacao prices, by the time of the survey application, they were on average 5,075 pesos per Kg in the case of dry cacao and 2,255 pesos per Kg of wet cacao. Although local experts indicated that prices of dry and wet cacao have a theoretical relation of 3:1, data show that selling wet cacao seems more profitable than dry, which explains the interest of farmers to sell in that way. As mentioned before, this entails they do not execute fermentation and drying practices, liberating time to allocate to other activities.

Recognition.

Although recognition was no questioned in the survey to Sierra Nevada farmers, many participants during qualitative data collection highlighted this factor as driver of technology adoption. Next, some selected quotes from participants are transcribed to illustrate what implications social recognition could have in the implementation of practices and technology by cacao farmers. For that, the 20-years expert shared a particular anecdotal:

> ...I see how recognition increasingly motivates cacao producers. For example, the Villa Gaby Farm in Arauca is more visited than a diplomat office because it is wonderful, very well maintained. That has made it known. Even the company that purchases their cacao launched a chocolate bar in Japan, named Elizabeth as the farmer ... this shows everybody that to be successful, farmers do not have to produce two tons per hectare. Even with 800 Kg it's OK since the price and recognition of quality compensate the farmer.

The member of the Consejo Nacional del Cacao shared an example related to indigenous people:

> I tell everyone it is the pat on the back strategy or acknowledgment works. Here everyone knows about the Arhuacos' cacao, that the Arhuacos go to Paris, and everyone wants to take pictures with them, which is a little more motivating than strictly the productivity... thus, quality is more relevant”.

Finally, the professional working in the Peace-Agreement programs implementation indicated:

> ... within the marketing support strategy, we have looked for ways to participate in fairs, contests, and create a social motivation to improve the producers' self-esteem. Let them be recognized. Because many of them do not even know what the cacao they produce is used for... when they realize which products can come from cacao, they change their behavior about practices".

Connectivity.

Participants of the focus group and interviews highlighted the relevance of connectivity to improve communication and access to the information and services useful to crop management. However, the lack of internet among other basic utilities for households, limits those potential benefits. According to survey responses, 82% of cacao farmers surveyed in Sierra Nevada reported having cellphone. From them, 59% indicated they use a conventional device while 41% have a smartphone. Although the Internet connectivity through the phone was not asked about, none of them reported Internet connection in their household, and barely 36% of them had electricity. These results confirm what participants said around connectivity; for instance, the agronomist-educator mentioned during the conversation about technical assistance quality:

> ... in terms of connectivity there is a huge void ... having that even would help us to other forms of technology transfer... it is not easy but that it is necessary; ... today many producers have to come to the town to connect their phones to the few open internet networks we have; they even have to charge the phone (battery) in the pharmacy or in the internet cafe, ... they don't have electricity on their farms".

One of her colleagues, working as a technical assistance in Montes de Maria provided another idea linked to new generations:

> ... the very old farmers almost do not give importance to it, although they do see the advantages of using the phone... those who are less old receive calls from businesses or messages about (opportunities of) waged labor, so, they feel that it is useful ... but for the young it is indispensable. That is a mandatory technical change. I would say we are behind on taking advantage of that to convince them to continue working in cacao. "

The expert working for a sectoral agency also contributed about his concerns related to ICT:

> Regarding ICT, as it happens to the entire agricultural sector ... it is difficult since cacao farmers access to information through *'flecha'* (conventional cellphone) phones; although they can get text and communicate, many new platform and apps works for smartphones. So, the maximum we can do is sending to them text messages with prices and short news.

How have these factors influenced the technology adoption process in that context?

This section presents the results obtained from the set of bivariate linear regression models utilized to analyze the relationship between the Degree of Technology Adoption (DTA) of relevant technologies for improving cacao productivity and quality and the socio-economic dimensions of cacao farmers in the Sierra Nevada de Santa Marta.

(1) $\hat{Y}i = \hat{\beta}o + \hat{\beta}1Xi$, where $0 \leq Yi \leq 1$

The quantitative data was collected by the survey applied in Santa Marta, Ciénaga, and Dibulla by 2019. Both technologies to be evaluated and explanatory variables (socioeconomic factors) were identified through the focus groups, interviews, the pilot survey, and workshops described previously. Although the survey incorporated a bigger set of questions, those described in Table 24 comply with an adequate level of responses to be examined through the bivariate linear regression models.

Table 24. Description of Explanatory Variables

Variable	Description	
Age	Age of farmer	Years
Gender	Gender of farmer	0.Male, 1. Female
Education	Schooling level	Years
Experience	Time of experience cultivating cacao	Years
Cacao area	Number of hectares cultivated in cacao at the time of interview	Hectares
Cacao relevance	Cacao relevance for household income according to Top 3	1, 2 or 3 (relevant for income=1) Other (No relevant for income = 0)
Membership	Membership to any type of farmers organization (cooperative or association)	No=0 Yes=1
Technical Assistance	Technical assistance access in the last 5 years	No=0 Yes=1
Market access	Market mechanism utilized for cacao commercialization	0.Directly, 1. Association, 2. Middlemen 3. Other
Market access	Cacao Price	$ per Kg of dry cacao
Transportation costs	Cost of transportation per sack	$ per sack

As I mentioned in Chapter Four, technologies analyzed in this section correspond to technologies divisible in a number of practices. Therefore, the DTA was calculated for each farmer and technology as a ratio between 0 and 1, describing the degree to which the farmer applied a certain number of practices from the total recommended practices that make up the technology. Thus, a farmer who applies no practices has a DTA=0, and a farmer who applies all practices that make up the technology has a DTA=1. In addition to the dependent variables and number of practices in each, Table 25 shows the municipal and regional average of DTAs by technology.

Table 25. DTA for Relevant Technologies in Sierra Nevada de Santa Marta

	DTAm	**DTApr**	**DTAsd**	**DTAf**	**DTApf**	**DTAd**
N	198					
# Practices	5	7	N.A.	5	12	6
Santa Marta	0.64	0.69	0.57	0.77	0.83	0.83
Ciénaga	0.84	0.85	0.66	0.82	0.87	0.88
Dibulla	0.82	0.81	0.79	0.88	0.91	0.90
Sierra Nevada region	**0.71**	**0.74**	**0.63**	**0.80**	**0.85**	**0.85**

DTAm: moniliasis cultural control; DTApr: maintenance pruning; DTAsd: sowing density; DTAp: pre-fermentation; DTAf: fermentation; DTApf: pre-fermentation + fermentation; and DTAd: drying.

In that sense, a set of 58 bivariate linear regressions was run using STATA 16.1 as software for data analysis; Table 26 summarize the coefficients estimate for each DTA and socioeconomic variables (Appendix B. provide detailed results).

Table 26. Results of Bivariate Linear Regression Models of Technology Adoption

	DTAm (Monilia control)	DTApr (Maint. pruning)	DTApf (Pre+Ferm.)	DTAf (Fermentation)	DTAd (Drying)	DTAsd (Sowing density)
Variables	β	β	β	β	β	β
Gender						
Female	-.0256 (0.628) ns	-.0145 (0.758) ns	-.0347 (0.100) ns	-.0725 (0.033) **	-.1139 (0.002) ***	-.0563 (0.126) ns
Age	.0038 (0.012) **	.0043 (0.001) ***	.0007 (0.241) ns	.0011 (0.256) ns	.0021 (0.055) *	.0018 (0.089) *
Experience	.0149 (0.000) ****	.0131 (0.000) ****	.0026 (0.055) *	.0046 (0.037) **	.0030 (0.224) ns	.0067 (0.005) ***
Membership						
Yes	.2835 (0.000) ****	.2442 (0.000) ****	.0636 (0.001) ***	.0681 (0.023) **	.0516 (0.141) ns	.2280 (0.000) ****
Tech. Assistance						
Yes	.2583 (0.000) ****	.2436 (0.000) ****	.0780 (0.000) ****	.0843 (0.004) ***	.0555 (0.078) *	.2202 (0.000) ****
Market Channel						
Organization	.2133 (0.003) ***	.1629 (0.013) **	.0834 (0.004) ***	.0888 (0.056) *	.0160 (0.768) ns	.0988 (0.056) *
Middleman	.0946 (0.179) ns	.0372 (0.556) ns	.0361 (0.191) ns	.0327 (0.468) ns	-.0289 (0.585) ns	-.0545 (0.278) ns
Cacao relevance						
Top3	.6621 (0.000) ****	.5625 (0.000) ****	.1613 (0.000) ****	.1692 (0.000) ****	.2008 (0.000) ****	.2403 (0.000) ****
Ln (Cacao area)	.2035 (0.000) ****	.1781 (0.000) ****	.0633 (0.000) ****	.0777 (0.000) ****	.0704 (0.000) ****	.0738 (0.001) ***
DTAsd	.5921 (0.000) ****	.5075 (0.000) ****				
Ln (Cacao price)	.5664 (0.019) **	.6819 (0.001) ***	-.0933 (0.336) ns	-.1675 (0.258) ns	-.1712 (0.324) ns	.0102 (0.954) ns
Prune training						
Yes	.6452 (0.000) ****	.5213 (0.000) ****				

* p < 0.10>;** p < 0.05; *** p < 0.01; **** p < 0.001; ns (no statictically significat): p > 0,10.

According to the results obtained by the set of bivariate statistical models (Table 26), the degree of technology adoption of the practices (5) for monilia control (DTAm) is positively and statistically significantly (p<0.001) correlated with years of experience in cacao cultivation, membership to a farmer association or cooperative, access to technical assistance, cacao relevance within household income, variation of cacao cultivation area, sowing density, and prune training.

With a statistical significance more moderated ($p<0.01$) is the positive relationship between DTA and the cacao commercialization through an organization (association or cooperative) compared to commercialize directly or through a middleman. All of these variables were tested for multicollinearity and normal distribution.

Those with showing higher estimates and relevant goodness of fit were the prune training which entails farmers have had the opportunity of acquired the capability to address the monilia control; the cacao relevance highlighting that farmer who consider cacao as 1 of the top 3 sources of household livelihoods can be more willing to implement the monilia control since allowing the disease to spread could mean high risk for the household. Followed by sowing density, membership, and access to technical assistance.

In the case of maintenance pruning, the degree of technology adoption of its seven practices (DTApr) is positively and statistically significantly ($p<0.001$ and $p<0.01$) correlated with almost all the variables analyzed except with the farmer gender and marketing channel utilized for cacao commercialization. This means that, in addition to those variables explaining the DTAm, the farmer's age and variations in cacao prices are also positively correlated to DTApr compared to DTAm. In fact, the higher coefficient was for the variable measuring changes in price. This was followed by the degree of adoption of sowing density (DTAsd), cacao relevance as an income source, the farmers' organization membership, and access to technical assistance.

On the other hand, the DTAs of post-harvest activities (pre-fermentation, fermentation, and drying) are positively and statistically significantly ($p<0.001$ and $p<0.01$) correlated with a small group of the socioeconomic factors identified compared with the technologies for cacao cultivation management. For DTApf, which describe the degree of adoption of 12 practices including pre-fermentation and fermentation processes, variables to which is positively and statistically correlated are membership to a farmer association or cooperative, access to technical assistance, the cacao commercialization through an organization (association or cooperative), the cacao relevance as an income source, and variation in cacao cultivation area.

Regarding fermentation practices (5) evaluated in DTAf, the positive and statistically significant ($p<0.001$ and $p<0.01$) correlation is limited even more since it only shows a relationship with access to technical assistance, cacao relevance in the household income, and the cacao area variations. As well the goodness of fit is lower compared to bivariate linear regression models for pre-harvest technologies. In the same sense, the degree of technology adoption for drying practices (6) DTAd is positive and significantly ($p<0.001$ and $p<0.01$) correlated to cacao relevance as an income source and the variation in the cacao area. In contrast, the DTAd is negatively correlated to gender when the cacao farmer is female.

Finally, DTAsd, which represents the percentage of the recommended tree density that effectively one farm has, is positive and significantly ($p<0.001$ and $p<0.01$) correlated with years of experience cultivating cacao, membership into cacao organization, technical assistance access, cacao relevance and variations in cacao area.

Like in DTAm and DTApr, the goodness of fit and the coefficient estimates are higher for the bivariate linear regression models showing the correlation between DTAsd and cacao relevance as an income source, the membership, and access to technical assistance, than the other variables, and even compared with the post-harvest technology analyzes.

How Might Those Factors Constraining Technology Adoption in Cacao Farms be Counteracted in Such Regions?

A I mentioned previously, in this section I return briefly to qualitative data analysis to recap the different proposals, recommendations, and experiences shared by participants around the third research question, *How might those factors constraining technology adoption in cacao farms be counteracted in such regions?*

After organizing the transcriptions of data collected during interviews and focus groups on both regions, I executed the two-cycle coding process mentioned in Chapter Four. Results are presented in Figure 19; however, I provide a brief description and meaningful quotes that reveal what participants shared during the fieldwork.

As Figure 19 shows, results are organized into five topics: 1) Flexibility of the technology adoption process; 2) Intervention strategies; 3) Diversification of markets (specialty, bulk, etc.) and marketing channels; 4) Supporting strategies; and 5) Strengthening farmer organizations.

Flexibility of technology adoption process

The category named flexibility of technology adoption process summarizes recommendations directed to address the technology adoption as a process that needs to be done in a phased manner considering the economic and physical resources available to farmers. This process also entails achieving objectives progressively, looking for resolution of technical issues from the most to the least critical. Therefore, the prioritization of technological incorporations should respond to an objective analysis that considers not only the problems faced by the farmers but also their resources and short-, medium-, and long-term objectives. Some participants even suggested that they have executed processes of incremental technology adoption in which the priority was the moniliasis control, then advancing on other practices and technologies that favor other objectives.

In this sense, several participants raised the need to “de-standardize” the technological package promoted by public and private agents; thus, technical recommendations should be adjusted to the context conditions and respond to farmers' expectations. For example, technology and practices adopted have to cope with the complementary crops or material that farmers want to sow. Regarding incorporate more flexibility to the technology adoption process, the 20-years expert indicated:

> It worked for us to focus on just one thing which was the monilia control... that is the big bottleneck. Why tell them (farmers) to fertilize if they have half of the trees full of monilia?... I think that (technology) adoption should be addressed in a scaling-up way, starting with critical aspects and incorporating the rest step-by-step.

In the same sense, the professional working for a local NGO in Montes de María declared:

> We [in Colombia] need to focus on a few high-impact and low-cost practices, a strategy more adjusted to the real budget of farmers ...

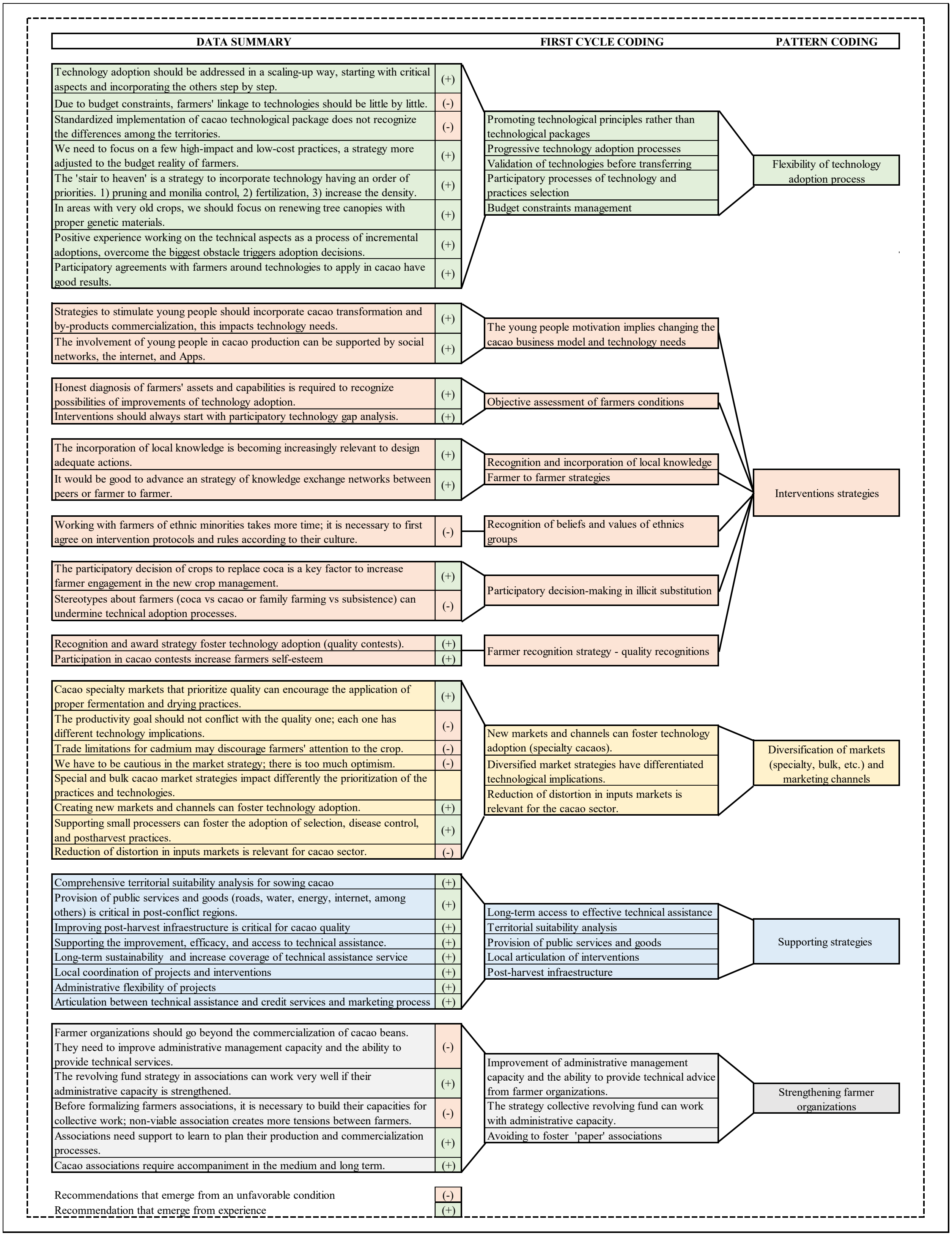
Likewise, the professional working in the Peace-Agreement programs implementation indicated:

> It is necessary to reevaluate the idea of the technological package to plant cacao. It is implemented in a highly standardized way that does not recognize the differences in the territories ... we can't continue combining cacao with plantain everywhere, ... we have to find other alternatives. Colombia is full of plantain with marketing problems".

Finally, the 20-years expert elaborated about his experience in a program that was running by that moment:

We have proposed what I call 'the stair to heaven,' which is to work first in pruning and monilia control to show farmers the benefits of doing that well ... when they get some productivity improvements, we can advance with fertilization... later increase the density, ... we cannot put more trees in a plot full of monilia or weeds, the key is in having an order of priorities.

Figure 19. Results of Two-cycle Coding of Recommendations and Suggestions.



Intervention Strategies

The second topic was summarized as intervention strategies, bringing together various recommendations directed to enhance the way in which cacao programs and projects are undertaken. To begin with, there are those related to generating a more favorable environment for the retention of young people in the cacao farms, understanding their interests in a different business model, and the implications of that on the technological needs for the farm. I also included the need to address interventions or projects aimed at technical change based on an objective participatory diagnosis of the technological gap; the third subtopic recognizes the importance of involving local, traditional, and ancestral knowledge in intervention strategies. Such knowledge would contribute a transdisciplinary and inclusive approach to the technology adoption processes; in addition to improve the alignment of the project design for the conditions of the participants. Precisely the professional working in the Peace-Agreement programs implementation indicated:

> It is necessary to have strategies to motivate young people but thinking beyond the cultivation ... they do not want produce cacao beans, they want to produce chocolates and other by-products, they want a different business, and this is also related to technology. For instance, require considering transformation practices".

On the other side, the 20-years expert made a comment calling for objective examination of real conditions of farmers as a requirement for project planning:

> I think projects have to do the real business numbers... they (farmers) are not having two harvests a year, maybe one and a half because the practices are delayed. In projects, you have to be honest with the reality of the available workforce; it is not true that they have the necessary labor for densities like the ones we propose of 1,200 [trees] or more.

As well, local knowledge has been seen as a mechanism to build projects more adequate for communities, in words of the professional of the agency for rural development:

> The incorporation of local knowledge is becoming relevant to design adequate actions to the communities, especially when we speak of agroforestry systems.

Subsequently, I included a subtopic of particular importance in the Sierra Nevada area. It refers to processes with ethnic communities (in this case indigenous people) because they should consider in their planning longer periods of execution that incorporate the time and resources required to address the conversation about the terms of working with those communities. Many of participants in the Sierra Nevada focus group agreed with the next statement from a technical assistant:

> Working with indigenous communities or other ethnic minorities takes more time; it is necessary to first agree on intervention protocols and negotiate the game rules... for instance, most of the indigenous do not accept grafting techniques.

The fifth subtopic refers in particular to regions involved in illicit crops substitution. Participants noted that certain stereotypes associated with such regions affect the entire development of cacao production: cacao growers' fear of being identified as former coca farmers, as well as participants highlighting that is a mistake that certain programs or interventions equate familiar agriculture to subsistence agriculture or poor communities. Such associations impact the behavior of farmers around cacao cultivation, in that regard the professional working in the Peace-Agreement programs implementation stated:

> ...farmers enrolled in projects for illicit crops substitution with cacao, after certain time, must overcome their condition of ex-coca growers and assume they are cacao farmers... we need to graduate these regions as cacao regions in few years and overcome the coca issue...

The condition of ex-coca farmers is affecting their relationship or association with other farmers.

Participants made several recommendations around participatory decision-making about crops or activities to replace coca: farmers themselves should define their alternatives. Besides, those farmers involved in substitution with cacao without having a historical or cultural link with that species use to have greater difficulties investing in technologies for cacao. Farmers in Montes de María indicated that Plan Colombia introduced cacao cultivation in the area; despite that they have been caring the crop, many of them prefer to spend their time and efforts on avocado or yam. The professional working for a local NGO in Montes de María claimed:

> Farmer's participation in decision of crops with which they want to replace coca is substantial; imposing products can be harmful to their engagement with the crop. Substitution programs (of illicit) must make this selection participatory.

Finally, social recognition of cacao quality (e.g., "Cacao de Oro" contest and other regional or international ones) and of the farmers themselves emerges as interesting element for generating self-esteem and motivation to improve practices or technologies along the production process. A coordinator of official programs for cacao trade and international promotion said:

> I believe in the recognition strategies for them (farmers); they are increasingly enthusiastic about the awards... the competition has worked well for the coffee sector, in cacao, we are doing well, but that has to become a culture.

Diversification of Markets (Specialty, Bulk, etc.) and Marketing Channels

The third topic identified in the coding process I named market diversification, in which the recommendations of the participants that are based on market development strategies and describe the effects that these would have on the adoption of technology are gathered. According to some participants, farmers' market strategies could have two different objectives: first those focusing on special cacaos and therefore rely on all the necessary tasks to achieve the quality requested by the processors. Second, the strategy of increasing volume to sell in the bulk market, which implies prioritizing different efforts and investments in technology.

Despite this, producers often aim to increase production volume, although some participants believe that this strategy is not viable for those producers that are too small (less than 1 hectare). For them, smallholders could benefit more from a strategy aimed at the specialty cacao market. A coordinator of official programs for cacao trade and international promotion said:

> Markets for specialty cacaos that prioritize quality can stimulate the best application of fermentation and drying practices, ... primarily in small ones that will not be able to increase productivity easily.

The 20-years expert also made a commented:

> Creating new marketing channels can foster (technology) adoption; the proliferation of small processors and artisan chocolatiers purchasing beans with a specific quality have made that farmer who wants to sell them need comply with certain practices ... a more direct market relationship seems to be beneficial.

Similarly, the expert working for a sectoral agency stated:

> …perhaps supporting small processors, we can foster a different business model which recognizes fair prices for the efforts on selection, disease control, and postharvest practices.

In contrast to the previous optimistic comments, the member of the Consejo Nacional del Cacao mentioned his concerns regarding expectations promoted by programs and experts:

> There is a race for new products, planting materials, and high expectations for the specialty market. You have to be cautious and not to be too optimistic. Those markets are still very restricted for us ... and we have to solve the cadmium issues.

Supporting Strategies

I identified the fourth topic as supporting strategies; it is focused on the provision of public goods and services as a mechanism to support cacao production, and therefore, the likelihood of adopting technologies and practices. Participants mentioned energy, water, internet, and road infrastructure as the most relevant services. In that sense, some participants pointed out that projects should examine the region's feasibility to grow cacao considering the availability of them or supply them by other projects.

In addition, this topic also includes other conditions that can offer support to the adoption of technologies, among them the improvement or access to post-harvest infrastructure, access to the technical assistance service for a period according to the crop cycle, and credit access. Participants also highlighted that technical assistance services enhancing is necessary; these should have the capacity to connect farmers with credit and market. In addition, such services should contract local extensionists, training them frequently and promoting technological principles instead of technological packages. In this topic the member of the Consejo Nacional del Cacao emphasized in public goods and services, he stated:

> We need that the PDETs effectively provide basic public goods and services in post-conflict areas; I would focus on improving roads, connecting energy, the internet ... and advancing land titling ...

While the agronomist-educator in Montes de María focused on infrastructure:

> Improving the post-harvest infrastructure would help us make a big leap in the quality of cacao beans.

And The 20-years expert stated:

> We cannot accept a trial-and-error technical assistance... remember what happened with the genetic materials planted in Putumayo and Caquetá a few years ago, they were not for the area ... so, after years expecting pods, people obtained nothing, this cost a lot ... time, effort, money, and institutional reputation.

Other recommendations were made by the expert working for multilateral agency focus on agricultural development and the professional from the agency for rural development, respectively:

1) I recommend working mainly with people from the territory... farmers feel better with technical assistants from the area ... if necessary, you have to train local technicians instead of bringing people from outside.
2) An important effort should be to carry out a national training program for cacao technicians, not to unify the knowledge of the (technological) package but to align technical principles, for example, the principles of agroforestry ...

Strengthening Farmers' Organizations.

Finally, the last topic covers the strengthening of farmers' organizations. In this, the proposals or recommendations aimed at building capacity in the producer associations were agglomerated. This task should focus on resource management capacity, revolving fund management, and technical assistance for the members of the organization. On the other hand, the participants expressed concern about the proliferation of associations that, although they are legally constituted in practice, do not operate due to their precarious capacity, making it difficult to take advantage of the collective activity to meet the technological needs of their crops. Some of the comment in this regard were made by the expert working for a sectoral agency and the member of the Consejo Nacional del Cacao:

1) Strengthening farmer organizations should be a key strategy to promote technology adoption, the association should go beyond the commercialization of cacao beans ... but they need help. Its main problem is administrative management capacity and the ability to provide technical advice to its farmers. It would be good to train in depth some members of each association.

1) The revolving fund strategy in associations can work very well if their administrative capacity is strengthened. I know some that emerged with the Productive Alliances that have worked well, for example, using fund resources for fertilization, and then farmers return the money when harvesting the cacao.

2) Before formalizing farmers' associations, it is necessary to build their capacities for collective work... the proliferation of 'paper' associations is not convenient, non-viable association creates more tensions between farmers ... initiatives can promote an association behavior in practice and then do the paperwork.

CHAPTER 6: DISCUSION AND CONCLUSIONS

This research ultimately sought to understand factors that influence the adoption of technologies and practices in cacao farms of post-conflict regions, from the standpoint of multiple actors involved with the Colombian cacao sector,

Understanding the relevance of contributing to the conciliation of development alternatives feasible, adequate, and respectful to communities of such regions, this research deepens on two cases: Montes de Maria and the Sierra Nevada de Santa Marta, reflecting the results of workshops, focus groups, interviews, and surveys that consulted more than 300 people, but mainly cacao farmers and other actors of the localities. Besides, I utilized a database from the APP for Cacao MdM (USAID) and diverse literature and secondary data to document both contexts widely.

Although selected regions are defined by more than 10 municipalities each, I executed the analysis for 3 municipalities in each region. They were Ciénaga, Santa Marta, and Dibulla in Sierra Nevada de Santa Marta region, and San Jacinto, San Juan Nepomuceno and El Carmen in Montes de María.

Contrary to most research on technology adoption in agriculture, this research is based on a participatory selection of the technologies and the identification of factors to be incorporated in the analysis. Therefore, there was no predefined set of technologies or variables nor an established unique technological package to test in the fieldwork or analysis. In that sense, this research contributes to recognize that farmers construct their own technology itineraries along with their experiences, which implies that they select a set of particular techniques, technologies, and practices over time; besides, their decisions respond to farmers' complexities and particular conditions.

As mentioned previously, I collected quantitative and qualitative data in all municipalities; they were analyzed and reported in detail in Chapter Five. Therefore, this section summarizes most of those findings critically, building conclusions and recommendations for future interventions and the research agenda required for cacao in post-conflict regions.

Like most of the research is undertaken in rural areas o Colombia, unexpected situations constrained the quantitative data collection in the Montes de María region; however, I utilized qualitative and secondary data to supplement the analysis.

General Findings

As the theoretical framework of this research proposed, technology adoption a dynamic and complex process that is ultimately is best explained by a group of variables that may be interrelated and feed each other at the local, regional, or national level; even international phenomena can courage or discourage decisions about technology adoption.

To examine the socioeconomic and technological factors that influence technology adoption in cacao farms in Montes de Maria and the Sierra Nevada, I addressed four questions in practice. 1) What are the relevant technologies for cacao farmers in those regions? 2) What socioeconomic factors influence the adoption of such technologies? 3) How these factors influence the adoption process? and 4) How do might those factors constraining technology adoption to be counteracted?

Regarding technologies to assess during the research, qualitative data collection and workshops allowed to gather information about the individual technological itineraries of cacao farmers, building the regional catalogs of technologies, and ranking the top five of relevant technologies using criteria of relative advantage, complexity, compatibility, trialability, observability (Rogers, 2003), ease of use and usefulness (Davis, 1985). The selection responded to two objectives that cacao farmers usually pursue, increasing productivity and improving cacao quality.

The rationality behind this methodology responded to the need to understanding what technologies and practices farmers accepted and considered relevant by their technological characteristics but exhibited non-desirable levels of adoption due to factors that prevented their use. Thus, I explored primarily the social and economic factors that might influence the degree of adoption of such technologies to find the constraints that could be tackled in the pathway construction for the illicit crops' substitution with cacao.

In the case of the Sierra Nevada de Santa Marta (Santa Marta, Ciénaga, and Dibulla), participants agreed that relevant technologies to increase cacao productivity are Moniliasis cultural control), maintenance pruning, and irrigation, while technologies and practices to improve quality are fermentation practices despite the growing trend towards selling wet cacao (baba). Sowing material and density are considered relevant for both goals.

On the other side, in Montes de María (San Jacinto, San Juan Nepomuceno, and El Carmen de Bolívar), participants considered that relevant technologies to improve productivity are maintenance pruning, diseases control also focused on cultural control of Moniliasis, control of squirrels, and adequate sowing material and density. They also agreed on post-harvest practices for quality improvement, especially because cacao in the region is sold entirely dry.

The data collection to identify socioeconomic factors and how they impact technology adoption was addressed by interviews and focus groups in Bogotá and the two regions of analysis. Both techniques were useful for hearing the national, regional, and local perspectives about such factors. The coding process revealed the topics or themes that emerged as socioeconomic conditions that might influence technology adoption in cacao farms.

The theoretical framework was the principal input to select preliminary topics for coding the qualitative data, but in turn, it was challenged regarding the validity of your proposals; in that sense, some of the resulting topics confirmed variables considered in the theoretical framework, others emerge as new topics to evaluate. In general, factors were classified as framers or context characteristics (factor):

Labor availability: labor availability emerged as a factor that potentially explains technology adoption, considering that cacao is labor-intensive cultivation. Besides, it competes for farmer time distribution with other crops and activities contributing to the short-term income generation. In both regions, labor in cacao is primarily family labor which is increasingly scarce due to young people's migration and the aging of farmers. Moreover, because cacao provides a long-term economic return of labor, farmers prioritized other activities during the growing season (36 to 48 months), looking for more immediate household sustenance. Waged labor is not a common solution for labor requirements in cacao; precisely, the long-term economic returns seem to prevent farmers from contracting external labor because they do not have any source for payment.

The labor availability in cacao farms in Montes de Maria is usually distributed with other relevant crops like yam, cassava, plantain, citrus, and avocado. According to harvest seasons and comparative prices, they motivate farmers to spend their time in one or another activity. In the Sierra Nevada de Santa Marta region, cacao farmers also have a diversified livelihoods portfolio competing for labor with cacao. Among farming activities contributing to household income and food security are plantain, coffee, beekeeping, avocado, and fruit trees, therefore some participants' comments about the need of implementing an objective assessment of farmers' assets, including labor, at the moment of design projects for fostering cacao.

Although the survey included some questions seeking to understand labor availability constraints, few responses were collected in that regard. Considering all the above, future analyses on technology adoption for cacao cultivation in post-conflict regions might pay more attention to the implication of having high-diversified household livelihoods and labor availability constraints.

Membership: the farmer membership to an association or cooperative organization was included as a potential predictor of technology adoption. Farmers' associations or cooperatives are useful for organizing collective action to provide technical assistance, inputs purchasing, commercialization, investments, etc.; In both regions, although primarily in Montes de Maria, farmers' organizations are identified by many actors as a mechanism to obtain resources and donations without increasing capability of solving problems.

In general, participants considered that cacao associations or cooperatives in Colombia and particularly in post-conflict regions are weak and lack of capability to provide services. Except for few highly recognized examples like Aprocasur in south Bolivar, Aprocampa in Boyacá, and Guardabosques de la Sierra, most cacao organizations need a big effort to support and provide the services and goods their members need.

In qualitative and quantitative analysis, organization membership appears as a predictor of the adoption of practices and technologies required for monilia control, maintenance pruning, pre-fermentation processes, and sowing density. The degree of technology adoption (DTAs) calculated for all of these technologies are positively and significantly correlated with belonging to an organization. In that sense, other efforts should be developed to study this topic in-depth, inquiring about the types of organizations, governance practices, and the desired conditions to be able of providing goods and services to the members.

Technical Assistance: like membership, access to technical assistance emerged as a predictor considering its potential to support and encourage farmers during technology adoption. However, in both regions, participants considered it could have low impact and capability to address technical change due to its resource scarcity, low quality, and short-term action compared to the crop-cycle of cacao (few projects and programs incorporate technical assistance or extension for cacao farmers for more than 1.5 years). In addition, poor labor conditions, lack of training, and scarce connections between technical assistants and R&D systems were also recognized as constraints to have adequate technical assistance for cacao cultivation.

In quantitative analysis, the degree of technology adoption (DTAs) calculated for all of the selected technologies are positively and significantly correlated with access to technical assistance in the implementation of monilia control practices, as well as in the maintenance pruning, the processes of pre-fermentation, and fermentation, and sowing density. Therefore, efforts to provide technical assistance or extension services in a timely and quality manner are required to improve cacao productivity and quality.

This finding could also be complemented with the results of quantitative analysis of pruning training as a predictor of DTAm (monilia control) and DTApr (maintenance pruning). The degree of practice implementation in both cases is positively and significantly correlated with access to pruning training. In that sense, actions for providing such capabilities should be considered a priority in post-conflict regions where cacao is fostered.

Cacao relevance and cultivation size: Most cacao cultivations are small (less than 3 hectares); therefore, cacao is considered an activity of the peasant and familiar economy. According to the survey, at the municipal level, the average farm size and cacao area in the Sierra Nevada and Montes de María were similar and lower, respectively, than the sizes accounted for in the FNC-FEDECACAO survey (cacao cultivation size in Magdalena: 2.17 hectares, La Guajira:1.99 hectares, and Bolivar: 2.58). The dimension of farms and cultivations has implications in farmers' capability and choices regarding technology adoption. In the same sense, cacao income dependence influences the decisions of farmers regarding crop management.

In the quantitative analysis, DTAs for all analyzed technologies showed a positive and statistically significant ($p<0.001$ and $p<0.01$) correlation with cacao relevance within household income and cultivation size. This finding is relevant considering that in the Sierra Nevada de Santa Marta, 71% of surveyed farmers indicated cacao as their first source of income, 7% ranked cacao as the second source, and 2% as of third. The remaining 20% have cacao cultivation as a minor item on their farm.

In addition, for participants, small cacao farmers are also vulnerable in the context of lack of public goods; many of them are located in municipalities with scarce access to basic utilities. Thus, both regions included in the study showed the precarious quality of life conditions for farmers.

Credit and Investment capacity: Cacao small farmers lack access to credit due to limited formal land property and low capacity to comply with requirements of the bank system. Associations are seen as enablers of access to collective credit. This was identified by participants as a factor constraining the investment in technology for cacao farmers. In general, it is considered they have a low capacity of investment in cultivation. Projects have promoted a culture of subsidizing fertilization in both regions. Finally, climate variability expressed as La Niña or El Niño diminishes the investment propensity of farmers, particularly in the Sierra Nevada de Santa Marta, particularly in areas with frequent periods of water scarcity (Santa Marta and Dibulla).

Age: Although cacao is also affected by the aging of rural areas, participants consider that age does not necessarily influence cacao farmers' attitude to learn or apply any practice or technology. Quantitative analysis confirmed such an idea cause age was not a statistically significant predictor for DTAs. Conversely, what was identified as a concern is that young people do not want to remain in agriculture in the countryside; therefore, participants considered urgent increase research and action to understand this problem and identify mechanisms to counteract it.

Gender: Although few participants made comments about gender as a factor influencing technology adoption, during workshops and pilot survey, I could notice the higher participation or recognition of men as cacao farmers even though the women (some of them present at the moment of the visit) use to be involved in many of the farming tasks. As mentioned above, survey data shows that in the Sierra Nevada de Santa Marta, 74% of farmers included in the study were men, and 26% were women. In the Montes de María region, participation was even more scarce, reaching barely 14% for women. These results are close to the data of the FNC-FEDECACAO survey for Bolivar and higher than those for Magdalena and La Guajira.

Gender as a predictor of technology adoption was only significantly correlated with drying practices adoption (DTAd). However, this correlation was negative. This predicts that women farmers have a lower DTAd than men regarding drying practices.

Transportation costs: the high cost of transportation also was indicated as a factor discouraging technology adoption. Lack of road infrastructure, transportation scarcity, and distances impact transportation costs, limiting access to inputs and making more expensive logistics for technical assistance and commercialization. Survey data showed that higher distance and/or time spent in transportation increase transportation costs per sack. Weather conditions like rainfall and floods also increase costs.

Cacao prices: Low prices and restrictive commercialization conditions discourage technology adoption. However, some participants consider that market diversification stimulates technology implementation, primarily for the special and certified cacaos. Other less optimistic think such opportunities are still market niches that should be carefully analyzed to rationalize the expectations.

Wet cacao commercialization seems a new strategy for buyers in the Sierra Nevada who want to control the postharvest process and increase quality. At the same time, for farmers, it appears a more profitable alternative since the price per Kg is one-third of dry cacao price, and farmers do not assume post-harvest costs and risks. In addition, some workshop participants indicated this might be an alternative for those lacking postharvest infrastructure and time constraining. However, a group of farmers expressed concern about the potential loss of knowledge regarding fermentation and drying, which could put them in difficulties if wet cacao buyers disappear.

Cacao quality recognition: this factor emerged as a new aspect to be considered. According to participants in this research, cacao quality recognition increases producers' self-esteem and encourages farmers to improve cacao productivity and quality. However, its effect could be temporary unless it is supported by stability in the purchase and the fair price. Thus, commercial agreements appeared as potential instruments to mitigate risk between buyers and farmers.

Lack of connectivity limits farmers' knowledge sharing and technology adoption process. Likewise, the lack of minimum conditions for public goods and services and security conditions affects processes that support innovation.

In general, results show that socioeconomic factors influencing the adoption of technologies oriented to increase yields in cacao farms (e.g., maintenance pruning and monilia cultural control) are partly different from those affecting technologies focused on beans quality (e.g., post-harvest practices). Therefore, understanding properly the production strategy pursued by farmers and objectives of programs and sectoral policies becomes substantial to design interventions to foster technology adoption effectively and suitably.

In that sense, sectoral strategies oriented to increase productivity should take into account socioeconomic variables like the years of experience of farmers, access to technical assistance, membership in an organization, sowing density, pruning training, and cacao price to guide the initiatives to cacao fostering in postconflict regions.

On the other hand, strategies focused on cacao quality have to consider the farmers' access to technical assistance in the case of fermentation practices adoption, and farmers' gender in the case of drying, besides the cacao relevance within household income (i.e., cacao as one of the top three sources of family income) and the cacao area. These latter two variables resulted positively correlated to all technologies' adoption degrees (DTAs) measured in this study.

Finally, I also found, both in Santa Marta and Montes de María, technology adoption difficulties linked to the low access to infrastructure and equipment for the proper post-harvest process. Although farmers know and apply many of the practices involved in the fermentation process, the use of plastic containers, fiber sacks, and buckets for fermentation is high, affecting cacao quality. Similarly, drying beans over patios, mats, or paved areas is common practice.

Finally, recommendations to improve technology adoption in cacao farms of post-conflict regions are associated to:

- Promoting technological principles rather than technological packages standardized, specialized considered the frame of cacao-based agroforestry systems.

- Undertake inclusive and participatory processes of technology and practices selection, especially in vulnerable communities.
- Fostering incremental technology adoption processes recognizing budget and knowledge constraints of farmers and local actors in general
- Focusing on addressing principal bottlenecks, "pains of the farmer," critical aspects based on particular technology gap analysis. The most generalized problem in both regions is the monilia disease.
- Motivating young people interested in cacao cultivation, recognizing that their expectations entail different technological needs than traditional ones.
- Recognition and incorporation of local knowledge, using strategies like a farmer-to-farmer technology transfer.
- Recognition of beliefs and values of ethnic groups (e.g., indigenous and black communities) and their implications in budgeting and project timeline.
- Farmer and cacao quality recognition strategy can improve technology adoption; however, it should be supported by adequate market conditions.
- Promote the diversification of markets (specialty, bulk, etc.) and market channels
- Long-term access to effective technical assistance and credit
- Local articulation of interventions associated with cacao, particularly in areas of projects and program proliferation.
- Elaborating detailed census of post-harvest infrastructure by regions, lacking infrastructure and tools imposes huge restrictions on technology implementation.
- Strengthening farmer organizations by improving administrative management capacity and the ability to provide technical advice from farmer organizations.

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Appendix A: Survey Questionnaire

Survey questionnaire: Socioeconomic factors influencing technology adoption in cacao farms in Colombia

Date: __ __ / __ __ / __ __ __ __   Questionnaire N°___________

Phone: ________________________________

1. Farmer's information

1.1 Name (s)	1.2 1st Last name	1.3 2nd Last name	1.4 ID Number and place of expedition	1.5 Birth date	1.6 Gender
			from:____________________	DD / MM / AAAA	[ ] Male [ ] Female [ ] Other

1.7 Schooling (Last grade coursed)	1.8 Experience in cacao cropping (# Years)	1.9 Does the farmer live in the farm?	1.9a (if 1.9 is No or by seasons) Specify time farmer spends between house and farm according to the most used transport mean	1.10 Family composition (# members living in the farmer's house)	1.11 Does the farmer have cellphone?	1.11a Type of Device
[ ] None Elementary: [ ] 1° [ ] 2° [ ] 3° [ ] 4° [ ] 5° Highschool: [ ] 6° [ ] 7° [ ] 8° [ ] 9° [ ] 10° [ ] 11° [ ] Technical [ ] Bachelor [ ] Other Specify ____________		[ ]Yes [ ] No [ ] By seasons	By walking:_______Min Moto:_______Min Bicycle: _______Min Car:_______ Min Bus:______Min Other:______Min	[ ] Spouse [ ] Children [ ] Grandchildren [ ] Parents [ ] Others Specify? ____________________ How many are older than 18: ________	[ ] Yes [ ] No	[ ] Conventional [ ] Smartphone

1.12 What are the crops or activities to which the farmer has dedicated in the last 5 years? (Multiple selections are possible, select other for those non-agricultural items)	1.12a What are the most important items in family income generation?	1.12b In the last 6 months, have the farmer or any member of his/her family been unable to take breakfast, lunch or dinner due to lack of resources or sufficient food?	Other comments:
[ ] Cacao [ ] Plantain [ ] Maize [ ] Yam [ ] Yucca [ ] Beans [ ] Avocado [ ] Citrics [ ] TreeFruits [ ] Vegetable [ ] Maderables [ ] Coffee [ ]Beekeeping [ ] Livestock [ ] Fish farming [ ] Pigs [ ] Poultry [ ] Milk [ ] Other Specify: ___________	1.________________ 2.________________ 3.________________	Very often [ ] Frequently [ ] Occasionally [ ] Rarely [ ] Never [ ]	

2. Farm (Location and characteristics)

2.1 Departament	2.2 Municipality	2.3 Corregimiento	2.4 Village	2.5 Farm name	2.6 Total farm area (Has)	2.7 Altitude (m.a.s.l)

2.8a Georeference - Latitude	2.8b Georeference - Longitude	2.9 Precipitation (mm/year)	2.10 What are the months of greatest rainfall in the region?	2.11 Type of soil	2.12 Mechanisms of access to land
N _____° _____' _____"	W _____° _____' _____"		[ ] Jan [ ] Feb [ ] Mar [ ] Apr [ ] May [ ] Jun [ ] Jul [ ]Aug [ ] Sep [ ] Oct [ ] Nov [ ] Dec	[ ] Alluvium [ ] Vega [ ] Sierra [ ] Piedmont [ ] Other Specify________________	[ ] Adjudication [ ] Inheritance [ ] Purchase [ ] Lease [ ] Other Specify ________

2.13 Land tenure type	2.14 Access to utilities in the farm	2.14a Have other energy source?	2.15 Construction materials of housing in the farm	2.16 ¿How the farmer considers the conditions of his/her house?
[ ] Legal possession [ ] Individual property [ ] Collective property [ ] Occupation [ ] Lease [ ] Sharecropping [ ] Comodato [ ] Other:	[ ] Clean water [ ]Sewerage [ ] Landline [ ] Electricity [ ] Internet [ ] Natural Gas [ ] Propane	[ ] Solar energy [ ] Wind energy [ ] Other:	Roof: _____________________ Walls:___________________ Floor: _____________________ Doors: ____________________ Bathroom:_________________ Kitchen:___________________	[ ] Excellent [ ] Good [ ] Fair [ ] Bad [ ] Poor

Other comments ____

3. Crop information

3.1 Type of production system	3.1a Total area for cacao	3.1b Planting design	3.2 Type of permanent shade	3.2a Number of plants	3.3 Type of temporary shade	3.3a Number of plants	3.4 Number of CACAO trees planted	3.4a Year of planting
[ ] Monoculture [ ] Agoforestry System (SAF) [ ] Intercroping no SAF [ ] Introduced in native forest [ ] Otro Cuál? ________________________	______Ha	[ ] Triangle [ ] Contours [ ] Square [ ] None	1.________________________ 2.________________________ 3. ________________________ [ ] No permanent shade	1.__________ 2.__________ 3.__________	1.________________________ 2.________________________ 3. ________________________ [ ] No temporary shade	1.__________ 2.__________ 3.__________	1.______________________ 2.______________________ 3. ______________________	1.__________ 2.__________ 3.__________

3.5 Type of planting material	3.5a Type of Hybrids		3.5b Rootsotck (genetic material)		3.5c Budwoods materials used (for grafting)				3.6 ¿Does the farmer select budwoods from his/her farm ?	3.7 ¿Has the farmer been trained for grafting cacao plants?
[ ] Hybrids [ ] Clones [ ] Both	a.	b.	a.	b.	a.	b.	c.	d.	[ ] Yes [ ] No	[ ] Yes [ ] No

3.7a ¿ Does he/she consider him/herself able to do grafting in cacao plants?	3.8 Source of cacao material planted	3.8a Project Name and year of sowing material reception	3.9 Cacao production (Dry)
[ ] Yes [ ] No	[ ] Project [ ] Donation of other farmer [ ] In house production [ ] Donation from Association [ ] Purchase [ ] Other _____________	Name of project: ___________________________ Year__________ Name of project: ___________________________ Year_________ Name of project: ___________________________ Year_________	a. Total production in 2018 ______Kg b. Volume produced in 2019 (To the date) _____Kg c. Production estimated to the end of 2019: ______Kg

Other comments____

4. Relevant technologies utilization

4.1 Maintenance Pruning			
4.1a This labor is carried out with:	[ ] Own labor [ ] Hired labor [ ] other ________________________		
4.1b Practices and components (Is the practice executed?)		Yes	No
a. Disinfection of tools		[ ]	[ ]
b. Elimination (cutting) of misdirected, disequilibrating and malformed main branches		[ ]	[ ]
c. Elimination (cutting) of diseased or dead parts of the tree.		[ ]	[ ]
d. Elimination of chupons and maintenance of the main trunk		[ ]	[ ]
e. Blunting and cuttinng of extended or crisscrossed brush branches		[ ]	[ ]
f. Elimination of low branches or directed towards the ground.		[ ]	[ ]
g. Application of pruning sealer		[ ]	[ ]
4.1c Frequency	[ ] more than twice a year [ ] twice a year [ ] once a year [ ] Never [ ] Other _______		
4.1d When is executed MP ?	[ ]Jan [ ] Feb [ ] Mar [ ] Apr [ ] May [ ] Jun [ ] Jul [ ] Aug [ ] Sep [ ]Oct [ ]Nov [ ] Dec		
4.1e Tools used for pruning:	[ ] Hand pruning shears [ ] Aerial pruning shears [ ] Handsaw [ ] Machete [ ] Knife [ ] Chainsaw [ ] Electric Scythe [ ] Gurbia [ ] Other Which? ________________		
4.1f Inputs used for pruning:	[ ] Pruning sealer bordaleza [ ] Immunization paint [ ] Sodium hypochlorite [ ] Copper oxychloride [ ] Ridomil [ ] Gloves [ ] Helmet + mask [ ] Other, Which? __________________		
4.1g What is the level of importance given by the farmer to the pruning?	[ ] Not important [ ] Moderately important [ ] Important [ ] Very important		

4.2 Disease Management (Monilia cultural control)			
4.2a This labor is carried out with:	[ ] Own labor [ ] Hired labor [ ] other ________________________		
4.2b Practices and components (Is the practice executed?)		Yes	No
a. Identification of pods with symptoms or signs of the disease (humps or deformations or black spots with spores)		[ ]	[ ]
b. Pods cutting before sporulation (release of contaminating spores)		[ ]	[ ]
c. Disposition of diseased pods on the ground		[ ]	[ ]
d. Coverage of diseased pods with leaf litter		[ ]	[ ]
e. Monitoring of the disease in rainy seasons and excess humidity		[ ]	[ ]
4.2c Control frecuency	Dry Season	[ ] Weekly [ ] Biweekly [ ] Monthly [ ] Other	
	Rainy Season	[ ] Weekly [ ] Biweekly [ ] Monthly [ ] Other	

4.3 Has he/she received training for pruning?	[ ] Yes [ ] No	4.4 Does he/she consider him/herself able to prune his/her crop?	[ ] Si [ ] No

4.6 Harvest and post-harvest			
4.6a This labor is carried out with:	[ ] Own labor [ ] Hired labor [ ] other ________________________		
4.6b Practices and components (Is the practice executed?)		Yes	No
a. Identification and cutting (with tool) of ripe pods in the field		[ ]	[ ]
b. Separation of suitable and unsuitable pods (sick, over-ripe, others)		[ ]	[ ]
c. Classification of pods by variety (clones or hybrids)		[ ]	[ ]
d. Opening of suitable pods without affecting the beans		[ ]	[ ]
Form of opening: [ ] cross section [ ] upper cut close to the peduncle [ ] blow with mallet [ ] Other			
e. Cleaning and disinfection of harvest containers		[ ]	[ ]
f. Shelling and removing the beans from the pod without placenta		[ ]	[ ]
g. Classification of beans by healthy and shape conditions		[ ]	[ ]
h. Stacking of empty pods for incorporation into the soil as fertilizer		[ ]	[ ]
4.6c Fermentation: Practices and components (Is the practice executed?)		Yes	No
a. Deposit of harvested beans in fermentation vessels on the same day of harvest		[ ]	[ ]
b. Rest of cacao mass without turning for 36 hours (anaerobic fermentation)		[ ]	[ ]
c. Turning cacao mass every 24 hours for 2 to 3 days (aerobic fermentation)		[ ]	[ ]
d. Measurement of cacao beans temperature in the fermentation vessel.		[ ]	[ ]
e.Control of food safety during the fermentation (control access of animals, polluting factors and impurities)		[ ]	[ ]
f. Mix of harvested beans on different days		[ ]	[ ]
4.6d Drying and preparing: Practices and components (Is the practice executed?)		Yes	No
a. Deposit of the fermented beans in layers with a maximum thickness of 5 cm		[ ]	[ ]
b. Drying with partial exposure to the sun (first hours of the morning and last of the afternoon) during the first day of drying		[ ]	[ ]
c. Gradual increase in sun exposure from the second day onwards		[ ]	[ ]
d. Frequent movement of beans for uniform distribution of heat and drying from the second day onwards		[ ]	[ ]
e. Evaluation of the degree of drying required (7% humidity)		[ ]	[ ]
f. Elimination of impurities, moldy beans, matches and voids.		[ ]	[ ]
g. Execution of the cutting test and classification according to the Technical Quality Standard - NTC 1252 of ICONTEC		[ ]	[ ]
4.6e Periodicity of harvesting in:	High harvest	[ ] Weekly [ ] Biweekly [ ] Monthly [ ] Other	
	Low harvest season	[ ] Weekly [ ] Biweekly [ ] Monthly [ ] Other	

4.5 Sowing material selection		
4.5a Which of the following factors does the producer analyze when selecting the materials he wants to plant?	Yes	No
a. Selection of materials according to the feasibility of the environmental conditions (soil, water, temperature, luminosity, etc.)	[ ]	[ ]
b. Selection of materials considering the cob index (number of corncobs required to obtain 1 kg of dry beans)	[ ]	[ ]
c. Selection of materials considering the beans index (weight of 100 fermented and dried cacao beans.)	[ ]	[ ]
d. Selection of materials for tolerance to diseases and pests	[ ]	[ ]
e. Selection according to the compatibility analysis between the planting materials	[ ]	[ ]
f. Selection of materials according to the market requirements (quality, price, demand)	[ ]	[ ]

4.5b Does the farmer know the Matrix of Compatibility between Clones? (Fedecacao 2013)	[ ] Yes [ ] No

4.7 Infrastructure, tools and equipment used:	
4.7a Harvest	
[ ] Hand scissors [ ] Short machete [ ] Wooden mallet [ ] Cutting blade or "donkey" [ ] Cutting machine [ ] Other Which one? ______________________________	
4.7b Fermentation	
[ ] Basket [ ] canvas or jute bags [ ] Posuelo or Canoa [ ] Rohan-type tray [ ] Wooden raised boxes with holes [ ] Wooden drawers without lifting [ ] Wooden boxes for staircase [ ] "pasera" [ ] plastic bucket [ ] Other ____________________	
Size and capacity of container	Length (_______Cm) x Width (______Cm) x Height (______ Cm) Capacity for dry beans (____Kg) Capacity for wet beans (____ Kg)
Type of wood used for boxes, trays and "paseras"	[ ] Campano (rain tree) [ ] Caracoli (Wild cashew) [ ] Other ____________________
What type of cover does the farmer use for fermentation?	[ ] Plantain leaves [ ] jute bags [ ] Other [ ] None
4.7c Drying	
[ ] Paseras [ ] Elba House [ ] Wood boxes [ ] Elba Sliding carriages [ ] Solar Canopy [ ] Wooden pallets [ ] Troja [ ] Jute bags [ ] Patio [ ] Drying tunnel [ ] Other: ____________________	
Tools for beans moving	[ ] wood rake [ ] wooden shovel [ ] other, which? ____________

4.8 Does the farmer use irrigation in the cacao crop?	4.8a Source of water and distance to the cacao crop (when apply) (mts)		4.8b What type of irrigation system he/she use in cacao crop?	4.8c Source of resources to fund the irrigation system?
[ ] Yes [ ] No	[ ] Groundwater (well) [ ] Reservoir [ ] Ponds [ ] Rainfall [ ] Canal [ ] River [ ] Streams [ ] Irrigation District [ ] Other ________________	________mts	[ ] Drip [ ] Gravity [ ] Sprinkling [ ] Micro-sprinkler [ ] Flood [ ] Manual (bucket) [ ] Other____________	[ ] Personal resources [ ] Credit [ ] Project (donation) [ ] Other ________________

4.9 Does the farmer make soil test?	4.9a Frecuency	4.10 In case the soil analysis has been made within a project, the farmer:					
[ ] Yes [ ] No	[ ] Every year [ ] Every other year [ ] By project [ ] Other ________________	Has received the results ?	[ ] Yes [ ] No	Has know the results ?	[ ] Yes [ ] No	Has utilized such information to take decisions over the cacao crop?	[ ] Yes [ ] No

4.11 Does the farmer know the Good Agricultural Practices for cacao production?	[ ] Yes [ ] No	Other comments ______________________________
4.11a Does the farmer apply the Good Agricultural Practices for cacao production?	[ ] Yes [ ] No	______________________________
4.11b Is the farm certified in Good Agricultural Practices for cacao production?	[ ] Yes [ ] No	______________________________

5. Participation in Organizations

5.1 Does the farmer belong to some type of farmer organization?	5.2 Name of organization and year of affiliation	5.3 Type of Organization	5.4 ¿Is the farmer involved in the directive board?
[ ] Yes [ ] No	Name ______________________ Year _______ Name ______________________ Year________	[ ] Association [ ] Cooperative [ ] Foundation [ ] Other ___________ [ ] Association [ ] Cooperative [ ] Foundation [ ] Other ___________	[ ] Yes [ ] No What role does he/she play? ________________

5.5 Fecuency of organization meetings	5.6 How would you rate the performance of the organization?	5.7 Do you have any kind of link with the following organizations?	
[ ] weekly [ ] biweekly [ ] monthly [ ] quarterly [ ] biannual [ ] annual [ ] never	[ ] Poor [ ] Fair [ ] Good [ ] Excellent	Fedecacao [ ] Yes [ ] No	Red Cacaotera [ ] Yes [ ] No

5.8 Which of the following activities are carried out by the organization?			5.8a In which activities does the producer participate?		Other comments
Actividad	Yes	No	Yes	No	______________________________
a. Agricultural Technical Assistance / Agricultural extension	[ ]	[ ]	[ ]	[ ]	______________________________
b. Training (conferences, technical tours, field days, method demonstrations, etc.)	[ ]	[ ]	[ ]	[ ]	______________________________
c. Production of plant material, clonal garden administration, nursery or grafting service	[ ]	[ ]	[ ]	[ ]	______________________________
d. Marketing of products (negotiation with clients or purchase of harvest)	[ ]	[ ]	[ ]	[ ]	______________________________
e. Acquisition of inputs collectively	[ ]	[ ]	[ ]	[ ]	______________________________
f. Cacao transformation and by-product processing	[ ]	[ ]	[ ]	[ ]	______________________________
g. Project management and execution	[ ]	[ ]	[ ]	[ ]	______________________________
h. Agricultural credit management	[ ]	[ ]	[ ]	[ ]	______________________________

6. Technical Assistance / Extension Services

6.1 Have you received any technical assistance or agricultural extension service for CACAO in the last 5 years?	6.2 How often did you receive or receive the service?	6.3 Which of the following agencies has been the main provider of the service?
[ ] Yes [ ] No	[ ] weekly [ ] biweekly [ ] monthly [ ] quarterly [ ] biannual [ ] annual [ ] other	[ ] Umata [ ] Epsagro [ ] Secretariat of Agriculture [ ] Guild (Fedecacao) [ ] NGO [ ] Input providers [ ] Association/Coorperative [ ] Buyer/Middlemen [ ] Other ________________________

6.4 The service received focuses on which of the following aspects? (You can mark more than one option	6.5 Which of the following methods have been used in the service received? (Can you mark more than one option)	6.6 Do you trust the recommendations of extension service and implement them in your crop?
[ ] Agronomic [ ] Commercial [ ] Administrative [ ] Environmental [ ] Food Quality/Food Safety [ ] Other ________________________	[ ] Field days [ ] Method demonstrations [ ] Demonstration plots [ ] Talks [ ] Visits [ ] Field Schools [ ] Technical tours [ ] Manuals [ ] Brochure / Handouts [ ] Other ________________________	[ ] Always [ ] Almost always [ ] Sometimes [ ] Almost never [ ] Never

6.7 What is the main limitation to adopting technology?	6.8 How would you rate the service you have received?	Other comments:
[ ] Resources [ ] Time [ ] Low labor availability [ ] Lack of training [ ] Difficult access to technology [ ] Other: ________________________	[ ] Poor [ ] Fair [ ] Good [ ] Excellent	________________________

7. Market Access

7.1 Harvesting marketing mechanism	7.2 Do you have pre-sale agreements for the harvest? (You can mark more than one option)	7.3 Information about the last cacao sale			
[ ] Directly [ ] Association/Cooperative [ ] Middlemen [ ] Other Which? ___________	[ ] Volume Agreement [ ] Price Agreement [ ] Quality Agreement [ ] None	Date	Price/Kg	Volume	[ ]Wet [ ] Dry Type

7.4 Name of the buyer and time of the business relationship (How long ago the farmer sell to this buyer?)	7.4a Have you sought to sell to another buyer?	7.5 Distance between the farm and the point of purchase or delivery of cacao	7.6 Cost of transport between the farm and the point of purchase ($ / Sack)	7.7 Type of transportation available to move the harvest from the farm to the point of purchase
Buyer: ____________________ Years: ________________	[ ] No [ ] Yes Which? __________________	In Kms ___________ Time (Min.)___________	Trip 1: __$________________ Trip 2: __$________________ Trip 3: __$________________	[ ] Animal [ ] Moto [ ] Car [ ] Bus [ ] Truck [ ] Other [ ] Animal [ ] Moto [ ] Car [ ] Bus [ ] Truck [ ] Other [ ] Animal [ ] Moto [ ] Car [ ] Bus [ ] Truck [ ] Other

7.7a Does he/she use own	7.8 How does he/she consider the transportation offer?	7.9 Type of road between the farm and the main or primary road	7.10 How does he/she consider the conditions of the road between the farm and
[ ] Yes [ ] No	[ ] Sufficient [ ] Fair [ ] Insufficient	[ ] Paved [ ] Paved footprint [ ] Gravel road [ ] Dirty Road [ ] Other ____________	[ ] Poor [ ] Bad [ ] Fair [ ] Good [ ] Excellent

Other comments

____

8. Other aspects

8.1 Have he/she used credit to finance the cacao crop during the last 5 years?	8.1a Where he/she has obtained the credit?
[ ] Yes , Year__________ [ ] No	[ ] Banco Agrario [ ] Other Bank [ ] Microfinance Institution [ ] Cooperative [ ] Revolving Fund [ ] Warehouse of supplies [ ] Other which? ____________

8.2 Workforce utilized in the cacao crop	People (number) for age range			
	< 18 years	18 - 35 years	36-50 years	> 50 years
Familiar				
Permanent external wage labor				
Temporary external wage labor				

8.3 There is a shortage of labor on the farm?
[ ] Yes [ ] No

8.3a In what months does the shortage of labor occur?		
January	May	September
February	June	October
March	July	November
April	August	December

8.5 Daily wage in the region	
$	
8.5a Daily wage includes meals [ ]Yes [ ]No	
8.5b Working hours	
AM	PM

Final comments of the farmer

The main objective of this instrument is to gather primary information to advance the various analyzes required by the study on socio-economic factors that influence technological innovation in cacao farms in two regions of Colombia, which will allow understanding which of these factors are explanatory elements of the decisions of technological adoption and therefore are the main focus for formulators and implementers of projects of promotion, support and improvement of cacao production. In no case the information collected through this will be presented individually or identifiable. The data collected will be treated anonymously.

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Thanks

Appendix B. Results of Bivariate Linear Regression Models of Technology Adoption

	DTAm (Monilia control)				DTApr (Maintenance pruning)				DTApf (Pre+Fermentation)			
	Coef	**Stand. Err.**	**R- Sq.**	**Obs.**	**Coef**	**Stand. Err.**	**R- Sq.**	**Obs.**	**Coef**	**Stand. Err.**	**R- Sq.**	**Obs.**
Gender												
Female	-.0256102	.0528404	0.001	198	-.0145201	.0469893	0.0005	198	-.0347472	.0210188	0.0138	198
	(0.628) ns				(0.758) ns				(0.100) ns			
Age	.0038519	.001524	0.0316	198	.0043883	.0013405	0.0518	198	.0007262	.0006177	0.0070	198
	(0.012) **				(0.001) ***				(0.241) ns			
Experience	.0149812	.003325	0.0947	196	.0131001	.0029596	0.0917	196	.0026666	.0013841	0.0188	196
	(0.000) ****				(0.000) ****				(0.055) *			
Membership												
Yes	.2835644	0.0440715	0.1878	181	.2442716	.0384808	0.1837	181	.0636964	.0182593	0.0637	181
	(0.000) ****				(0.000) ****				(0.001) ***			
Tech. Assistance												
Yes	.2583148	.0420418	0.1629	196	.2436975	.0372706	0.1806	196	.0780427	.0175438	0.0926	196
	(0.000) ****				(0.000) ****				(0.000) ****			
Market Channel												
Organization	.2133333	.071746	0.0568	191	.1629814	.06468	0.0567	191	.08343	.0282211	0.0569	191
	(0.003) ***				(0.013) **				(0.004) ***			
Middleman	.0946237	.0700975			.0372671	.0631939			.036154	.0275726		
	(0.179) ns				(0.556) ns				(0.191) ns			
Cacao relevance												
Top3	.6621234	.0390097	0.5951	198	.5625641	.0368163	0.5436	198	.1613462	.0216669	0.2205	198
	(0.000) ****				(0.000) ****				(0.000) ****			
Ln (Cacao area)	.2035657	.027332	0.2206	198	.1781136	.0244039	.2137	198	.0633526	.0115373	0.1333	198
	(0.000) ****				(0.000) ****				(0.000) ****			
DTAsd	.5921655	.0948967	0.1686	194	.5075358	.084589	0.1579	194				
	(0.000) ****				(0.000) ****							
Ln (Cacao price)	.5664253	.2389653	0.0279	198	.6819828	.2098713	0.0511	198	-.0933521	.0967896	0.0047	198
	(0.019) **				(0.001) ***				(0.336) ns			
Prune training												
Yes	.6452381	.0327797	0.6641	198	.5213239	.0337797	0.5486	198				
	(0.000) ****				(0.000) ****							

* p < 0.10>;** p < 0.05; *** p < 0.01; **** p < 0.001; ns (no statictically significat): p > 0,10.

	DTAf (Fermentation)				DTAd (Drying)				DTAsd (Sowing density)			
	Coef	**Stand. Err.**	**R- Sq.**	**Obs.**	**Coef**	**Stand. Err.**	**R- Sq.**	**Obs.**	**Coef**	**Stand. Err.**	**R- Sq.**	**Obs.**
Gender												
Female	-.072589	.0338038	0.0230	198	-.1139706	.0370865	0.0466	195	-.0563069	.0366399	0.0122	194
	(0.033)	**			(0.002)	***			(0.126)	ns		
Age	.0011385	.0009984	0.0066	198	.0021247	.0011015	0.0198	195	.0018255	.0010667	0.0150	194
	(0.256)	ns			(0.055)	*			(0.089)	*		
Experience	.0046823	.0022308	0.0222	196	.0030593	.0025078	0.0077	193	.0067751	.0023827	0.0408	192
	(0.037)	**			(0.224)	ns			(0.005)	***		
Membership												
Yes	.0681436	.029655	0.0287	181	.0516239	.0348668	0.0123	178	.2280426	.0293877	0.2560	177
	(0.023)	**			(0.141)	ns			(0.000)	****		
Tech. Assistance												
Yes	.0843455	.0287065	0.0426	196	.0555312	.031338	0.0162	193	.2202063	.0280539	.2449	192
	(0.004)	***			(0.078)	*			(0.000)	****		
Market Channel												
Organization	.0888116	.0460965	0.0274	191	.0160595	.0542481	0.0088	188	.0988671	.0513729	0.1048	187
	(0.056)	*			(0.768)	ns			(0.056)	*		
Middleman	.0327256	.0450374			-.0289855	.053033			-.054512	.0501293		
	(0.468)	ns			(0.585)	ns			(0.278)	ns		
Cacao relevance												
Top3	.1692253	.0377679	0.0929	198	.2008891	.0415507	0.1080	195	.2403338	.0392535	0.1633	194
	(0.000)	****			(0.000)	****			(0.000)	****		
Ln (Cacao area)	.0777162	.0192404	.0768	198	.0704722	.0217458	0.0516	195	.0738467	.0209377	0.0608	194
	(0.000)	****			(0.000)	****			(0.001)	***		
Ln (Cacao price)	-.1675712	.1563106	0.0058	198	-.1712427	.1732255	0.0050	195	.0102327	.1788219	0.000	194
	(0.258)	ns			(0.324)	ns			(0.954)	ns		

* p < 0.10>;** p < 0.05; *** p < 0.01; **** p < 0.001; ns (no statictically significat): p > 0,10.